Semisimple Deformations of Contraction Algebras for Smooth Threefold Flops

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Abstract

Building on the work of Bryan–Katz–Leung, Hua–Toda and Kawamata, we study deformations of contraction algebras and the local geometry of the scheme of associative algebra laws. Let A be the basic contraction algebra of a completed projective flopping contraction of a smooth complex threefold. We construct a finite free polynomial family that becomes a constant semisimple algebra away from the origin, with the original curve projectors fixed. For each restricted positive root β, the split generic fibre of this family has n_β matrix blocks of size $\sum_i \beta_i$, and the labelled projector e_i has rank β_i on each block. Here n_β is the corresponding genus-zero Gopakumar–Vafa invariant. For every finite free associative formal deformation of A with geometrically semisimple generic fibre, that geometric generic fibre has the same matrix factors and the same ranks for any lifts of the original projectors. The scheme of associative algebra laws is smooth at the point defined by A, and its unique irreducible component through that point is the closure of the corresponding semisimple orbit.

The proof uses the curve divisors described by Katz–Morrison and Nabijou–Wemyss, universal algebra and cocenter presentations, and the orbit method of Jelisiejew–Smoktunowicz. The presentations also prove the three geometric Hua–Toda flatness assertions over arbitrary complete Noetherian local complex bases.

Near A, the centre and cocenter form dual vector bundles compatible with arbitrary base change. We also examine which commutative deformations of the centre arise from deformations of the algebra. The type D algebras $D_{n,2}$ have first-order centre deformations that do not lift, showing that the centre morphism can fail to be smooth even at an unobstructed algebra.

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1 Introduction

Does a contraction algebra determine the semisimple geometric generic fibres of its associative deformations? For several exceptional curves, this asks for both the matrix factors and the ranks of the original curve projectors. The geometric families of Hua–Toda [16, Theorem 4.2] and Kawamata [22, Theorem 3.1(2)–(3)] provide candidate semisimple algebras against which to compare the geometric generic fibres of arbitrary finite free associative formal deformations.

The cocenter $HH_0(A) = A/[A, A]$ supplies the numerical comparison: when A is semisimple, $\dim HH_0(A)$ is the number of matrix factors of A. Together with the orbit method of Jelisiejew–Smoktunowicz [19, Section 1 and Theorem 2.14], the cocenter calculation proves smoothness of the scheme of associative algebra laws at A and determines which semisimple geometric generic fibres can occur. The resulting local geometry also controls how centres vary in associative families.

Let $f : X \to \operatorname{Spec} R$ be the completion at the exceptional image point of a simple flopping contraction of a smooth complex threefold, with exceptional curve $C \cong \mathbb{P}^1$. The contraction algebra of Donovan–Wemyss [7] records noncommutative deformations of $\mathcal{O}_C(-1)$. For the basic Van den Bergh generator $R \oplus N$, put

$$\Lambda = \operatorname{End}_R(R \oplus N)^{\mathrm{op}}, \qquad A = A_{\mathrm{con}} = \Lambda/\Lambda e_0 \Lambda,$$

where e_0 projects onto the free summand. Thus the quotient removes endomorphisms that factor through finite sums of that free summand. Bryan, Katz and Leung deform the neighbourhood of C to a family with isolated $(-1,-1)$-curves and identify the number of such curves in each homology class with the corresponding Hilbert-scheme multiplicity [6, Section 2, Proposition 2.3 and Lemma 2.8]. For a contraction of length ℓ, these multiplicities are the genus-zero GV invariants $n_1, \dots, n_\ell$; n_j counts curves of class $j[C]$ [6, 32]. Katz's local DT–GV construction recovers the same multiplicities for contractible curves [34, Proposition 3.3 and Corollary 3.4]. The invariants n_j define the semisimple algebra

$$S_{\mathrm{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \tag{1.1}$$

Toda's width formula gives $\dim_{\mathbb{C}} A = \sum_j j^2 n_j = \dim_{\mathbb{C}} S_{\mathrm{GV}}$ [32, Theorem 1.1]. A semisimple deformation realizes this numerical equality at the level of multiplication.

For finite-dimensional algebras over an algebraically closed field k of characteristic zero, we say that A *deforms to* S if a finite free unital $k[[s]]$-algebra has special fibre A and geometric generic fibre $S \otimes_k \overline{k((s))}$. Jelisiejew–Smoktunowicz prove that this deformation notion agrees with the polynomially split and orbit-closure formulations [19, Theorem 2.20]. In the polynomially split formulation, the family is defined over $k[u]$ and becomes the constant algebra S after inverting u. Thus, if the geometric generic fibre is semisimple, S is a product of matrix algebras over k, and its block sizes and multiplicities agree with those of that fibre.

Conjecture 1.1 (Wemyss; cf. [19, Questions 1.1–1.2])**.** *The algebra A deforms to S_{GV}, and every complex semisimple algebra to which A deforms is isomorphic to S_{GV}.*

Semisimple generic fibres and smoothness of algebra laws

For the general statements, let $f_0 : X_0 \to Y_0$ be the completion at an exceptional image point of a projective flopping contraction of a smooth complex threefold. Fix the normalized basic

Van den Bergh generator and its labelled nonfree projectors $(e_i)_{i\in J}$. A general surface section gives an ADE diagram with retained subset J. We use the simultaneous ADE resolutions of Katz–Morrison [26, Theorem 1], together with the restricted-root formulation and perturbation construction of Nabijou–Wemyss [28, Theorem 2.6 and Section 2.6]. A nonzero restricted positive root $\beta = (\beta_i)_{i\in J}$ is the vector of retained coefficients of a positive root; distinct proportional vectors remain distinct. Let n_β be the corresponding genus-zero GV invariant, using the curve-counting description in [28, Section 3.1], and set

$$s_\beta = \sum_i \beta_i, \qquad N = \sum_\beta n_\beta, \qquad E = \bigoplus_{i\in J} \mathbb{C}e_i.$$

A marking is the fixed embedding $E \hookrightarrow A$. The embedding records which projector belongs to which curve. The associated Peirce spaces are the vector spaces e_iAe_j; together they decompose the underlying vector space of A. On a matrix factor, the marking is described by the ranks of the orthogonal projectors e_i, whose sum is the identity of that factor.

Write X_d for the scheme of associative laws on a fixed d-dimensional vector space with fixed identity, and let $G = \mathrm{Stab}_{\mathrm{GL}_d}(1)$ be the group of allowed changes of basis. The fixed-idempotent scheme X_E also fixes the Peirce spaces e_iAe_j and the labelled vectors e_i; denote its change-of-basis group by G_E. Orbits in these schemes represent isomorphism classes with the indicated data fixed.

Theorem 1.2 (Semisimple generic fibres and smoothness)**.** *For the basic contraction algebra A of f_0, put*

$$S_E = \prod_\beta \mathrm{End}_{\mathbb{C}} \left(\bigoplus_{i\in J} \mathbb{C}^{\beta_i} \right)^{\times n_\beta}.$$

(i) *There is a finite free marked $\mathbb{C}[u]$-algebra $\mathscr{P}$ with $\mathscr{P}/(u) \cong A$ and $\mathscr{P}[u^{-1}] \cong S_E \otimes_{\mathbb{C}} \mathbb{C}[u, u^{-1}]$. On each β-factor, e_i has rank β_i.*

(ii) *Every finite free formal associative deformation of A over $\mathbb{C}[[t]]$ with geometrically semisimple generic fibre has, on that geometric generic fibre and with respect to any lifts of the original projectors, the matrix factors and ranks prescribed by S_E. Forgetting the projectors, the corresponding complex semisimple algebra is $\prod_m M_m(\mathbb{C})^{\times N_m}$, where $N_m = \sum_{s_\beta = m} n_\beta$.*

(iii) $\dim HH_0(A) = \dim Z(A) = N$. *Both X_d and X_E are smooth at A, each with unique component the closure of its corresponding semisimple orbit.*

For one curve, β consists of the single coefficient j, so the theorem proves Conjecture 1.1. Theorem 5.1 constructs the family with split generic fibre S_E, and Theorem 5.12 proves that all geometrically semisimple generic fibres have these matrix factors and projector ranks. The vector of projector ranks retains information lost when one records only the matrix size: its entries specify how each original curve projector acts on the corresponding factor. The theorem determines the geometric generic fibre with these projectors; the integral families can still differ.

Relation to earlier work

For a single curve, the recovery of GV invariants from the abstract contraction algebra is already a theorem of Hua–Toda: their noncommutative Donaldson–Thomas formula determines these invariants [16, Theorem 4.4 and Corollary 4.6]. We ask in addition which semisimple

algebras can occur as geometric generic fibres of associative deformations. Hua–Toda's flatness conjecture [16, Conjecture 4.3] concerns the relative algebra, centre and cocenter in geometric families. We treat these three objects over complete Noetherian local bases in Section 4.

Kawamata's proof records the individual summand multiplicities [22, Theorem 3.1(3) and its proof], which give the projector ranks in the generic matrix factors. Our construction through the universal quotient makes the comparisons over formal classifying arcs explicit and produces a polynomially split family with fixed projectors. These steps are proved in Section 5.

Explicit algebraic approaches include the deformation of a nine-dimensional contraction algebra to $M_2(\mathbb{C}) \times \mathbb{C}^5$ by Bowman–Puljić–Smoktunowicz [3, Section 4], and Smoktunowicz's study of strongly flat deformations [30]. Using the local normal forms of Brown–Wemyss [5], Jelisiejew–Smoktunowicz prove semisimple existence and uniqueness for Types A and D [19, Theorem 1.3]. We use their comparison of tangent dimensions with a known semisimple orbit. The cocenter calculation supplies the missing numerical equality for the exceptional cases, and relative Hochschild cochains give the version with every projector fixed. Thus the proof of uniqueness applies to all the finite free associative formal deformations in the main theorem.

Flatness for several exceptional curves

For several curves, the universal retained algebra and its cocenter have square presentations. Their determinants convert intersection orders with the curve divisors into ranks of the resulting flat family. A geometric family pulls these matrices back along classifying maps to power-series rings $B[[w]]$, where B is the deformation base and w is the surface-section parameter. The resulting cokernels are free over every complete Noetherian local deformation base, including bases with nilpotents. This proves geometric flatness even when the generic fibre is not semisimple.

Theorem 1.3 (Geometric Hua–Toda flatness)**.** *Let f_B be a flat deformation of f_0 over a complete Noetherian local complex algebra $(B, \mathfrak{m}, \mathbb{C})$. Form the relative contraction quotient $\mathscr{A}_B$ using the lift of the fixed marked basic generator and the full finite moving support. Then the algebra, cocenter, and centre are finite free, with ranks*

$$\operatorname{rank}_B \mathscr{A}_B = \sum_{\beta} n_\beta s_\beta^2, \qquad \operatorname{rank}_B HH_0(\mathscr{A}_B/B) = \operatorname{rank}_B Z(\mathscr{A}_B) = N.$$

The centre and cocenter commute with every commutative scalar extension $B \to B'$, and every symmetrizing trace of the central algebra lifts to a perfect relative trace. The corresponding local statement holds over Noetherian algebraic or analytic bases when the relative quotient is finite over the base.

We obtain the algebra presentation from retained corners of the Etingof–Rains central extension [9, Theorem 3.2]. For the cocenter, we combine the full-algebra trace results of Etingof–Latour–Rains [10, Theorem 3.1 and Corollary 3.5] with a vertex-deletion argument. The resulting Theorem 4.14 relates the corner, the full central extension and the deleted subdiagram. Pulling back the algebra and cocenter presentations gives the ranks for the relative quotient defined in Remark 4.1. Flatness and local base change are proved in Theorem 4.22 and Corollary 4.23.

Consequences for centres

The main theorem gives a smooth open neighbourhood of A in the scheme of associative algebra laws. On this neighbourhood, centres and cocenters vary in vector bundles of constant rank. The vector-bundle statement applies to arbitrary finite free associative families through A. There is also a converse problem. Given a deformation of the commutative algebra $Z(A)$, can it be obtained as the centre of a deformation of A? The tangent map of the centre morphism detects first-order obstructions to this lifting problem. For a commutative algebra Z, let $T^1_{\mathrm{com}}(Z)$ be its Harrison space of first-order commutative deformations [14].

Theorem 1.4 (The centre deformation problem)**.** *For A as above, with $d = \dim A$, the following hold.*

(i) $\widehat{\mathcal{O}}_{X_d,A} \cong \mathbb{C}[[t_1, \dots, t_{d(d-1)-d+N}]]$*; in particular A is unobstructed.*

(ii) *Near A in X_d, the centre and cocenter of the tautological algebra are dual vector bundles of rank N, compatible with arbitrary base change. Every finite free associative deformation of A over a Noetherian local complex base inherits these properties and a lift of any chosen symmetrizing trace of the closed fibre.*

(iii) *The centre morphism is smooth at an algebra law B in this neighbourhood if and only if $\chi_B : HH^2(B) \to T^1_{\mathrm{com}}(Z(B))$ is surjective. This holds on a dense open subset containing the semisimple laws.*

(iv) *For $D_{n,2}$, $n \geq 2$, a polynomial family of centres inside the smoothable component has first-order class outside the image of χ. Thus the centre morphism is not smooth at $D_{n,2}$. For $D_{2,2}$, nonlifting also holds without a chosen identification of the special centre.*

These consequences establish unobstructedness and local constancy of the centre dimension in [19, Conjecture 1.4] for the contractions considered here. The type D obstruction shows that smoothness of the centre morphism imposes an additional condition, even when its target is the smoothable component of commutative algebra laws.

Outline of the proof

The proof uses root divisors to connect the geometric construction to the dimensions needed in the orbit argument. Write C_J for the universal contraction quotient with the projectors indexed by J retained, and R_J for its parameter ring. Let $P_\beta = 0$ be the divisor along which a curve of restricted class β appears. The zeroth Fitting ideals of the following modules are generated by products of these divisor equations. Their exponents record three different measurements:

module	exponent of P_β	information recorded
$e_i C_J e_j$	$\beta_i \beta_j$	individual Peirce dimensions
C_J	s_β^2	dimension of the matrix algebra
$HH_0(C_J/R_J)$	1	number of simple blocks

Restricting to a classifying arc turns these exponents into ranks by multiplying them by the divisor intersection orders n_β. The one-curve exponents j^2 and 1 are the scalar version of this description. The rank formulas provide inputs for uniqueness; the marked product itself comes from the geometric construction. The perturbation of the classifying arc uses the Katz–Morrison

and Bryan–Katz–Leung picture [26, 6]. The subsequent identification of matrix factors uses determinant degrees of the transported generator, following Hua–Toda and Kawamata [16, Theorem 4.2 and its proof] [22, Theorem 3.1(3) and its proof].

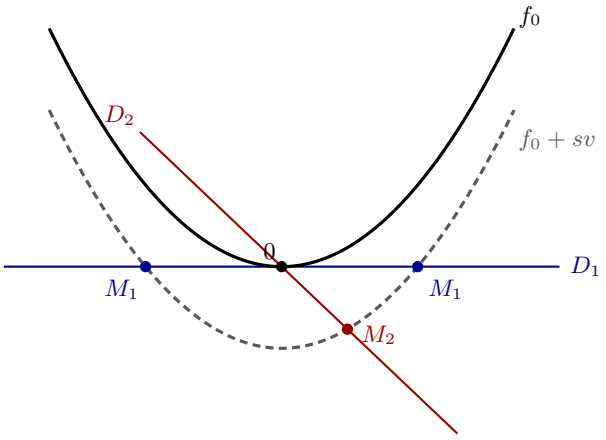


Figure 1: A schematic slice of the marked base with two curve divisors. The central classifying arc is tangent to D_1; a small translation separates that tangency into transverse intersections. An intersection with D_j gives the local factor M_j, while the conserved intersection number determines how many such factors occur in the generic product of matrix algebras. This figure depicts the one-curve divisor-count construction of Bryan–Katz–Leung [6, Section 2.1, Proposition 2.3, and Lemma 2.8].

August's symmetry theorem [2, Theorem 2.6], the bimodule sequence of Jasso–Keller–Muro [18, Theorem 3.2.5], and Linckelmann's Tate duality [27, equation (2.11)] give $\dim HH^1(A) = \dim HH^2(A)$. The cocenter rank then makes the tangent dimension equal to the dimension of the constructed semisimple orbit. This equality forces smoothness and a unique component through A, and hence determines the possible semisimple geometric generic fibres. Relative Hochschild cochains give the corresponding tangent formula with every projector fixed, which proves the full marked statement.

In the one-curve argument, the coefficient-five E_8 corner is computed from the exact relations and rank certificate in Subsection 3.6 and Appendix B, using the results of Etingof–Rains and Etingof–Latour–Rains [9, 10]. The exact sequence in Theorem 4.14 subsequently computes the cocenter dimension for arbitrary retained subsets. The appendices record the exact root enumeration, the E_8 calculation, and the $D_{2,2}$ tangent data; the source distribution includes their verification programs.

Conventions and organization. All algebras are associative and unital. The geometry is over $\mathbb{C}$, while the general orbit criterion works over an algebraically closed field of characteristic zero. We write $A^e = A \otimes A^{\mathrm{op}}$, $HH^i(A) = \mathrm{Ext}^i_{A^e}(A, A)$, and $D = \mathrm{Hom}_{\mathbb{C}}(-, \mathbb{C})$. The symbol $[A, A]$ denotes the linear span of commutators, and $HH_0(A) = A/[A, A]$. The parameters w, s, z denote respectively the surface slice, the external deformation, and the central-extension element. In the multi-curve construction $\mathscr{E}$ denotes the central extension; $E = \bigoplus_i \mathbb{C}e_i$ denotes the separable algebra of labelled idempotents.

Sections 2 and 3 construct a family with semisimple generic fibre for one curve. They also prove that every finite free associative formal deformation of A with geometrically semisimple generic fibre has the same matrix factors as this family. Section 4 begins with a criterion for freeness after pullback and the lifting of symmetrizing traces. It then constructs the universal presentations and identifies their geometric pullbacks, first for one curve and then for several curves. Section 5 determines the semisimple geometric generic fibres with the curve projectors fixed and proves smoothness of the corresponding schemes of algebra laws. Section 6 derives the consequences for centres in associative families and studies the centre-lifting obstruction.

2 Constructing a semisimple deformation for one curve

For a simple one-curve flop, the construction combines three geometric inputs. Katz–Morrison's length classification and construction are recalled in [20, Theorems 1.1–1.2]; Bryan–Katz–Leung describe the curve-divisor count in a one-parameter analytic deformation [6, Section 2.1, Proposition 2.3, and Lemma 2.8]; and Toda relates the resulting GV multiplicities to the contraction-algebra dimension [32, Section 2.4 and Theorem 1.1]. We combine them with a fixed tilting generator to construct a finite free algebra family. Root-divisor intersections determine the number of local factors, and the determinant degree of the generator determines their matrix sizes. Throughout this section the ground field is $\mathbb{C}$. Let

$$f\colon X \longrightarrow Y = \operatorname{Spec} R$$

be the completion at the contraction point of a simple flopping contraction of a smooth complex threefold. Thus $\operatorname{Exc}(f) = C \cong \mathbb{P}^1$, the scheme-theoretic length ℓ belongs to $\{1, \dots, 6\}$, and the contraction algebra is denoted by A_{con}. We write n_j for the genus-zero Gopakumar–Vafa invariant in class $j[C]$.

Theorem 2.1 (Semisimple existence)**.** *Let f be any completed simple one-curve flop of a smooth complex threefold, of length ℓ, with contraction algebra A_{con} and genus-zero GV invariants n_j. There exists a finite free $\mathbb{C}[[s]]$-algebra $\mathscr{A}$ such that*

$$\mathscr{A}/(s) \cong A_{\mathrm{con}}.$$

After a finite splitting extension $K'/\mathbb{C}((s))$,

$$\mathscr{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \tag{2.1}$$

Definition 2.2 (Marked partial-resolution data)**.** Let Φ be the primitive ADE root system of a general hyperplane section, with complex Cartan space $\mathfrak{h}$, Weyl group W, and simple roots $\Delta = \{\alpha_i\}$, which determine the positive roots Φ^+. A marking fixes the retained vertex α_* and the corresponding summand projector in the endomorphism algebra. For a root $\alpha = \sum_i c_i(\alpha)\alpha_i$, the marked coefficient is $c_*(\alpha)$; the Levi roots are those with $c_*(\alpha) = 0$. Put

$$W_0 = \langle s_i : i \neq * \rangle, \qquad H = \mathfrak{h}/W_0,$$

and let $\pi_W : H \to \mathfrak{h}/W$ be the induced finite map to the semiuniversal ADE base. Choose affine coordinates on H centred at the origin.

If θ is the highest root, the length classification gives

$$\ell = c_*(\theta)$$

[26]. The curve divisors and symbolic thickenings are indexed by $1 \le j \le \ell$ [6, Proposition 2.3 and Lemma 2.4]. Modulo the Levi root lattice, $[\alpha] = c_*(\alpha)[\alpha_*]$. Degrees of extended line bundles express the comparison with curve classes; see Lemma 2.8.

The retained root coefficient labels a curve class, whereas the quotient by the Levi Weyl group parametrizes the marked partial resolution. The six primitive cases are shown in Figure 2.

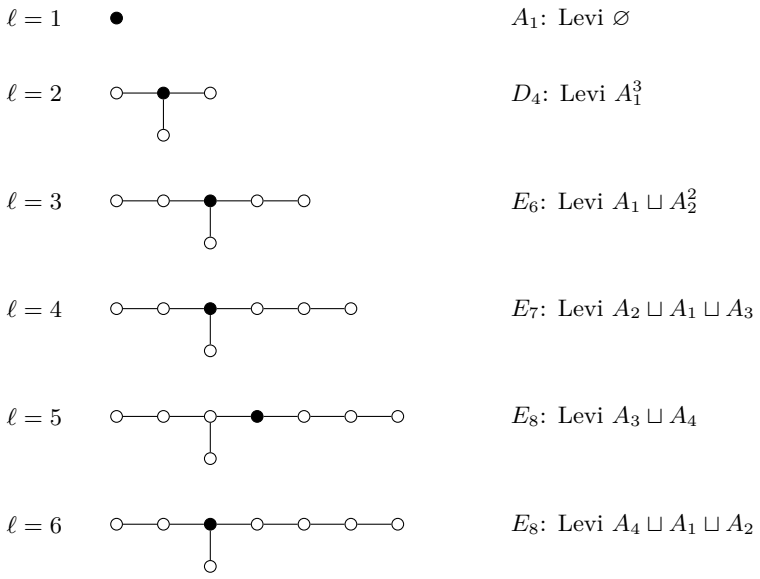


Figure 2: The six primitive marked ADE pairs. The filled vertex is the retained vertex α_*; the highest root has coefficient $\ell = c_*(\theta)$ at this vertex, and the Levi type is the diagram left after deleting it. This is the primitive one-curve situation in the Bryan–Katz–Leung divisor picture [6, Section 2.1 and Proposition 2.3]. For lengths $2, 3, 4, 6$, the retained vertex is the branching vertex used in the corner computations of Etingof–Latour–Rains. The coefficient-five corner is computed in Proposition 3.13.

Lemma 2.3 (The marked analytic classifying arc)**.** *After replacing the source and target of the contraction by sufficiently small analytic neighbourhoods, there is a holomorphic map*

$$f_0 \colon \mathbb{D}_w \longrightarrow H, \qquad f_0(0) = 0,$$

for which the given analytic contraction is isomorphic, as a marked contraction, to the pullback of the simultaneous partial resolution over H. The completion of f_0 at $w = 0$ is a formal classifying arc

$$\nu_0 \colon \operatorname{Spf} \mathbb{C}[[w]] \longrightarrow H,$$

whose pullback is isomorphic to the original completed marked contraction.

Proof. Choose a general analytic hyperplane parameter w through the contraction point. The central fibre of the resulting hyperplane-section family is the marked partial resolution of the

ADE surface section. The semiuniversal deformation spaces of the full resolution, the marked partial resolution, and the ADE surface are, respectively,

$$\mathfrak{h}, \qquad H = \mathfrak{h}/W_0, \qquad \mathfrak{h}/W.$$

Bryan–Katz–Leung set up the compatible analytic classifying maps and their curve-divisor geometry [6, Section 2.1, especially Propositions 2.2–2.3]. The local universal-flop formulation used here is also stated in [20, Theorem 1.2]. These results identify the resolved threefold neighbourhood with the Cartesian pullback of the simultaneous marked partial resolution. Completing that diagram at $w = 0$ gives the stated isomorphism of marked formal contractions. The later subsection compares the contraction quotient after transporting the tilting generator. □

Let S_0 be the central partial-resolution surface, and write $\mathcal{I}_C \subset \mathcal{O}_{S_0}$ for the ideal of C in that surface.

Proposition 2.4 (The marked family and GV multiplicities)**.** *For the classifying arc of Lemma 2.3, the simultaneous partial resolution has the following properties.*

(i) *The base H is smooth affine space, the quotient $\pi_W \colon H \to \mathfrak{h}/W$ is finite, and the universal simultaneous partial resolution*

$$\pi_\ell \colon \mathcal{Y}_\ell \longrightarrow \mathcal{X}_\ell = \operatorname{Spec} \mathcal{R}_\ell$$

is projective with source and target flat over H. Its fibres have dimension at most one and

$$R\pi_{\ell *}\mathcal{O}_{\mathcal{Y}_\ell} \cong \mathcal{O}_{\mathcal{X}_\ell}.$$

The universal flopping algebra is finite over its centre $\mathcal{R}_\ell$.

(ii) *The curve discriminant is a union of precisely ℓ reduced irreducible divisors*

$$D^{\mathrm{curv}} = D_1 \cup \cdots \cup D_\ell.$$

At a general point of D_j, the partial-resolution surface contains one smooth (-2)-curve in class $j[C]$. These curves form a projective flat family $\mathcal{C}_j \to D_j$, and its central fibre is the symbolic thickening C_j defined by $\mathcal{I}_C^{(j)}$.

(iii) *A sufficiently small boundary-proper generic perturbation of f_0 meets D_j in k_j points, counted with multiplicity, where k_j is the length of the local Hilbert scheme at C_j.*

(iv) *Toda's pure thickening is*

$$\mathcal{O}_{C^{(j)}} = (\mathcal{O}_{S_0}/\mathcal{I}_C^j)/Q,$$

where Q is the maximal zero-dimensional subsheaf, and n_j is the length of its local Hilbert scheme. A generic geometric deformation has exactly n_j disjoint $(-1,-1)$-curves in class $j[C]$.

(v) *The contraction algebra satisfies*

$$\dim_{\mathbb{C}} A_{\mathrm{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Proof. Bryan–Katz–Leung obtain the curve divisors and their generic curve classes in their analytic simultaneous-partial-resolution picture [6, Propositions 2.2–2.3 and Lemma 2.4]; they identify the transverse intersection number with the Hilbert-scheme multiplicity k_j [6, Lemma 2.8]. Nabijou–Wemyss give the restricted-root formulation for general flopping contractions [28, Theorem 2.6]. Toda identifies the integers n_j through the pure-thickening/GV interpretation and proves the width formula in (v) [32, Section 2.4 and Theorem 1.1]. The equality of the two central thickenings is proved here in Lemma 2.11. The universal-flop construction supplies the flat simultaneous family, its relative-dimension bound, and $R\pi_{\ell *}\mathcal{O} = \mathcal{O}$ [20, Section 2.3, Assumption 3.1, and Section 3.3]. Karmazyn's endomorphism-ring description [20, Proposition 3.10] shows that the universal algebra is finite over $\mathcal{R}_\ell$, since its defining module is finite over that Noetherian ring. □

2.1 Root divisors and a deformation into ordinary double points

The marked root coefficient defines a divisor on the deformation base. Bryan–Katz–Leung's simultaneous-partial-resolution construction gives the analytic divisor picture [6, Section 2.1 and Proposition 2.3]. That construction builds on Katz–Morrison's root-hyperplane description of simultaneous resolutions [26, Theorem 1]. This divisor geometry permits a formal constant translation while retaining its finite moving support. Let $q : \mathfrak{h} \to H$ be the quotient map, and for a root α let $H_\alpha \subset \mathfrak{h}$ be its root hyperplane. The group W_0 preserves the marked coefficient and permutes the non-Levi roots with a fixed coefficient. If $\mathcal{O}$ is one such orbit, the square-free product

$$p_{\mathcal{O}} = \prod_{\alpha \in \mathcal{O}} \alpha$$

is W_0-invariant and descends to an equation on H. The roots of marked coefficient j form a single W_0-orbit $\mathcal{O}_j$ (Lemma 2.5), so $p_{\mathcal{O}_j}$ descends to a polynomial $P_j \in \mathcal{O}(H)$ with $q^* P_j = p_{\mathcal{O}_j}$. This polynomial cuts out the reduced irreducible divisor D_j of Proposition 2.4(ii). Put

$$D^{\mathrm{curv}} = \bigcup_{j=1}^{\ell} D_j;$$

This is the curve discriminant, the locus where the marked partial resolution has a complete exceptional curve. Its good points have one vanishing root; the constant translation below separates the corresponding root intersections away from the higher-codimension locus.

2.1.1 Root divisors and transverse local models

Lemma 2.5 (The coefficient strata are single W_0-orbits)**.** *For each of the six primitive marked ADE pairs and each $1 \le j \le \ell$, the set*

$$\mathcal{O}_j := \{\alpha \in \Phi^+ : c_*(\alpha) = j\}$$

is a single W_0-orbit. Its cardinalities are:

length ℓ	*marked pair*	$\lvert\mathcal{O}_1\rvert, \dots, \lvert\mathcal{O}_\ell\rvert$
1	A_1	1
2	D_4	8, 1
3	E_6	18, 9, 2
4	E_7	24, 18, 8, 3
5	E_8, $c_*(\theta) = 5$	40, 30, 20, 10, 4
6	E_8, $c_*(\theta) = 6$	30, 30, 20, 15, 6, 5

In particular $\sum_j \lvert\mathcal{O}_j\rvert$ is the number of positive roots with $c_*(\alpha) > 0$, written N_p in Subsection 3.5, and each D_j is irreducible.

Proof. Bryan–Katz–Leung identify a marked coefficient with a curve-divisor component [6, Proposition 2.3]. The group W_0 preserves c_*, hence acts on every $\mathcal{O}_j$. The reflection calculation in Appendix A enumerates these roots and their W_0-orbits, giving the displayed cardinalities and transitivity. A proper factor of the descended product $p_{\mathcal{O}_j}$ would select a proper nonempty W_0-stable subset of $\mathcal{O}_j$, contradicting transitivity. Thus D_j is irreducible. Lemma 2.6 supplies its reducedness. □

Lemma 2.6 (Reduced root divisors and the higher-root locus)**.** *Let α be a non-Levi root. The quotient $\mathfrak{h} \to H = \mathfrak{h}/W_0$ is étale at a general point of H_α. Consequently the divisor descended from every W_0-orbit of non-Levi root hyperplanes is generically reduced.*

There is a closed subset $\Sigma^{\mathrm{bad}} \subset D^{\mathrm{curv}}$ of codimension at least two in H, containing:

(i) *the singular loci of all curve-discriminant components;*

(ii) *intersections of distinct curve-discriminant components;*

(iii) *the images of intersections $H_\alpha \cap H_\beta$, where α is non-Levi and $\beta \neq \pm\alpha$.*

We call the points of $D^{\mathrm{curv}} \setminus \Sigma^{\mathrm{bad}}$ good. At each Cartan lift of a good point of D_j, exactly one root pair $\{\pm\alpha\}$ vanishes.

Proof. The root-hyperplane input from Katz–Morrison makes the first assertion a stabilizer calculation upstairs. Choose

$$x \in H_\alpha \setminus \bigcup_{\beta \neq \pm\alpha} H_\beta.$$

Its stabilizer in W is $\{1, s_\alpha\}$. Since α is not a Levi root, $s_\alpha \notin W_0$, and the W_0-stabilizer of x is trivial. A finite quotient in characteristic zero is étale at a point with trivial stabilizer. Therefore the product of the distinct linear root equations descends with generic multiplicity one.

For the bad locus, distinct root hyperplanes meet in codimension two. A finite morphism preserves the dimension of an irreducible closed subset, so all images in (iii) have codimension at least two in H. Distinct reduced curve divisors have no common codimension-one component, and the singular locus of a reduced divisor in the smooth space H also has ambient codimension at least two. Their finite union has the required codimension. Outside it, every Cartan lift lies on exactly one root hyperplane. For A_1, there are no intersections of distinct root hyperplanes and the bad locus is empty. □

Lemma 2.7 (Quadratic reflection coordinate)**.** *Let $\bar{x} \in D_j \setminus \Sigma^{\mathrm{bad}}$, and choose a Cartan lift $x \in H_\alpha$. There are completed local coordinates $(\zeta, z_2, \dots, z_r)$ at $\bar{x}$ and $(t, z_2, \dots, z_r)$ at its image in $\mathfrak{h}/W$ such that*

$$\widehat{\mathcal{O}}_{H,\bar{x}} = \mathbb{C}[[\zeta, z_2, \dots, z_r]], \qquad \widehat{\mathcal{O}}_{\mathfrak{h}/W,\pi_W(\bar{x})} = \mathbb{C}[[t, z_2, \dots, z_r]],$$

and $\pi_W^* t = \zeta^2$, while $D_j = (\zeta = 0)$.

Proof. At the chosen lift, the W_0-stabilizer is trivial and the full Weyl stabilizer is the order-two group generated by s_α. The reflection acts by $\zeta \mapsto -\zeta$ on the normal line to H_α and trivially along the hyperplane. The resulting invariant completed local ring is therefore $\mathbb{C}[[\zeta^2, z_2, \dots, z_r]]$, which gives the asserted quadratic parameter t. □

Lemma 2.8 (Specialization of the marked degree)**.** *Let $\mathscr{L}$ be a line bundle on the marked partial-resolution family, and let E_b be its exceptional curve over a good point $b \in D_j$. Then*

$$\deg_{E_b} \mathscr{L}_b = j \deg_C \mathscr{L}_0.$$

Proof. At the generic point of C, the normal central surface has a DVR as local ring. The symbolic j-thickening has generic length j, hence fundamental cycle $[C_j] = j[C]$. Thus $\deg_{C_j} \mathscr{L}_0 = j \deg_C \mathscr{L}_0$. In the projective flat curve family $\mathcal{C}_j \to D_j$ of Proposition 2.4, both $\mathcal{O}_{\mathcal{C}_j}$ and the restriction of $\mathscr{L}$ are flat over the base. Their fibrewise Euler characteristics are locally constant. Their difference is the degree of $\mathscr{L}$ on a fibre, so it is constant on the connected base germ. This gives the equality; compare [32, Section 2.4, equation (15)]. □

Lemma 2.9 (The transverse threefold is an ordinary double point)**.** *At a point of $D_j \setminus \Sigma^{\mathrm{bad}}$, the transverse surface deformation is*

$$xy - z^2 = t.$$

After pullback to the marked partial-resolution base it is

$$xy - z^2 = \zeta^2,$$

and hence, after $\xi_+ = z + i\zeta$, $\xi_- = z - i\zeta$, it is the ordinary double point

$$xy - \xi_+ \xi_- = 0.$$

If a one-dimensional classifying curve meets D_j transversely, its partial resolution contains one $(-1,-1)$-curve in numerical class $j[C]$.

Proof. At a good point, the vanishing root subsystem is $\{\pm\alpha\} \cong A_1$. Bryan–Katz–Leung identify the transverse simultaneous-resolution model with the semiuniversal A_1-deformation [6, Proposition 2.2]. The A_1 singularity is $xy - z^2 = 0$, and its one-dimensional T^1 is generated by the constant class; hence a miniversal equation is

$$xy - z^2 = t.$$

Lemma 2.7 gives $t = \zeta^2$. Since $(z + i\zeta)(z - i\zeta) = z^2 + \zeta^2$, the pulled-back equation is exactly $xy - \xi_+\xi_- = 0$. On a transverse classifying curve, ζ is a local parameter. Using it as the curve coordinate absorbs the higher terms in the original parametrization and gives the same completed ODP equation. The partial resolution is one of the two small resolutions of this ODP, so its exceptional curve has normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. The degree comparison in Lemma 2.8 identifies its class as $j[C]$. □

2.1.2 Perturbing the classifying arc

We perturb the classifying arc transversely to the curve divisors in order to realize the local factors simultaneously. Bryan–Katz–Leung use a boundary-proper analytic generic perturbation [6, Section 2.1 and Lemma 2.8]. The constant translation below stays inside a fixed disk, so the boundary contributes no new intersections and the total divisor multiplicities remain visible.

Lemma 2.10 (The central arc is not contained in the curve discriminant)**.** *The image of f_0 is contained in no D_j. After shrinking to a closed disk $\overline{\mathbb{D}}_\rho$,*

$$f_0^{-1}(D^{\mathrm{curv}}) \cap \overline{\mathbb{D}}_\rho = \{0\}, \qquad f_0(\partial\mathbb{D}_\rho) \cap D^{\mathrm{curv}} = \varnothing.$$

If P_j is a reduced equation of D_j, then

$$0 < m_j := \mathrm{ord}_{w=0}\, P_j(f_0(w)) < \infty.$$

Proof. If a classifying arc were contained in a curve divisor D_j, the exceptional locus would contain a surface. Indeed, pulling back the projective flat family $\mathcal{C}_j \to D_j$ would give a two-dimensional closed subset of the exceptional locus of the threefold contraction, contrary to $\mathrm{Exc}(f) = C$. Thus every reduced equation P_j pulls back to a nonzero holomorphic function whose zero at the origin is isolated. There are finitely many j, so one common shrinking gives the boundary statement and the finite, positive orders. □

Lemma 2.11 (Symbolic and pure thickenings coincide)**.** *The Bryan–Katz–Leung symbolic thickening C_j equals Toda's pure one-dimensional thickening $C^{(j)}$. Consequently their Hilbert multiplicities satisfy $k_j = n_j$.*

Proof. Work in a stalk $\mathcal{S}$ of the normal surface S_0, and let $\mathfrak{p} \subset \mathcal{S}$ be the height-one prime of C. By definition,

$$\mathfrak{p}^{(j)} = \mathfrak{p}^j\mathcal{S}_{\mathfrak{p}} \cap \mathcal{S}.$$

The quotient $\mathfrak{p}^{(j)}/\mathfrak{p}^j$ is supported at closed points and hence has finite length. The ideal $\mathfrak{p}^{(j)}$ is $\mathfrak{p}$-primary, so $\mathcal{S}/\mathfrak{p}^{(j)}$ has only the associated prime $\mathfrak{p}$ and therefore has no nonzero finite-length submodule. In the exact sequence

$$0 \longrightarrow \mathfrak{p}^{(j)}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^{(j)} \longrightarrow 0,$$

the first term is the maximal finite-length submodule of $\mathcal{S}/\mathfrak{p}^j$, namely Toda's Q. Thus $(\mathcal{O}/I^j)/Q = \mathcal{O}/I^{(j)}$, so the local Hilbert rings are identical. Proposition 2.4(iii)–(iv) then gives $k_j = n_j$. □

Lemma 2.12 (Parametric transversality with boundary)**.** *Let M be a compact smooth manifold with boundary, let $\mathcal{U}$ and N be smooth manifolds, and let $S_1, \dots, S_m \subset N$ be smooth locally closed submanifolds. If a smooth map $\Xi\colon M \times \mathcal{U} \to N$ and its boundary restriction are transverse to every S_i, then the parameters for which both slice maps are simultaneously transverse to all S_i form a residual dense subset of $\mathcal{U}$. If $\dim_{\mathbb{R}} M = 2$, such a slice misses every stratum of real codimension at least four, and its boundary misses every stratum of real codimension at least two.*

Proof. For one stratum, put $Z = \Xi^{-1}(S)$. Transversality makes Z a smooth incidence manifold, with boundary incidence Z_∂. A parameter is a regular value of $Z \to \mathcal{U}$ exactly when its slice is transverse to S, and the same equivalence holds on the boundary. Apply Sard's theorem on a countable exhaustion by relatively compact coordinate pieces, then intersect the residual dense sets obtained for the finitely many strata. The dimension formula for transverse inverse images gives the two avoidance assertions. □

Lemma 2.13 (A boundary-controlled constant translate)**.** *There is an arbitrarily small vector $v \in H \cong \mathbb{C}^{\dim H}$ such that $f_v(w) = f_0(w) + v$ is transverse to every smooth codimension-one curve-discriminant stratum, avoids Σ^{bad}, and has no curve-discriminant point on $\partial\mathbb{D}_\rho$. The vector can be chosen so that, for some $\epsilon > 0$,*

$$f_0(w) + sv \notin D^{\mathrm{curv}} \quad \text{for } w \in \partial\mathbb{D}_\rho, \quad |s| \le 1 + \epsilon.$$

Proof. Choose a finite Whitney stratification of the curve discriminant adapted to Σ^{bad}. The translation map

$$\overline{\mathbb{D}}_\rho \times H \longrightarrow H, \qquad (w, \xi) \longmapsto f_0(w) + \xi$$

and its boundary restriction are submersions. Lemma 2.12 therefore gives translations for which the disk avoids the codimension-at-least-four bad strata and its boundary avoids every curve-discriminant stratum. The compact boundary of the central disk is disjoint from the closed curve discriminant, so reduced local equations have a positive minimum there. Uniform continuity supplies a ball of translations preserving boundary nonvanishing; choose the transverse parameter in a slightly smaller ball. □

2.1.3 Root counts and the formal generic fibre

The translation preserves each divisor intersection number. Weierstrass polynomials make the moving intersections into finite families over the translation parameter; their formal generic fibre supplies the points of the local matrix blocks.

Lemma 2.14 (Conservation of the root intersections)**.** *Put $F(s, w) = f_0(w) + sv$ and $G_j(s, w) = P_j(F(s, w))$. On a connected s-disk U containing 0 and 1, the zero scheme of $G_j(s, \cdot)$ is finite flat of degree m_j and is represented by a monic polynomial*

$$W_j(s, T) \in \mathcal{O}(U)[T], \qquad \deg_T W_j = m_j, \qquad W_j(0, T) = T^{m_j}.$$

Proof. Boundary nonvanishing makes the argument-principle integral

$$\frac{1}{2\pi i}\int_{\partial\mathbb{D}_\rho}\frac{\partial_w G_j(s,w)}{G_j(s,w)}\,dw$$

constant in s, with value m_j. For $r \geq 1$, the analogous integral with an extra factor w^r is holomorphic in s and gives the r-th power sum of the zeros, counted with multiplicity. Newton's identities then give holomorphic elementary symmetric functions on the whole parameter disk. Let $W_j(s,T)$ be the monic polynomial with these coefficients.

For each s, the roots of W_j, with multiplicities, are exactly the zeros of $G_j(s,\cdot)$ in the w-disk. Local Weierstrass preparation at every zero makes G_j/W_j extend holomorphically across the zero locus as a unit. Thus $G_j = 0$ and $W_j = 0$ define the same analytic scheme. The quotient defined by the monic polynomial is locally free with basis $1, T, \ldots, T^{m_j-1}$, and boundary nonvanishing makes its support proper over the parameter disk. At $s = 0$, all m_j zeros lie at the origin, giving $W_j(0,T) = T^{m_j}$. □

Lemma 2.15 (Passage from one good fibre to the formal generic fibre)**.** *Put $B = \mathbb{C}[[s,w]]$ and $K = \mathbb{C}((s))$. The polynomials $W_j \in K[T]$ are separable and pairwise coprime. The classifying map on* $\operatorname{Spec} B[1/s]$ *avoids Σ^{bad}, and every root is transverse to the smooth part of its unique divisor D_j.*

Proof. The good fibre at $s = 1$ controls the formal generic fibre. At $s = 1$, the discriminant of each W_j and every pairwise resultant are nonzero. These are therefore nonzero holomorphic functions of s. Their Taylor series may be divisible by s, but become units after passing to K. This proves separability and coprimality.

For an irreducible component $S \subset \Sigma^{\mathrm{bad}}$, put $I_S = F^{-1}(S)$. Since $S \subset D^{\mathrm{curv}}$, this is a closed subspace of the finite root incidence $\bigcup_j \{G_j = 0\}$ from Lemma 2.14. Its image in U is a proper analytic subset because its fibre at $s = 1$ is empty. The completed direct image of $\mathcal{O}_{I_S}$ at $s = 0$ therefore has finite length over $\mathbb{C}[[s]]$ and vanishes after inverting s. There are finitely many such components. Finally, at a smooth point of D_j, separability is the condition $\partial_w G_j \neq 0$, which is precisely transversality. □

Proposition 2.16 (The conserved multiplicities are the GV invariants)**.** *For each j,*

$$m_j = n_j. \tag{2.2}$$

Thus the formal constant-translation family has exactly n_j coefficient-j roots.

Proof. Write $\widetilde{f}$ for the boundary-proper generic perturbation of f_0 used by Bryan–Katz–Leung. We compare it with the constant translation. For $\widetilde{f}$, their Lemma 2.8 identifies the Cartier intersection number with the Hilbert-scheme multiplicity, $m_j = \#\widetilde{f}^{-1}(D_j) = k_j$ [6, Lemma 2.8]. Lemma 2.11 gives $k_j = n_j$, and Lemma 2.14 gives the same intersection number for the constant-translation family. □

Proposition 2.17 (A formal marked deformation into ordinary double points)**.** *There is a finite extension K'/K with valuation ring $\mathbb{C}[[u]]$, where $K' = \mathbb{C}((u))$ and $s = u^e$ for some $e \geq 1$. The completed local models at the generic curve points are small resolutions of split ordinary double points. Moreover,*

$$W_j(u^e, w) = \prod_{a=1}^{n_j} (w - w_{j,a}(u)), \qquad w_{j,a}(u) \in u\mathbb{C}[[u]],$$

with all roots pairwise distinct, including roots belonging to different coefficients. The pullback of the universal partial resolution has exactly n_j pairwise disjoint $(-1,-1)$-curves in class $j[C]$, and no other complete curves.

Proof. Split the finitely many separable polynomials W_j. The resulting curve points are geometrically ordinary double points by Lemma 2.9. Enlarge the field by a finite extension that splits their quadratic tangent cones. The formal Morse lemma then gives split ODP equations, and the small resolutions correspond to their two rulings. Denote the resulting field by K'. The normalization of $\mathbb{C}[[s]]$ in K' is a complete equicharacteristic DVR with residue field $\mathbb{C}$, hence is $\mathbb{C}[[u]]$; after a unit adjustment of u, its ramified parameter is $s = u^e$. Each root is integral because W_j is monic, and $W_j(0,T) = T^{n_j}$ forces its residue to be zero. Thus $w_{j,a}(u) \in u\mathbb{C}[[u]]$. Separability and coprimality give all the asserted distinctness. Lemma 2.9 supplies one $(-1,-1)$-curve of class $j[C]$ at each root; different roots lie in different w-fibres and are disjoint.

It remains to exclude further complete curves. The completed target is affine, so every complete curve in the source maps to a point and lies in the exceptional locus; the regular function w is constant on it. The curve discriminant D^{curv} is the full locus of complete exceptional curves. Every generic point of the formal disk on that locus is a good point of one unique D_j, where the unique curve has already been counted, and the disk avoids the codimension-two bad locus. Hence the list is exhaustive. □

Remark 2.18. The relative family retains all moving roots over $\mathbb{C}[[s,w]]$. Its generic fibre is formed over $\mathbb{C}[[s,w]][1/s]$ and finite extensions of that ring, as in Lemma 2.15. Over $\mathbb{C}((u))[[w]]$, the element $w - w_{j,a}(u)$ is a unit whenever $w_{j,a} \neq 0$. Thus one must localize the finite relative family before making any further completion; completing first at $w = 0$ would discard moving root points.

2.2 Tilting generators and the local matrix blocks

The deformed classifying arc locates the exceptional curves and records their classes. We now identify the algebra at each curve. The fixed universal generator survives the relevant classifying base changes; at a split ordinary double point, its determinant degree counts the nonfree projective summands, whose stable endomorphisms produce the matrix block.

2.2.1 The universal generator and base change

Karmazyn's universal-flop construction supplies the simultaneous geometry and its universal algebra [20, Sections 2.3 and 3.3]. Van den Bergh's perverse generators select a normalized basic generator on that family [33, Propositions 3.2.5–3.2.7]. The lemmas below establish how

this generator pulls back along an arbitrary Noetherian classifying base. The universal length-ℓ partial resolution carries a basic zero-perverse generator and its positive dual

$$\mathcal{T} = \mathcal{O} \oplus \mathcal{T}', \qquad \mathcal{P} = \mathcal{T}^{\vee} = \mathcal{O} \oplus \mathcal{P}'.$$

Set

$$\mathcal{A}_\ell = \operatorname{End}_{\mathcal{Y}_\ell}(\mathcal{T})^{\mathrm{op}} \cong \operatorname{End}_{\mathcal{Y}_\ell}(\mathcal{P}), \qquad e_0 = [\mathcal{O}].$$

The universal source, target, centre, and algebra are flat over the marked base $H = \operatorname{Spec} \mathbb{H}_\ell$ by the universal-flop construction [20, Sections 2.3 and 3.3], and the universal target is affine of finite type over H. The following lemmas establish the Tor-independence and ordinary endomorphism base change needed for an arbitrary Noetherian classifying algebra. The positive nonfree summand is the normalized maximal extension

$$0 \longrightarrow \mathcal{O}^{\oplus(\ell-1)} \longrightarrow \mathcal{P}' \longrightarrow \mathcal{L} \longrightarrow 0, \qquad \deg_C \mathcal{L} = 1. \tag{2.3}$$

Thus $\operatorname{rk} \mathcal{P}' = \ell$ and $\deg_C \det \mathcal{P}' = 1$. The generator $\mathcal{P}$ belongs to the (-1)-perverse heart, while $\mathcal{T}$ belongs to the zero-perverse heart. Duality preserves the distinguished free projector and replaces the endomorphism algebra by its opposite.

Lemma 2.19 (Tor-independence for arbitrary classifying pullback)**.** *Let B be a Noetherian $\mathbb{H}_\ell$-algebra, and form*

$$\begin{array}{ccc} \mathcal{Y}_B & \xrightarrow{g} & \mathcal{Y}_\ell \\ \pi_B \downarrow & & \downarrow \pi \\ \mathcal{X}_B & \xrightarrow{h} & \mathcal{X}_\ell. \end{array}$$

This square is Tor-independent. Consequently

$$R\pi_{B*} L g^* E \simeq L h^* R\pi_* E \tag{2.4}$$

for every perfect E on $\mathcal{Y}_\ell$, and $R\pi_{B*}\mathcal{O}_{\mathcal{Y}_B} = \mathcal{O}_{\mathcal{X}_B}$.

Proof. Write the local rings as $C \to A \to Q$, where A and Q are flat over $C = \mathcal{O}_H$. Then

$$Q \otimes^L_A (A \otimes_C B) \simeq Q \otimes^L_C B \simeq Q \otimes_C B.$$

Thus the displayed Cartesian square is Tor-independent. The morphism π is projective, hence quasi-compact and quasi-separated. Independently, a perfect complex E has quasi-coherent cohomology. The derived base-change isomorphism of [31, Lemma 36.22.5, Tag 08IB] therefore applies: its canonical map

$$L h^* R\pi_* E \longrightarrow R\pi_{B*} L g^* E$$

is an isomorphism, giving (2.4). Taking $E = \mathcal{O}$ and using $R\pi_*\mathcal{O} = \mathcal{O}$ gives the final assertion. □

Lemma 2.20 (Base change of the perverse generator)**.** *The pullbacks $\mathcal{P}_B = g^*\mathcal{P}$ and $\mathcal{T}_B = g^*\mathcal{T}$ are projective generators of the (-1)- and zero-perverse hearts of π_B respectively, and they induce the corresponding equivalences on bounded coherent derived categories.*

Proof. We verify the hypotheses of Van den Bergh's projective-generator criterion after pullback. His category $\mathfrak{V}$ consists of globally generated bundles M satisfying $R^1\pi_*(M^\vee) = 0$. A finite evaluation surjection for $\mathcal{P}$ remains surjective after pullback. Since $\mathcal{P} \in \mathfrak{V}$, $R\pi_*\mathcal{T} = (\pi_*\mathcal{T})[0]$. By Lemma 2.19,

$$R\pi_{B*}\mathcal{T}_B \simeq Lh^*(\pi_*\mathcal{T}).$$

The left side lies in $D^{\geq 0}$, while the right side lies in $D^{\leq 0}$. Both are therefore concentrated in degree zero, giving $R^1\pi_{B*}\mathcal{T}_B = 0$. Relative ampleness of $\det \mathcal{P}$, global generation, and the direct $\mathcal{O}$-summand also survive. Van den Bergh's intrinsic projective-generator criterion now applies [33, Propositions 3.2.5–3.2.7], and the derived equivalences follow from [33, Corollary 3.2.8]. □

Proposition 2.21 (Ordinary endomorphism base change)**.** *There is a canonical isomorphism of marked algebras*

$$\mathrm{End}_{\mathcal{Y}_B}(\mathcal{T}_B)^{\mathrm{op}} \cong \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B. \tag{2.5}$$

Moreover, all higher self-Ext groups vanish. The isomorphism preserves e_0, and the bundle pullbacks preserve ranks and determinants. The algebra identification remains valid after central localization and Noetherian completion.

Proof. Apply derived base change to the vector bundle $\mathcal{E}nd(\mathcal{T})$. The tilting property gives

$$R\pi_*\mathcal{E}nd(\mathcal{T}) = \pi_*\mathcal{E}nd(\mathcal{T})[0].$$

Its module of sections is the $\mathbb{H}_\ell$-flat module underlying $\mathcal{A}_\ell^{\mathrm{op}}$. Thus its derived pullback is its ordinary pullback in degree zero. Taking sections proves (2.5) and the Ext vanishing. The distinguished free summand is pulled back, and rank and determinant commute with pullback. Localization and Noetherian completion are flat, so the same argument applies after either operation. □

Write $\Lambda_B = \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B$. The marked endomorphism comparison precedes formation of the relative contraction quotient $\Lambda_B/\Lambda_B e_0 \Lambda_B$.

Proposition 2.22 (The special fibre is the basic NCCR)**.** *Let $B = \mathbb{C}[[s, w]]$ carry a classifying map whose restriction modulo s is the central arc. Localize Λ_B at the central contraction point of its centre and complete there. The reduction $\widehat{\Lambda_B}/(s)$ is isomorphic to the basic NCCR of the original flop, preserving the free idempotent e_0.*

Proof. The algebra is finite over its Noetherian centre, so completion commutes with reduction modulo s. Lemma 2.3 then identifies the completed geometry with the original flop.

It remains to identify the transported generator. On the completed one-curve flop, Van den Bergh's classification gives exactly two indecomposable positive perverse projectives: $\mathcal{O}$ and the maximal extension $\mathcal{M}$, with $\mathrm{rk}\,\mathcal{M} = \ell$ and $\deg_C \det \mathcal{M} = 1$ [33, Proposition 3.5.4 and Theorem 3.5.5]. Write the reduced nonfree universal summand as $\mathcal{O}^{\oplus a} \oplus \mathcal{M}^{\oplus b}$. Determinant degree gives $b = 1$, while rank gives $\ell = a + \ell$, hence $a = 0$. The reduced positive generator is therefore $\mathcal{O} \oplus \mathcal{M}$. Dualizing gives the selected zero-perverse basic generator and fixes e_0. □

2.2.2 The generator at a transverse root

After the finite extension of Proposition 2.17, let λ be a geometric generic coefficient-j point, let $\widehat{Y}_\lambda \to \operatorname{Spec} \widehat{R}_\lambda$ be its completed small resolution, and let C_λ be its exceptional curve. The local model is split over K'.

Lemma 2.23 (The split ODP projectives over the splitting field)**.** *Let F be any characteristic-zero field and put*

$$R_F = F[[a_1, a_2, b_1, b_2]]/(a_1a_2 - b_1b_2), \qquad I = (a_1, b_1).$$

On the small resolution $\pi_Z \colon Z = \operatorname{Proj}_{R_F} \bigoplus_{n\geq 0} I^n \to \operatorname{Spec} R_F$, let $\mathcal{Q} = \mathcal{O}_Z(1)$. Then $\mathcal{O}_Z \oplus \mathcal{Q}$ is the basic positive perverse projective generator. Every positive perverse projective vector bundle is a direct sum of $\mathcal{O}_Z$ and $\mathcal{Q}$, and

$$\operatorname{rk} \mathcal{Q} = 1, \qquad \deg_C \mathcal{Q} = 1, \qquad \pi_{Z*}\mathcal{Q} = I.$$

These assertions remain true after completion along the exceptional curve.

Proof. Write $\theta = b_1/a_1$ on the a_1-chart and $\theta^{-1} = a_1/b_1$ on the b_1-chart. After removing the Rees torsion and completing along the exceptional curve, the charts and their gluing are

$$\widehat{U}_{a_1} = \operatorname{Spf} F[\theta][[a_1, b_2]], \quad b_1 = a_1\theta, \quad a_2 = \theta b_2,$$

$$\widehat{U}_{b_1} = \operatorname{Spf} F[\theta^{-1}][[b_1, a_2]], \quad a_1 = \theta^{-1}b_1, \quad b_2 = \theta^{-1}a_2.$$

Thus Z is the split Atiyah small resolution. Its exceptional curve is given by $a_1 = b_2 = 0$ on $\widehat{U}_{a_1}$ and $b_1 = a_2 = 0$ on $\widehat{U}_{b_1}$, so $C \cong \mathbb{P}^1_F$. The transition of the tautological quotient is that of $\mathcal{O}_{\mathbb{P}^1}(1)$, hence $\mathcal{Q}|_C = \mathcal{O}_C(1)$. The quotient $R_F/I \cong F[[a_2, b_2]]$ and the depth lemma show that I is maximal Cohen–Macaulay, hence reflexive. The pushforward of $\mathcal{Q}$ is reflexive by Hartogs extension across C in the smooth small resolution. It agrees with I away from the vertex, so $\pi_{Z*}\mathcal{Q} = I$.

To verify the perverse-generator conditions, use $N_{C/Z} \cong \mathcal{O}_C(-1)^{\oplus 2}$. On the n-th infinitesimal graded piece of the formal neighbourhood of C, the restrictions of $\mathcal{O}_Z$ and $\mathcal{Q}^{-1}$ are respectively direct sums of

$$\operatorname{Sym}^n N_{C/Z}^\vee \cong \mathcal{O}_C(n)^{\oplus(n+1)} \quad \text{and} \quad \mathcal{O}_C(n-1)^{\oplus(n+1)}.$$

Their first cohomology vanishes for every $n \geq 0$. Induction over the infinitesimal thickenings, followed by the theorem on formal functions, gives

$$R^1\pi_{Z*}\mathcal{O}_Z = R^1\pi_{Z*}\mathcal{Q}^{-1} = 0.$$

The algebraic small resolution of the affine quadratic cone is the total space of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$; the charts above describe its completion along the zero section. The bundle $\mathcal{Q}$ is relatively ample and globally generated. Together with the cohomology vanishings and the structure-sheaf summand, these properties verify Van den Bergh's projective-generator criterion [33, Propositions 3.2.5–3.2.7], both over the complete local target and on the formal completion.

For the classification of projectives, the endomorphism algebra of $\mathcal{O}_Z \oplus \mathcal{Q}$ is finite over the complete local ring R_F and is semiperfect. The two summands have local endomorphism rings

$\mathrm{End}(\mathcal{O}_Z) = R_F = \mathrm{End}(\mathcal{Q})$, and their degrees on C distinguish them. Their projectors form a complete set of pairwise inequivalent primitive idempotents. Projectives over a semiperfect ring are direct sums of the corresponding indecomposable projectives. The tilting equivalence therefore gives the decomposition of every projective in the heart. Both summands and their idempotents are defined over F, so the same decomposition holds on the formal completion. □

Proposition 2.24 (Decomposition of the local projective generator)**.** *Let $\mathcal{Q}$ be the unique nonfree indecomposable positive perverse projective on the completed ODP small resolution, normalized by* $\mathrm{rk}\,\mathcal{Q} = 1$ *and* $\deg \det \mathcal{Q}|_{C_\lambda} = 1$*. Then*

$$\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus(\ell-j)} \oplus \mathcal{Q}^{\oplus j}. \tag{2.6}$$

After pushforward to the ODP centre,

$$N_\lambda \cong \widehat{R}_\lambda^{\oplus(\ell-j)} \oplus I_\lambda^{\oplus j},$$

where I_λ is the rank-one Atiyah MCM module.

Proof. Proposition 2.21 and Lemma 2.20 transport the universal positive generator to the completed local model after field specialization and completion. The roots have been split over K', so the local model is the split ODP of Lemma 2.23 with $F = K'$. That lemma and Krull–Schmidt give $\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus a} \oplus \mathcal{Q}^{\oplus b}$. Rank gives $a + b = \ell$. Determinant commutes with pullback, and Lemma 2.8 gives

$$\deg_{C_\lambda} \det \widehat{\mathcal{P}}'_\lambda = j \deg_C \det \mathcal{P}' = j.$$

The degree of the decomposition is b. Thus $b = j$ and $a = \ell - j$. Lemma 2.23 identifies the pushforward of $\mathcal{Q}$ with the Atiyah module I_λ. □

2.2.3 The local contraction quotient

The preceding decomposition identifies the local contraction quotient with stable endomorphisms of the Atiyah module. For a module over a local ring R, write $\underline{\mathrm{End}}_R$ for endomorphisms modulo maps factoring through finite free R-modules.

Lemma 2.25 (Stable endomorphisms of the Atiyah module)**.** *Let*

$$R_{\mathrm{odp}} = F[[a_1, a_2, b_1, b_2]]/(a_1a_2 - b_1b_2), \qquad I = (a_1, b_1),$$

where F has characteristic zero. Then

$$\underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I) \cong F, \qquad \underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I^{\oplus j}) \cong M_j(F).$$

Proof. Write $R = R_{\mathrm{odp}}$. Since I is rank-one reflexive over the normal domain R, its endomorphisms are multiplications by fractions. At each height-one prime, I is free of rank one; hence $(I : I) \subseteq \bigcap_{\mathrm{ht}\, p=1} R_p = R$. Thus $\mathrm{End}_R(I) = R$.

To identify the maps that factor through a free module, we claim that

$$I^{-1} = R + R(b_2/a_1).$$

The displayed right-hand side lies in I^{-1}, since $(b_2/a_1)a_1 = b_2$ and $(b_2/a_1)b_1 = a_2$. Conversely, if $qI \subset R$, put $r = qa_1 \in R$. Then $b_1 r \in a_1 R$. In $R/(a_1) = F[[a_2, b_1, b_2]]/(b_1 b_2)$, the annihilator of b_1 is (b_2). Thus $r = a_1 r_0 + b_2 r_1$, and $q = r_0 + (b_2/a_1)r_1$, proving the claim. Endomorphisms factoring through a free module form the ideal

$$II^{-1} = (a_1, b_1) + (b_2, a_2) = (a_1, a_2, b_1, b_2) = \mathfrak{m}.$$

Therefore $\underline{\mathrm{End}}_R(I) = R/\mathfrak{m} = F$; taking matrices gives the second assertion. □

Proposition 2.26 (Comparison with the MCM generator)**.** *For the completed ODP small resolution, pushforward gives an isomorphism*

$$\mathrm{End}_{\widehat{Y}_\lambda}(\widehat{\mathcal{P}}_\lambda) \cong \mathrm{End}_{\widehat{R}_\lambda}(\widehat{R}_\lambda^{\oplus(\ell-j+1)} \oplus I_\lambda^{\oplus j}) \tag{2.7}$$

which preserves the projector onto the distinguished free summand.

Proof. The completed ODP is a normal complete local Gorenstein threefold, and its small resolution is smooth, projective, birational, and crepant, with fibres of dimension at most one and $R\widehat{\pi}_*\mathcal{O} = \widehat{R}$. The positive bundle is the perverse projective generator by Lemma 2.20; its two required $R^1\widehat{\pi}_*$-vanishings follow from the defining $\mathfrak{V}$-condition and global generation. Van den Bergh's comparison theorem identifies the resolved endomorphism algebra with that of the pushed-forward MCM generator [33, Lemma 3.2.9 and Proposition 3.2.10]. Naturality sends the direct-summand projector to the distinguished free projector. □

Theorem 2.27 (The completed coefficient-j factor)**.** *At every transverse coefficient-j point produced by the formal marked deformation of the classifying arc, the completion of the relative contraction quotient is the matrix algebra $M_j(K')$. The maximal ideal of the completed centre acts trivially on this factor.*

Proof. After the arbitrary classifying pullback, central localization and Noetherian completion are flat. For a finite algebra Λ with idempotent e_0,

$$\widehat{\Lambda/\Lambda e_0 \Lambda} \cong \widehat{\Lambda}/\widehat{\Lambda} e_0 \widehat{\Lambda}.$$

By Proposition 2.26, the completed quotient is the endomorphism algebra of $\widehat{R}^{\ell-j+1} \oplus I^j$ modulo the ideal generated by one distinguished free projector. Every other free projector factors through that copy, so the quotient is the stable endomorphism algebra of I^j. Lemma 2.25 gives $M_j(K')$. The centre acts through the residue field, proving the last assertion. □

2.3 Assembling a finite free deformation

The preceding two subsections provide the geometric points and their local matrix factors. The root multiplicities come from the Bryan–Katz–Leung divisor count and the GV interpretation used above [6, Lemma 2.8]; Toda's formula supplies the final dimension comparison [32, Theorem 1.1]. We assemble the local factors inside one finite relative quotient. Put $B = \mathbb{C}[[s, w]]$, and let $\nu : \operatorname{Spec} B \to H$ be the completion of the constant-translation family $F(s, w) = f_0(w) + sv$ from Lemmas 2.13 and 2.14. Set

$$R_B = \mathcal{R}_\ell \otimes_{\mathbb{H}_\ell} B.$$

Let $\mathfrak{n} \subset R_B$ be the maximal ideal of the central contraction point (it contains s), and define the central completion

$$S_\nu = \widehat{(R_B)_{\mathfrak{n}}}.$$

Let $\mathcal{T}_\nu$ be the completed pullback of the tilting bundle. The endomorphism comparison identifies the corresponding algebra before forming its free-vertex quotient:

$$\Lambda_\nu = \mathcal{A}_\ell \otimes_{\mathcal{R}_\ell} S_\nu \cong \mathrm{End}(\mathcal{T}_\nu)^{\mathrm{op}}, \qquad \mathscr{A} = \Lambda_\nu / \Lambda_\nu e_0 \Lambda_\nu. \tag{2.8}$$

Proposition 2.28 (Identification of the special fibre)**.** *There is a marked algebra isomorphism*

$$\mathscr{A}/(s) \cong A_{\mathrm{con}}.$$

Proof. The generator comparison is made before the idempotent quotient. Proposition 2.22 gives $\Lambda_\nu/(s) \cong \Lambda$, the selected basic NCCR of the original flop, preserving e_0. Right exactness then gives

$$\mathscr{A}/(s) \cong \frac{\Lambda}{\Lambda e_0 \Lambda} = A_{\mathrm{con}}.$$

□

Lemma 2.29 (Topological finiteness)**.** *Let $R_0 = \mathbb{C}[[s]]$, and define S as follows:*

$$S = R_0[[x_1, \dots, x_m]]/J.$$

Then S is complete Noetherian. Let M be finite over S. If M/sM is finite-dimensional over $\mathbb{C}$, then M is finite over R_0.

Proof. Choose lifts $m_1, \dots, m_d$ of a $\mathbb{C}$-basis of M/sM. For $m \in M$, successive reduction modulo sM gives

$$m = \sum_{i=1}^{d} \left(\sum_{a=0}^{q-1} c_{ia} s^a \right) m_i + s^q m^{(q)}, \qquad c_{ia} \in \mathbb{C}.$$

The series $c_i(s) = \sum_{a \geq 0} c_{ia} s^a$ belong to R_0. For each q, their tails are divisible by s^q, so

$$m - \sum_i c_i(s) m_i \in s^q M.$$

The ring S is Noetherian local, s lies in its maximal ideal, and M is finite over S. Krull intersection gives $\bigcap_q s^q M = 0$. Hence the m_i generate M over R_0. □

Proposition 2.30 (Finiteness of the relative contraction algebra)**.** *The algebra $\mathscr{A}$ is a finite $\mathbb{C}[[s]]$-module.*

Proof. The universal target is of finite type over H, so its pullback is of finite type over $B = \mathbb{C}[[s, w]]$. Completion at the central contraction point gives a ring of the form

$$S_\nu \cong \mathbb{C}[[s, x_1, \dots, x_m]]/J$$

for finitely many generators. Proposition 2.4(i) states that $\mathcal{A}_\ell$ is finite over $\mathcal{R}_\ell$. Base change and completion make Λ_ν finite over S_ν, and its idempotent quotient $\mathscr{A}$ remains finite over S_ν.

By Proposition 2.28, $\mathscr{A}/(s) \cong A_{\mathrm{con}}$ is finite-dimensional. The topological finiteness lemma now gives finiteness over $\mathbb{C}[[s]]$. □

Lemma 2.31 (Fullness away from the root support)**.** *Over the geometric generic marked slice, the relative contraction quotient vanishes away from the finitely many transverse curve-root points. Consequently it is a finite algebra supported on exactly those closed points.*

Proof. By Proposition 2.4(ii), D^{curv} is the full locus where a complete curve occurs. Away from D^{curv}, a positive-dimensional fibre of the projective morphism would contain a complete curve. The morphism is therefore quasi-finite and hence finite. Since $R\pi_{\ell *}\mathcal{O}_{\mathcal{Y}_\ell} = \mathcal{O}_{\mathcal{X}_\ell}$, a finite base-changed morphism is an isomorphism. Proposition 2.17 meets D^{curv} only at the good transverse roots and supplies exactly the curves already counted. Away from those points, every summand of the pushed-forward vector bundle is locally free and, after shrinking, finite free. Every matrix unit in the endomorphism algebra of the pushed-forward vector bundle then factors through the distinguished rank-one free summand. Thus e_0 is full and the idempotent quotient vanishes.

At every transverse point, Theorem 2.27 supplies a nonzero matrix block. By Proposition 2.30, the generic quotient is a finite-dimensional algebra over the generic point, finite over its Artinian centre. Its support is consequently a finite set of closed points, and the preceding argument identifies those points with the transverse curve-root points. □

Lemma 2.32 (Assembly of the local factors)**.** *Let S be a Noetherian K'-algebra and A a finite S-algebra supported at K'-rational maximal ideals $p_1, \dots, p_r$, so $S/p_i = K'$. Suppose*

$$\widehat{A_{p_i}} \cong M_{d_i}(K')$$

and the maximal ideal of $\widehat{S_{p_i}}$ acts trivially. Then

$$A \cong \prod_{i=1}^{r} M_{d_i}(K').$$

In particular its scheme-theoretic support is reduced.

Proof. Completion $S_{p_i} \to \widehat{S_{p_i}}$ is faithfully flat. Since $p_i\widehat{A_{p_i}} = 0$, faithful flatness gives $p_i A_{p_i} = 0$. Hence A_{p_i} already equals its completion and is $M_{d_i}(K')$. Put $J = \bigcap_i p_i$. The finite module JA localizes to zero away from the support and also at every p_i, where all other maximal ideals become units. Thus $JA = 0$. The Chinese remainder theorem gives $S/J \cong (K')^r$; its central idempotents split A, and the local factors are the stated matrix algebras. □

Proposition 2.33 (The geometric generic fibre)**.** *After the finite splitting extension $K'/\mathbb{C}((s))$ of Proposition 2.17,*

$$\mathscr{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \tag{2.9}$$

Proof. Propositions 2.16 and 2.17 give exactly n_j coefficient-j points. In the normalized ramified model of that proposition, $K' = \mathbb{C}((u))$, $s = u^e$, and every root lies in $u\mathbb{C}[[u]]$. The split points are therefore K'-rational points of the formal w-disk, all specializing to its central maximal ideal. Lemma 2.31 gives the full support, and Theorem 2.27 identifies every completed factor with $M_j(K')$. Apply Lemma 2.32 simultaneously to all support points. □

Lemma 2.34 (Freeness from the fibre dimensions)**.** *Let M be a finite module over the DVR $\mathbb{C}[[s]]$. If*

$$\dim_{\mathbb{C}} M/sM = \dim_{\mathbb{C}((s))} M[1/s] = r,$$

then M is free of rank r.

Proof. The structure theorem gives

$$M \cong \mathbb{C}[[s]]^{\oplus r} \oplus \bigoplus_{h=1}^{m} \mathbb{C}[[s]]/(s^{e_h}).$$

The generic dimension is the free rank, while reduction modulo s has dimension $r+m$. Equality forces $m = 0$. □

Proof of Theorem 2.1. Take the relative contraction quotient defined in (2.8). Proposition 2.28 gives its special fibre, and Proposition 2.33 gives its geometric generic product. In particular

$$\dim_{\mathbb{C}((s))} \mathscr{A}[1/s] = \sum_{j=1}^{\ell} j^2 n_j.$$

Proposition 2.30 gives finiteness over $\mathbb{C}[[s]]$. On the other hand, Proposition 2.4(v) and the special-fibre identification give

$$\dim_{\mathbb{C}} \mathscr{A}/(s) = \dim_{\mathbb{C}} A_{\mathrm{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Lemma 2.34 now removes the possible torsion and proves that $\mathscr{A}$ is finite free of this rank. □

3 The cocenter and uniqueness of the semisimple generic fibre

Section 2 constructs a deformation whose geometric generic fibre is a specified product of matrix algebras and puts the contraction algebra in the closure of the corresponding change-of-basis orbit. Here we prove that this orbit closure is the unique irreducible component of the algebra-law scheme through the contraction algebra. Comparing the orbit and tangent dimensions reduces the claim, by symmetry and periodicity, to a cocenter count on the marked ADE base.

3.1 Statement and proof strategy

For the completed simple one-curve flop $f\colon X \to \operatorname{Spec} R$, let $A = A_{\mathrm{con}}$ be its contraction algebra with $d = \dim_{\mathbb{C}} A$, and let n_j be its genus-zero GV invariants. Put

$$S_{\mathrm{GV}} := \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}.$$

Theorem 3.1 (Uniqueness of the semisimple generic fibre for one curve)**.** *For every completed simple one-curve flop of a smooth complex threefold,*

$$\dim_{\mathbb{C}} HH_0(A) = \dim_{\mathbb{C}} Z(A) = \sum_{j=1}^{\ell} n_j. \tag{3.1}$$

Consequently A is a smooth point of the fixed-unit associative-law scheme X_d, the orbit closure $\overline{G \cdot S_{\mathrm{GV}}}$ is the unique irreducible component of X_d through A, and every complex semisimple algebra to which A deforms is isomorphic to S_{GV}. More generally, if $\mathscr{D}$ is a finite free unital $\mathbb{C}[[u]]$-algebra with $\mathscr{D}/u\mathscr{D} \cong A$ and geometrically semisimple generic fibre, then $\mathscr{D} \otimes_{\mathbb{C}[[u]]} \overline{\mathbb{C}((u))} \cong S_{\mathrm{GV}} \otimes_{\mathbb{C}} \overline{\mathbb{C}((u))}$.

The theorem identifies the split generic fibre. Section 2 separately constructs a finite free integral $\mathbb{C}[[u]]$-deformation that realizes it.

The orbit criterion requires equality of the tangent and orbit dimensions. Symmetry, periodicity, and centre–cocenter duality reduce that equality to $HH_0(A)$. We compute this cocenter from a universal presentation on the Cartan cover, descend the presentation to the marked base, and restrict it to the original classifying arc. Each reduced curve divisor occurs once in its determinant, so the arc length counts the simple factors in the resulting product of matrix algebras.

3.2 The orbit criterion

Jelisiejew–Smoktunowicz study the unframed scheme of algebra laws; their orbit analysis gives local closedness and openness for semisimple laws [19, Section 1 and Theorem 2.14]. We need the corresponding statements for a fixed identity. Let k be algebraically closed of characteristic zero, let $V = k \cdot 1 \oplus W$ have dimension d, and let X_d be the affine scheme of associative multiplications on V for which the distinguished vector 1 is the identity. The scheme X_d is defined by the associativity equations. The group preserving the identity is

$$G = \mathrm{Stab}_{\mathrm{GL}(V)}(1).$$

For a unital algebra Λ, the normalized Hochschild cochains are

$$C^n_{\mathrm{norm}}(\Lambda, \Lambda) = \mathrm{Hom}_k((\Lambda/k \cdot 1)^{\otimes n}, \Lambda).$$

We write Z^n_{norm} and B^n_{norm} for the cocycles and coboundaries of this complex.

Lemma 3.2 (Fixed-unit tangent spaces)**.** *For $\Lambda \in X_d(k)$,*

$$T_\Lambda X_d = Z^2_{\mathrm{norm}}(\Lambda, \Lambda), \qquad T_\Lambda(G \cdot \Lambda) = B^2_{\mathrm{norm}}(\Lambda, \Lambda),$$

and

$$\mathrm{Lie}\,\mathrm{Stab}_G(\Lambda) = \mathrm{Der}(\Lambda).$$

Moreover, G is smooth, connected, and irreducible of dimension $d(d-1)$, and its orbits are precisely the unital isomorphism classes.

Proof. A dual-number multiplication $m + \varepsilon\phi$ has the prescribed unit precisely when ϕ is normalized. Linearized associativity is then the normalized Hochschild cocycle equation. Differentiating transport by $1 + \varepsilon f$, with $f(1) = 0$, gives the normalized coboundaries; the kernel of the resulting linear map is the derivation space.

Relative to $k \cdot 1 \oplus W$, every element of G has the unique form

$$\begin{pmatrix} 1 & \lambda \\ 0 & h \end{pmatrix}, \qquad \lambda \in W^*, \quad h \in \mathrm{GL}(W).$$

Thus $G \cong W^* \rtimes \mathrm{GL}(W)$, which proves the assertions about the group. A unital algebra isomorphism fixes the identity, so its change of basis lies in G. □

Lemma 3.3 (A family gives fixed-unit orbit closure)**.** *Let $\mathfrak{o}$ be a complete equicharacteristic DVR with algebraically closed residue field k, coefficient field $k \subset \mathfrak{o}$ and fraction field K. Let $\mathscr{C}$ be a finite free unital $\mathfrak{o}$-algebra with special fibre A. If S is a d-dimensional k-algebra and*

$$\mathscr{C}_K \otimes_K \overline{K} \cong S \otimes_k \overline{K},$$

then

$$A \in \overline{G \cdot S} \subset X_d.$$

Proof. A uniformizer identifies $\mathfrak{o}$ with $k[[\varpi]]$. The unit of $\mathscr{C}$ is unimodular because it reduces to the nonzero unit of A. Extend it to an $\mathfrak{o}$-basis lifting the prescribed basis of A. The resulting structure constants define a morphism $\operatorname{Spec} \mathfrak{o} \to X_d$ with closed point A.

The geometric generic-fibre isomorphism has finitely many entries algebraic over K, so it descends to a finite extension K'/K. The integral closure $\mathfrak{o}'$ of $\mathfrak{o}$ in K' is a finite complete DVR. Its residue field is a finite extension of the algebraically closed field k, hence equals k. Since the generic-fibre isomorphism preserves the unit, the base-changed generic point is a $G(K')$-translate of S.

Let $Y = \overline{G \cdot S} \subset X_d$ with its reduced closed structure. Every equation of Y vanishes at the generic point of the base-changed arc and hence vanishes in the domain $\mathfrak{o}'$. Thus the whole arc factors through Y, and its closed point is A. This is the required orbit-closure relation. □

The next lemma gives the boundary form of the orbit argument. The algebraic-group action makes the orbit smooth, and a local dimension comparison controls a point in its closure.

Lemma 3.4 (Smoothness at an open-orbit boundary)**.** *Let a smooth irreducible algebraic group H act on a finite-type scheme X over an algebraically closed field of characteristic zero. Write $O = H \cdot y$.*

(i) *If $\dim T_y X = \dim O$, then y is smooth and O is open in X.*

(ii) *Suppose O is open. If a closed point $x \in \overline{O}$ satisfies $\dim T_x X = \dim O$, then x is smooth and $\overline{O}$ is its unique irreducible component. Every open orbit whose closure contains x equals O.*

Proof. The orbit is a smooth locally closed subvariety: stabilizers in characteristic zero are smooth, and the orbit is the corresponding homogeneous space [4, Proposition 1.11]. For (i), let $R = \mathcal{O}_{X,y}$ and write the orbit local ring as R/I. With $\delta = \dim O$,

$$\delta = \dim(R/I) \leq \dim R \leq \operatorname{edim} R = \delta.$$

Thus R is regular, hence a domain. The prime ideal I has height zero, so it is zero. The locally closed orbit therefore contains an open neighbourhood of y; its translates show that O is open in the original scheme X. For (ii), irreducibility of H makes $\overline{O}$ an irreducible component of dimension δ. At the closed point x, the same dimension inequalities give regularity and hence smoothness. A regular local ring has one minimal prime, so this is the unique component through x. The closure of any other open orbit through x is the same component. The two nonempty open orbits must intersect, and hence coincide. □

Proposition 3.5 (Openness of semisimple orbits)**.** *If $S \in X_d(k)$ is split semisimple, then S is a smooth point of the scheme X_d and $G \cdot S$ is open in X_d.*

Proof. The algebra S is separable, hence projective as an S^e-module, so $HH^i(S,S) = 0$ for $i > 0$. In particular its normalized cocycles and coboundaries in degree two agree. By Lemma 3.2, the tangent dimension at S is the orbit dimension. Lemma 3.4(i) proves smoothness and openness. □

Theorem 3.6 (Orbit-theoretic smoothness and uniqueness)**.** *Let $A, S \in X_d(k)$, with S split semisimple. Assume*

$$A \in \overline{G \cdot S}, \qquad \dim T_A X_d = \dim(G \cdot S). \tag{3.2}$$

Then:

(i) *A is a smooth point of X_d;*

(ii) *$\overline{G \cdot S}$ is the unique irreducible component of X_d through A;*

(iii) *if S' is split semisimple and $A \in \overline{G \cdot S'}$, then $S' \cong S$.*

Proof. Proposition 3.5 makes both semisimple orbits open, and G is smooth and irreducible by Lemma 3.2. Apply Lemma 3.4(ii). The equality of the two orbits yields a unital algebra isomorphism, again by Lemma 3.2. □

Corollary 3.7 (Hochschild numerical criterion)**.** *Let $A, S \in X_d(k)$, with S semisimple and $A \in \overline{G \cdot S}$. The dimension equality in Theorem 3.6 is equivalent to*

$$\dim \mathrm{Der}(A) - \dim HH^2(A) = d - \dim Z(S). \tag{3.3}$$

In particular, it holds if

$$\dim Z(A) = \dim Z(S), \qquad \dim HH^1(A) = \dim HH^2(A).$$

Proof. Lemma 3.2 gives

$$\dim T_A X_d = d(d-1) - \dim \mathrm{Der}(A) + \dim HH^2(A).$$

For split semisimple S, all derivations are inner and $\dim \mathrm{Der}(S) = d - \dim Z(S)$. Therefore

$$\dim(G \cdot S) = d(d-1) - (d - \dim Z(S)),$$

where characteristic zero ensures that the stabilizer group scheme is smooth, so its dimension equals the dimension of its Lie algebra. This proves the equivalence. Finally, $\dim \mathrm{Der}(A) = d - \dim Z(A) + \dim HH^1(A)$. □

3.3 Symmetry, periodicity, and Tate duality

The deformation of Section 2 supplies the orbit-closure hypothesis. For the remaining numerical hypothesis, symmetry and four-periodicity give $\dim HH^1(A) = \dim HH^2(A)$, and a symmetrizing trace identifies the centre with the dual cocenter. The orbit criterion is therefore reduced to one cocenter-dimension calculation. The next results supply these two identifications for the contraction algebra.

Proposition 3.8 (Symmetry)**.** *The contraction algebra is symmetric:* $A_{\mathrm{con}} \cong DA_{\mathrm{con}}$ *as* A_{con}*-bimodules.*

Proof. Let N be the nonfree Van den Bergh summand. The completed base R is an isolated cDV singularity and $A = \underline{\mathrm{End}}_R(N)^{\mathrm{op}}$. The stable endomorphism algebra is symmetric by [2, Theorem 2.6], and the opposite of a symmetric algebra is symmetric. □

Proposition 3.9 (Bimodule four-periodicity)**.** *In the stable category of* A*-bimodules,*

$$\Omega^4_{A^e}(A) \simeq A.$$

Proof. Jasso–Keller–Muro's Theorem 3.2.5 applies because R is an isolated cDV singularity with a crepant resolution. That theorem furnishes the exact sequence of A-bimodules

$$0 \longrightarrow A \longrightarrow P_3 \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow A \longrightarrow 0$$

with projective middle terms [18, Theorem 3.2.5]. This sequence represents the restricted universal Massey product in $\mathrm{Ext}^4_{A^e}(A, A)$ and yields the stated stable bimodule isomorphism. □

Alternative cluster-tilting argument. Dugas's theorem provides a second route to four-periodicity from a cluster-tilting object in a Frobenius category. Let $\mathcal{B} = \mathrm{CM}(R)$, whose projective-injective objects are $\operatorname{add} R$. Write the Van den Bergh tilting bundle on the smooth source X as $\mathcal{O}_X \oplus \mathcal{N}$, and put $N = f_*\mathcal{N}$. By [18, Theorem 1.3.1], N is a $2\mathbb{Z}$-cluster-tilting object of $\underline{\mathrm{CM}}(R)$, and $\underline{\mathrm{End}}_R(N)^{\mathrm{op}} = A^{\mathrm{op}}$. Thus $R \oplus N$ is a 2-cluster-tilting object of the Frobenius category $\mathcal{B}$. The hypersurface identity $\Omega^2_R \simeq \mathrm{Id}$ gives the parameters $d_{\mathrm{D}} = 2$ and $r_{\mathrm{D}} = 1$ in [8, Theorem 1.1(3)]. If $A \neq \mathbb{C}$, the one-curve contraction algebra is local and nonsemisimple, so it has no semisimple block, and Dugas's theorem gives a periodic projective bimodule resolution of A^{op}, hence of A, with period dividing $(d_{\mathrm{D}} + 2)r_{\mathrm{D}} = 4$. If $A = \mathbb{C}$, its stable bimodule class is zero and its positive Hochschild groups vanish.

Lemma 3.10 (Tate–Hochschild duality)**.** *For every symmetric finite-dimensional algebra* A *over* $\mathbb{C}$*,*

$$(\widehat{HH}^{-n}(A))^\vee \cong \widehat{HH}^{n-1}(A) \qquad (n \in \mathbb{Z}).$$

If A *is four-periodic in the stable bimodule category, then*

$$DHH^1(A) \cong HH^2(A), \qquad \dim HH^1(A) = \dim HH^2(A). \tag{3.4}$$

Proof. For a symmetric algebra A, the enveloping algebra A^e is symmetric, and $\widehat{HH}^*(A) = \widehat{\mathrm{Ext}}^*_{A^e}(A, A)$. The displayed duality is the Tate–Hochschild duality of [27, equation (2.11)]. Taking $n = 3$ and using the period-four bimodule isomorphism gives

$$DHH^1(A) \cong D\widehat{HH}^{-3}(A) \cong \widehat{HH}^2(A) = HH^2(A),$$

because positive Tate degrees agree with ordinary Hochschild cohomology. □

Etingof–Latour–Rains use the same trace-space duality for their central extensions [10, Corollary 2.2]. The following symmetric-algebra form is the one needed here and later for relative traces.

Lemma 3.11 (Centre–cocenter duality)**.** *If A is finite-dimensional symmetric with symmetrizing trace $\tau\colon A \to \mathbb{C}$, then*

$$Z(A) \xrightarrow{\sim} HH_0(A)^\vee, \qquad z \mapsto (\bar{a} \mapsto \tau(za)).$$

Consequently,

$$\dim Z(A) = \dim HH_0(A). \tag{3.5}$$

Proof. For a central element z,

$$\tau(z[a,b]) = \tau(zab - zba) = \tau(zab - azb) = 0,$$

so the functional factors through $A/[A,A]$. Conversely, identify $A \cong DA$ by the symmetrizing form. If the functional associated to $c \in A$ annihilates every commutator, then for all a, b,

$$0 = \tau(c[a,b]) = \tau((ca - ac)b).$$

Nondegeneracy forces $ca = ac$ for all a, so $c \in Z(A)$. The annihilator of the commutator subspace is therefore exactly the centre, which proves perfection. □

3.4 Reduction to the cocenter dimension

For $G = \mathrm{Stab}_{\mathrm{GL}_d}(1)$, Lemma 3.2 and Corollary 3.7 give the tangent and orbit formulas below. The second equality uses $\dim \mathrm{Der}(A) = d - \dim Z(A) + \dim HH^1(A)$ and Lemma 3.10; the last uses $\dim Z(S_{\mathrm{GV}}) = \sum_j n_j$ and the fact that all derivations of a semisimple complex algebra are inner:

$$\dim T_A X_d = d(d-1) - \dim \mathrm{Der}(A) + \dim HH^2(A) \tag{3.6}$$

$$= d(d-1) - d + \dim Z(A), \tag{3.7}$$

$$\dim(G \cdot S_{\mathrm{GV}}) = d(d-1) - d + \sum_j n_j. \tag{3.8}$$

Subtracting and applying Lemma 3.11 gives

$$\dim T_A X_d - \dim(G \cdot S_{\mathrm{GV}}) = \dim HH_0(A) - \sum_j n_j. \tag{3.9}$$

Theorem 2.1 and Lemma 3.3 already give $A \in \overline{G \cdot S_{\mathrm{GV}}}$. Thus the only remaining hypothesis of Theorem 3.6 is that the right-hand side of (3.9) vanish. This reduces Theorem 3.1 to the cocenter count $\dim HH_0(A) = \sum_j n_j$.

3.5 Universal marked cocenter on the Cartan cover

The Cartan cover separates the individual root factors; descent to the marked base will group them by curve coefficient. Fix one of the six primitive marked ADE pairs (Φ, p). Here p is the retained vertex α_* of Section 2, so $m_p(\alpha) = c_*(\alpha)$. Let $\mathfrak{h}$ be the Cartan space of rank r, and put

$$W_0 = \langle s_i : i \neq p \rangle, \qquad H = \mathfrak{h}/W_0.$$

Write

$$\widetilde{R} = \widehat{\mathcal{O}}_{\mathfrak{h},0} \cong \mathbb{C}[[\lambda_1, \dots, \lambda_r]].$$

For a positive root $\alpha = \sum_i m_i(\alpha)\alpha_i$, put

$$N_p := \#\{\alpha \in \Phi^+ : m_p(\alpha) > 0\}. \tag{3.10}$$

Choose an orientation Q of the finite Dynkin diagram, and let $\overline{Q}$ be its double. Over $\widetilde{R}$, set

$$\Pi_\lambda = \widetilde{R}\,\overline{Q}/\left(\sum_{a\in Q_1}[a,a^*] - \sum_i \lambda_i e_i\right).$$

We use the convention of [20, Proposition 4.15] and write $(\lambda,\alpha) = \sum_i m_i(\alpha)\lambda_i$. Let

$$\mathcal{D}_p = e_p \Pi_\lambda e_p.$$

Theorem 3.12 (The marked cocenter on the Cartan cover)**.** *For every primitive marked ADE pair (Φ, p), there is an exact sequence*

$$0 \longrightarrow \widetilde{R}^{N_p} \xrightarrow{\Psi_p} \widetilde{R}^{N_p} \longrightarrow HH_0(\mathcal{D}_p/\widetilde{R}) \longrightarrow 0 \tag{3.11}$$

such that

$$\det \Psi_p = \eta\Delta_p, \qquad \Delta_p := \prod_{\substack{\alpha\in\Phi^+\\ m_p(\alpha)>0}} (\lambda,\alpha), \qquad \eta \in \widetilde{R}^\times. \tag{3.12}$$

To prove Theorem 3.12, we realize its presentation as multiplication by the central element on the cocenter of a finite central extension. We first compute the generic rank and the closed-fibre dimension of that cocenter. The coefficient-five E_8 case is completed in the next subsection; the proof of the presentation then uses only the resulting equality.

The central extension. Take $\mu = \rho$, normalized by $\rho_i = (\rho,\alpha_i) = 1$. In the notation of Etingof–Rains [9], the universal parameters x_i satisfy $\sum_a [a,a^*] = \sum_i x_i e_i$, and z is adjoined centrally. The coefficient ring for the regular weight ρ is

$$K_\rho = \mathbb{C}[x_1 - \rho_1 z, \dots, x_r - \rho_r z].$$

The fibre at a parameter λ is the algebra Π^ρ_λ with relations

$$\sum_a [a,a^*] = \sum_i (\rho_i z + \lambda_i) e_i.$$

Via the coordinates $\lambda_i := x_i - \rho_i z$, completing K_ρ at its augmentation gives the formal power-series ring $\mathbb{C}[[\lambda_1,\dots,\lambda_r]]$, namely the completed Cartan parameter ring $\widetilde{R}$. The λ_i are base coordinates, whereas z remains a central element of the algebra. The finite-free statement over K_ρ is exactly [9, Theorem 3.2(i)]; passing to the completed base and the marked corner is part of the present construction.

Define the marked corner of the completed universal central extension by

$$E_p := e_p(\Pi[z]\widehat{\otimes}_{K_\rho}\widetilde{R})e_p, \qquad \mathsf{M}_p := HH_0(E_p/\widetilde{R}).$$

Since the Etingof–Rains central extension is finite free over K_ρ, the corner E_p is finite over $\widetilde{R}$. Its cocenter M_p, being a quotient of E_p, is finite as well.

The closed fibre of E_p is $e_p\Pi_0^\rho e_p$. Since HH_0 is the cokernel of the commutator map, it commutes with arbitrary base change. In particular,

$$\mathsf{M}_p \otimes_{\widetilde{R}} \mathbb{C} \cong HH_0(e_p\Pi_0^\rho e_p).$$

Generic rank. Etingof–Rains prove generic semisimplicity in [9, Theorem 3.2(v) and the display immediately following it]. Their positive-root-indexed decomposition has ordinary deformed preprojective factors. The Crawley–Boevey criterion supplies a simple module of dimension vector α [36, Theorem 1.2]. Combined with the Etingof–Rains dimension calculation, this criterion identifies these factors as matrix algebras. After passing to a splitting field $L/\operatorname{Frac}(\widetilde{R})$,

$$(E_p)_L \cong \prod_{m_p(\alpha)>0} M_{m_p(\alpha)}(L). \tag{3.13}$$

Each marked matrix factor contributes one dimension to its cocenter, so

$$\dim_L(\mathsf{M}_p)_L = N_p. \tag{3.14}$$

The closed-fibre dimension. To prove freeness, we also need the special-fibre equality:

$$\dim_{\mathbb{C}} HH_0(e_p\Pi_0^{\rho}e_p) = N_p. \tag{3.15}$$

At $\lambda = 0$, our normalization of ρ gives the central-extension relation used by Etingof–Latour–Rains (ELR) [10]. Their case analysis at $\mu = \rho$ computes the nodal corners [10, Section 3.2]. ELR state the calculation for nodal vertices, so the comparison with the six primitive flop markings and their deleted Levi diagrams uses Karmazyn's presentations below.

ELR call the branching vertex of a Dynkin diagram of type D or E the *nodal vertex*. Karmazyn's six primitive flop presentations determine the retained vertex and the Levi Weyl subgroup W_0. The retained vertices and Levi types follow from [20, Examples 4.22–4.27]. With that numbering, they are:

length	Dynkin type	retained vertex	W_0 in Karmazyn	deleted diagram	ELR status
1	A_1	unique	trivial	empty	immediate
2	D_4	$p = 4$	S_2^3	A_1^3	nodal D
3	E_6	$p = 6$	$S_2 \times S_3^2$	$A_1 \sqcup A_2^2$	nodal E_6
4	E_7	$p = 7$	$S_3 \times S_2 \times S_4$	$A_2 \sqcup A_1 \sqcup A_3$	nodal E_7
5	E_8	$p = 4$	$S_4 \times S_5$	$A_3 \sqcup A_4$	non-nodal
6	E_8	$p = 8$	$S_5 \times S_2 \times S_3$	$A_4 \sqcup A_1 \sqcup A_2$	nodal E_8

For a positive root α, ELR's phrase "the root contains the nodal simple root" is equivalent to $m_p(\alpha) > 0$. Hence their root count is N_p.

For type D, ELR [10, Section 3, Lemma 3.4] give degreewise upper bounds for the cocenter of the nodal corner. Summing them gives

$$\dim_{\mathbb{C}} HH_0(e_p\Pi_0^{\rho}e_p) \leq N_p.$$

The generic decomposition (3.14) gives a finite $\widetilde{R}$-module M_p of generic rank N_p. For a finite module over the local domain $\widetilde{R}$, the closed-fibre dimension is at least the generic rank. This supplies the reverse inequality and hence equality.

For the nodal E_6, E_7, E_8 corners, ELR [10, Section 3, Case 2b] compute the same graded cocenter dimensions by exact Magma calculations and obtain equality with the positive-root count. These are the primitive flop lengths $3, 4, 6$ in the table.

For A_1, the closed corner $e_p\Pi_0^\rho e_p = \mathbb{C}[z]/(z)$ has one-dimensional cocenter. The remaining coefficient-five E_8 corner is not a nodal ELR case. Subsection 3.6 and Appendix B establish its special-fibre cocenter dimension 104 by a direct relation calculation and exact certificate. The following subsection supplies this remaining case before we construct the universal cocenter presentation.

3.6 The coefficient-five E_8 computation

The coefficient-five corner supplies the final closed-fibre dimension needed to prove the single-curve cocenter formula. Unlike the nodal cases, it is computed here from explicit homogeneous relations. The exact sequence for the cocenters of a retained corner and of the quotient after deleting vertices, proved in Subsection 4.3.3, later gives the same value in the multi-curve setting. The direct calculation here establishes the special-fibre equality used in the current single-curve proof.

Deleting the coefficient-five vertex of E_8 leaves $A_3 \times A_4$, so

$$N_p = |\Phi^+(E_8)| - |\Phi^+(A_3)| - |\Phi^+(A_4)| = 120 - 6 - 10 = 104. \tag{3.16}$$

Let $B_5 = e_p\Pi_0^\rho e_p$ be the closed marked corner. Appendix B.1 derives its relations directly from the preprojective relations on the ρ-line. It constructs elements $\beta, \gamma \in B_5$ that, together with z, generate B_5 (Lemma B.2). The three relations

$$\begin{aligned}
\mathsf{P} &= (\beta - 4z)(\beta - 5z)(\beta - 6z)(\beta - 7z),\\
\mathsf{R}_4 &= (\gamma + \beta^2)^2 + \beta\gamma\beta - 5z^2(\gamma + \beta^2) + 4z^4,\\
\mathsf{R}_5 &= \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 - 5z^2\gamma\beta
\end{aligned}$$

vanish in B_5 (Lemma B.3). Give $\mathbb{C}[z]\langle\beta,\gamma\rangle$ and B_5 the half-path grading

$$\deg z = \deg \beta = 1, \qquad \deg \gamma = 2.$$

The three relations are homogeneous, so there is a surjection of graded algebras

$$Q_\rho := \frac{\mathbb{C}[z]\langle\beta,\gamma\rangle}{(\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5)} \twoheadrightarrow B_5. \tag{3.17}$$

The induced surjection $HH_0(Q_\rho) \twoheadrightarrow HH_0(B_5)$ transfers cocenter upper bounds from Q_ρ to B_5. Thus it is enough first to bound the quotient defined by the three displayed relations. Remark B.4 explains why the signs of the parameter terms differ from a direct substitution into the displayed universal relations of [20, Example 4.26].

Proposition 3.13 (Exact modular calculation)**.** *Let $h_m = \dim_{\mathbb{C}} HH_0(B_5)_m$. In degrees $0, \dots, 23$, the following sequence gives upper bounds for h_m:*

$$(1, 2, 4, 6, 7, 7, 7, 6, 6, 6, 6, 5, 5, 4, 4, 4, 4, 3, 3, 2, 2, 2, 2, 1).$$

The sum of these bounds is 99*.*

Proof. We bound each cocenter piece using exact modular row ranks. In degree m, use cyclic-word columns $z^a[w]_{\mathrm{cyc}}$ with $a + \mathrm{wt}(w) = m$. The rows are cyclic classes of homogeneous right

multiples of $\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5$. These span the relations in the cocenter: a two-sided multiple $u\mathsf{R}v$ has the same cyclic class as $\mathsf{R}vu$, and central powers of z can be moved to the front. The cokernel in degree m is therefore $HH_0(Q_\rho)_m$.

The relation matrices have integer entries. Exact Gaussian elimination over $\mathbb{F}_{1000003}$ gives the ranks recorded in Appendix B.2; running the same computation over $\mathbb{F}_{1000033}$ gives the same ranks as an additional modular check. The conclusion over characteristic zero uses each nonzero modular minor: that minor is also nonzero over $\mathbb{Q}$, so the modular rank is a lower bound for the characteristic-zero rank. Subtracting from the column count gives an upper bound for $\dim HH_0(Q_\rho)_m$, and the surjection (3.17) gives the stated upper bound for h_m. □

3.6.1 Tail bound and exact grading comparison

The finite calculation controls degrees through 23. To bound the tail, write h for the Coxeter number. In the Etingof–Rains grading, arrows have degree 1 and z has degree 2. Their Hilbert-series formulas [9, Section 2.1, equation (1), and Theorem 3.2(iii)–(iv)] show that the ordinary preprojective algebra has top arrow degree $h-2$, while the ρ-special central extension has top degree $2(h-2)$. On the marked corner (z, β, γ) have ER degrees $(2, 2, 4)$. The finite computation uses exactly half this grading. For E_8, $h = 30$, so the top ER degree is $2(h-2) = 56$, which is computational half-degree 28. Setting $z = 0$ leaves the ordinary preprojective corner. Because the E_8 graph is bipartite, its closed paths have even arrow length; the top arrow-degree bound $h - 2 = 28$ therefore forces the corner to vanish in half-degrees above 14.

Since z is central,

$$HH_0(B_5)/(z) \cong HH_0(B_5/(z)).$$

Consequently multiplication by z is surjective

$$HH_0(B_5)_{m-1} \twoheadrightarrow HH_0(B_5)_m \qquad (m > 14).$$

The finite calculation gives $\dim HH_0(B_5)_{23} \le 1$, so every degree $24, \dots, 28$ has dimension at most one. Together with the degreewise bounds of Proposition 3.13, this gives

$$\dim HH_0(B_5) \le 99 + 5 = 104.$$

The finite module M_p has generic rank $N_p = 104$ by (3.14), and its closed fibre is $HH_0(B_5)$. The closed-fibre dimension is at least the generic rank, which supplies the opposite inequality:

$$\dim HH_0(B_5) \ge 104.$$

The two inequalities therefore give

$$\dim HH_0(B_5) = 104 = N_p. \tag{3.18}$$

This proves (3.15) in the remaining case. Only after the tail bound and generic-rank comparison do the degreewise upper bounds become equalities. Since their sum is the exact dimension, every bound of Proposition 3.13 is attained, each degree $24, \dots, 28$ contributes exactly one, and $HH_0(Q_\rho)_m \to HH_0(B_5)_m$ is an isomorphism for $m \le 23$.

3.7 Constructing the cocenter presentation

Proof of Theorem 3.12. The preceding case analysis, completed by (3.18), gives (3.15) for all six primitive markings.

Freeness. Choose lifts of a basis of the special fibre in (3.15). By (3.15), the special fibre of M_p has dimension N_p. Nakayama gives a surjection

$$\widetilde{R}^{N_p} \twoheadrightarrow \mathsf{M}_p.$$

After tensoring with $\mathrm{Frac}(\widetilde{R})$, (3.14) shows that this map is an isomorphism. Its kernel is a submodule of a free module over the domain $\widetilde{R}$, hence torsion-free; since its generic fibre is zero, the kernel is zero. Therefore

$$\mathsf{M}_p \cong \widetilde{R}^{N_p}. \tag{3.19}$$

Multiplication by the central element. Using the free $\widetilde{R}$-module in (3.19), multiplication by z induces

$$\mu_z \colon \mathsf{M}_p \to \mathsf{M}_p.$$

Setting $z = 0$ recovers the ordinary deformed preprojective corner. Since z is central,

$$HH_0(\mathcal{D}_p/\widetilde{R}) \cong \mathsf{M}_p/z\mathsf{M}_p = \operatorname{coker} \mu_z. \tag{3.20}$$

On the root factor α, the decomposition displayed immediately after [9, Theorem 3.2(v)] gives

$$z|_{V_\alpha} = -\frac{(\lambda, \alpha)}{(\rho, \alpha)}.$$

Therefore μ_z has the same eigenvalue on the one-dimensional cocenter of that factor, and over the fraction field

$$\det \mu_z = \prod_{m_p(\alpha)>0} \left(-\frac{(\lambda, \alpha)}{(\rho, \alpha)} \right).$$

The denominators are nonzero complex constants, so

$$\det \mu_z = \eta \Delta_p, \qquad \eta \in \widetilde{R}^\times.$$

This determinant is nonzero, so μ_z is injective over the domain $\widetilde{R}$. Together with (3.19)–(3.20), this gives the required presentation and determinant. □

3.8 Descent to the marked base

The Cartan-cover calculation distinguishes individual roots. Descending through the Levi Weyl group records roots with the same marked coefficient by one divisor on the marked base.

Write

$$\mathbb{H}_\Gamma = \mathcal{O}(\mathfrak{h}), \qquad \mathbb{H}_\ell = \mathbb{H}_\Gamma^{W_0}, \qquad R_H = \widehat{\mathbb{H}}_\ell, \qquad \widetilde{R} = \widehat{\mathbb{H}}_\Gamma,$$

with completions at the graded origins. The extension $R_H \to \widetilde{R}$ is finite and faithfully flat. Indeed, W_0 is a reflection group, so both rings are regular local of the same dimension. A system of parameters of R_H is a system of parameters, hence a regular sequence, in the finite Cohen–Macaulay ring $\widetilde{R}$. Auslander–Buchsbaum implies that $\widetilde{R}$ is free over R_H; its rank is positive.

3.8.1 The marked algebra on the Cartan cover

Let $\mathcal{A}_\Gamma$ be Karmazyn's universal extended-Dynkin algebra and put $e_{0p} = e_0 + e_p$. Write $\mathcal{T}_{\Gamma,0p}$ for the two-summand generator on the Cartan partial resolution associated to this corner [20, Proposition 4.21], and $\mathcal{T}_\ell$ for the generator of Subsection 2.2. Completions below are taken at the central graded origins.

Proposition 3.14 (Comparison of the marked generators)**.** *On the completed Cartan partial resolution there is an isomorphism*

$$\theta : \widehat{\mathcal{T}}_\ell \widehat{\otimes}_{R_H} \widetilde{R} \xrightarrow{\sim} \widehat{\mathcal{T}}_{\Gamma,0p} \tag{3.21}$$

preserving the structure-sheaf and retained-vertex summands. It induces an isomorphism

$$\widehat{\mathcal{A}}_\ell \widehat{\otimes}_{R_H} \widetilde{R} \xrightarrow{\sim} \widehat{e_{0p}\mathcal{A}_\Gamma e_{0p}} \tag{3.22}$$

sending e_0 to e_0 and e_p to e_p.

Proof. Karmazyn's Cartesian comparison [20, Proposition 4.21] supplies the algebraic base-change comparison before completion. We use it to place the two generators on one completed partial-resolution family over $\widetilde{R}$. The family is projective over an affine target, and both source and target are flat over the complete local Noetherian base. On its central fibre, both generators restrict to

$$\mathcal{T}_0 = \mathcal{O} \oplus \mathcal{T}_{0,p},$$

where the summands are indexed by the affine and retained vertices.

Each summand of $\mathcal{T}_0$ has vanishing higher self-Ext. It remains to compare the two lifts one summand at a time. On the truncations of $\widetilde{R}$ by powers of its maximal ideal, the obstruction to lifting a bundle lies in Ext^2, whereas the obstruction to extending an isomorphism lies in Ext^1. Both vanish. The compatible lifts therefore determine a lift over the completed family, which is projective over an affine base [21, Lemma 3.2]. Hence there are isomorphisms between the two lifts of $\mathcal{O}$ and between the two lifts of $\mathcal{T}_{0,p}$; we fix one of each. Their direct sum is θ and preserves both projectors. The choices are determined up to automorphisms of the two summands. Taking opposite endomorphism algebras and applying Proposition 2.21 gives (3.22), with both idempotents preserved by the summand-wise bundle comparison. □

For an algebra Λ with idempotent e_0, scalar extension commutes with the quotient $\Lambda/\Lambda e_0\Lambda$ by right exactness. The corner identity

$$\frac{e_{0p}\mathcal{A}_\Gamma e_{0p}}{(e_{0p}\mathcal{A}_\Gamma e_{0p})e_0(e_{0p}\mathcal{A}_\Gamma e_{0p})} \cong e_p(\mathcal{A}_\Gamma/\mathcal{A}_\Gamma e_0 \mathcal{A}_\Gamma)e_p \tag{3.23}$$

then identifies the marked quotient upstairs. By [20, Proposition 4.15], $\mathcal{A}_\Gamma$ is the universal extended-Dynkin deformed preprojective algebra. Scalar extension and the idempotent quotient therefore identify the completed marked algebra as follows. Killing the affine vertex of $\mathcal{A}_\Gamma$ leaves the finite-Dynkin algebra Π_λ: the affine parameter relation eliminates t_0, and the remaining parameters are $\lambda_i = t_i$. Thus

$$\left(\widehat{\mathcal{A}}_\ell/\widehat{\mathcal{A}}_\ell e_0 \widehat{\mathcal{A}}_\ell\right) \widehat{\otimes}_{R_H} \widetilde{R} \cong e_p \Pi_\lambda e_p = \mathcal{D}_p. \tag{3.24}$$

3.8.2 The cocenter on the marked base

Put

$$\mathcal{C}_H = \widehat{\mathcal{A}}_\ell / \widehat{\mathcal{A}}_\ell e_0 \widehat{\mathcal{A}}_\ell, \qquad \mathsf{M}_H = HH_0(\mathcal{C}_H / R_H).$$

The completion is at the graded origin. Since $\widetilde{R}$ is finite free over R_H, tensoring a complete module with $\widetilde{R}$ gives a finite direct sum of copies of that module and is already complete. Thus (3.24) gives

$$\mathcal{C}_H \otimes_{R_H} \widetilde{R} \cong \mathcal{D}_p. \tag{3.25}$$

Theorem 3.15 (The universal marked cocenter)**.** *The algebra $\mathcal{C}_H$ is finite over R_H. Its cocenter has a free resolution*

$$0 \longrightarrow R_H^q \xrightarrow{\Theta} R_H^q \longrightarrow \mathsf{M}_H \longrightarrow 0 \tag{3.26}$$

with

$$\mathrm{Fitt}_0^{R_H}(\mathsf{M}_H) = (\det \Theta) = (P_1 \cdots P_\ell),$$

where $P_j = 0$ defines the reduced coefficient-j curve divisor.

Proof. We descend the Cartan-cover presentation in two stages: first its square resolution, then its determinant. The algebra $\mathcal{D}_p$ is finite over $\widetilde{R}$, so finiteness descends along the faithfully flat Cartan cover to $\mathcal{C}_H$ and M_H over R_H. The commutator-cokernel description gives

$$\mathsf{M}_H \otimes_{R_H} \widetilde{R} \cong HH_0(\mathcal{D}_p / \widetilde{R}). \tag{3.27}$$

Take a finite free surjection onto M_H, with kernel K. After base change, Theorem 3.12 makes $K \otimes_{R_H} \widetilde{R}$ projective. Finite projectivity descends, and a finite projective module over the local ring R_H is free. The Cartan pullback has a square presentation with nonzero determinant, so M_H has generic rank zero. The two free ranks therefore agree, giving (3.26).

For the determinant, Fitting ideals commute with base change, so

$$\mathrm{Fitt}_0^{R_H}(\mathsf{M}_H)\widetilde{R} = (\Delta_p). \tag{3.28}$$

For $i \neq p$, the simple reflection s_i preserves m_p and sends every positive root other than α_i to a positive root. Thus W_0 permutes the factors of Δ_p without signs, and $\Delta_p \in \widetilde{R}^{W_0} = R_H$. Contracting the ideal in (3.28) along the faithfully flat extension gives

$$\mathrm{Fitt}_0^{R_H}(\mathsf{M}_H) = (\Delta_p). \tag{3.29}$$

Grouping the roots by marked coefficient and using Lemmas 2.5 and 2.6 gives

$$\Delta_p = \eta P_1 \cdots P_\ell, \qquad \eta \in R_H^\times. \tag{3.30}$$

□

The square presentation expresses the cocenter as a torsion module whose determinant records every reduced curve divisor once. Restriction to the classifying arc therefore converts its determinant into the required number of simple factors.

3.9 Restriction to the central arc

Let $R_1 = \mathbb{C}[[w]]$, and let $f_0 : \operatorname{Spec} R_1 \to H$ be the classifying arc of the original flop. Pulling the descended presentation back to this arc will compute the cocenter length of the original contraction algebra as the sum of the orders of intersection with the curve divisors. Set

$$\mathcal{C}_0 = \mathcal{C}_H \otimes_{R_H, f_0} R_1, \qquad \mathsf{M}_0 = HH_0(\mathcal{C}_0/R_1).$$

By Lemma 2.10 and Proposition 2.16, the arc meets the curve discriminant only at the origin, and $\operatorname{ord}_w P_j(f_0(w)) = n_j$.

3.9.1 The contraction algebra on the arc

Let $\widehat{\mathcal{R}}_\ell$ be the completion, at the graded origin, of the affine target ring $\mathcal{R}_\ell$ of the universal family, and put

$$Z_0 = \widehat{\mathcal{R}}_\ell \otimes_{R_H} R_1, \qquad \Lambda_0 = \widehat{\mathcal{A}}_\ell \otimes_{R_H} R_1.$$

Then $\mathcal{C}_0 = \Lambda_0/\Lambda_0 e_0 \Lambda_0$. Let $\mathfrak{m}$ denote the contraction maximal ideal of Z_0.

Lemma 3.16 (The central-arc quotient)**.** *The natural completion map $\mathcal{C}_0 \to \widehat{\mathcal{C}_{0\mathfrak{m}}}$ is an isomorphism, and*

$$\mathcal{C}_0 \cong \widehat{\mathcal{C}_{0\mathfrak{m}}} \cong A. \tag{3.31}$$

In particular,

$$HH_0(\mathcal{C}_0) \cong HH_0(A). \tag{3.32}$$

Proof. The classifying arc is nonconstant: the power series $g(w) = P_1(f_0(w))$ has positive finite order. Weierstrass division makes $\mathbb{C}[[w]]$ finite over $\mathbb{C}[[g(w)]]$. The latter subring lies in the image of R_H, so R_1 is finite over R_H.

Consequently Z_0 is finite over the complete Noetherian local ring $\widehat{\mathcal{R}}_\ell$. Modulo the maximal ideal of $\widehat{\mathcal{R}}_\ell$, the fibre is $R_1/\mathfrak{m}_H R_1$, where $\mathfrak{m}_H$ is the maximal ideal of R_H. This fibre is local Artinian with residue field $\mathbb{C}$. Thus Z_0 is local and complete at $\mathfrak{m}$. The finite Z_0-algebra Λ_0 and its quotient $\mathcal{C}_0$ are also complete.

The ordinary tensor products above therefore already compute the completed pullback along the central arc. Proposition 2.22 identifies Λ_0 with the original basic NCCR, preserving e_0. Taking the idempotent quotient gives $\mathcal{C}_0 \cong A$, and (3.32) follows on passing to cocenters. □

3.9.2 The length calculation

Base change gives $\mathsf{M}_0 = \mathsf{M}_H \otimes_{R_H} R_1$. The square presentation therefore becomes a finite-length presentation over the DVR R_1. Write $\Theta_0 = f_0^* \Theta$. Its determinant is

$$\det \Theta_0 = \eta(w) \prod_{j=1}^{\ell} P_j(f_0(w)) \neq 0. \tag{3.33}$$

Thus the base change of (3.26) is exact:

$$0 \longrightarrow R_1^q \xrightarrow{\Theta_0} R_1^q \longrightarrow \mathsf{M}_0 \longrightarrow 0. \tag{3.34}$$

Smith normal form over R_1 yields

$$\dim_{\mathbb{C}} \mathsf{M}_0 = \operatorname{length}_{R_1} \mathsf{M}_0 = \operatorname{ord}_w \det \Theta_0. \tag{3.35}$$

The relative and absolute cocenters have the same underlying $\mathbb{C}$-vector space, since $r[a,b] = [ra,b]$ for central r. Combining Lemma 3.16, (3.33), and Proposition 2.16 gives

$$\dim_{\mathbb{C}} HH_0(A) = \sum_j \operatorname{ord}_w P_j(f_0(w)) = \sum_j n_j. \tag{3.36}$$

Centre–cocenter duality then gives

$$\dim_{\mathbb{C}} Z(A) = \sum_j n_j. \tag{3.37}$$

3.10 Completion of the uniqueness proof

The cocenter count gives the remaining tangent-dimension equality. The orbit-closure relation from the existence theorem, together with this equality, allows the orbit criterion to identify the semisimple algebra obtained by splitting the generic fibre.

Proof of Theorem 3.1. Equations (3.9) and (3.36) give

$$\dim T_A X_d = \dim(G \cdot S_{\mathrm{GV}}). \tag{3.38}$$

The integral family from the general existence theorem of Section 2, combined with Lemma 3.3, supplies

$$A \in \overline{G \cdot S_{\mathrm{GV}}}. \tag{3.39}$$

Theorem 3.6 now shows that A is smooth, that $\overline{G \cdot S_{\mathrm{GV}}}$ is its unique component, and that any split semisimple algebra whose orbit closure contains A is isomorphic to S_{GV}.

Let $\mathscr{D}$ be any finite free formal deformation of A with geometrically semisimple generic fibre, and let S' be a product of complex matrix algebras such that $\mathscr{D} \otimes_{\mathbb{C}[[u]]} \overline{\mathbb{C}((u))} \cong S' \otimes_{\mathbb{C}} \overline{\mathbb{C}((u))}$. Thus S' records the matrix sizes and multiplicities of the split generic fibre. Lemma 3.3 gives $A \in \overline{G \cdot S'}$, and the orbit criterion yields

$$S' \cong S_{\mathrm{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \tag{3.40}$$

Tensoring this isomorphism with $\overline{\mathbb{C}((u))}$ gives the asserted description of the geometric generic fibre. □

Corollary 3.17. *Let $\mathscr{B}$ be a finite free $\mathbb{C}[[s]]$-algebra with special fibre A and geometrically semisimple generic fibre. Then $HH_0(\mathscr{B}/\mathbb{C}[[s]])$ is free of rank $\sum_j n_j$.*

Proof. The relative cocenter is finite and commutes with arbitrary base change. Theorem 3.1 gives dimension $\sum_j n_j$ on the closed fibre and, after a splitting extension, on the generic fibre. Lemma 2.34 proves freeness. □

The divisor presentation computes the single-curve cocenter, and orbit geometry turns its dimension into the generic-fibre statement. For several retained curves, Section 4 uses the exact cocenter sequence associated with deleting vertices. Lemma 4.2 controls pullback of the square presentations along classifying maps $R \to B[[w]]$, yielding the numerical statements used in Sections 5 and 6.

4 Universal presentations and flat families

Let a smooth threefold flop vary over a complete Noetherian local complex base. Using a lift of its fixed basic generator, we form the relative contraction quotient. We prove that this quotient, its cocenter, and its centre are finite free over the base, with ranks determined by the central flop. This is the content of the three flatness assertions in [16, Conjecture 4.3], in the completed smooth-threefold setting. Kawamata established algebra flatness over a disk for configurations of exceptional curves [22, Theorem 3.1(2)].

Under the hypotheses verified below, the simultaneous partial resolutions recalled in [20, Section 2.3] and the tilting-bundle theorem of [21, Theorem 3.4] identify the geometric family and its endomorphism algebra. The quotient by the free projector and its cocenter have square presentations derived from the Etingof–Rains central extension [9, Theorem 3.2]. The determinants of these two square presentations generate the zeroth Fitting ideals, with factors given by the curve-divisor equations.

Pullback gives matrices over $B[[w]]$, where B is the external base and w is the surface parameter. Lemma 4.2 shows that finiteness of the cokernel on the closed arc forces freeness over B, even when B has nilpotents. The rank of that cokernel is the order of the closed-arc determinant. We apply the lemma separately to the algebra and its cocenter, then use a symmetrizing trace to obtain the centre.

The one-curve cocenter presentation is available from Section 3. The pullback and trace arguments below are independent of the number of retained curves. For several curves, the corresponding presentations are constructed in Subsection 4.3; the Peirce and cocenter ranks of the multi-curve contraction quotient enter Section 5.

Remark 4.1 (Relative-algebra convention). A geometric deformation over $(B, \mathfrak{m}, \mathbb{C})$ means a projective contraction with affine Noetherian target, source and target flat over B, and the prescribed central contraction as closed fibre. We complete along the central contraction point and its inverse image. The relative algebra is the idempotent quotient of a lift of the fixed normalized basic Van den Bergh generator, with its structure-sheaf summand and all curve labels retained.

The finite relative algebra carries all support points that move away from the central point. Base change is performed on this algebra. For example, $B[[w]]/(w^2 - t)$, with $B = \mathbb{C}[[t]]$, is free of rank two over B, whereas $\mathbb{C}((t))[[w]]/(w^2 - t) = 0$. Thus the generic fibre of the finite relative algebra is obtained by localizing that algebra; completing again at $w = 0$ would remove its moving points. Fixing the generator is equally significant: its original summand multiplicities give the matrix blocks in the split fibres. A block M_j contributes j^2 to the algebra dimension, while its basic Morita representative $\mathbb{C}$ has dimension one.

4.1 Freeness and traces over local bases

Two algebraic facts turn the universal presentations into flat families. A square matrix over $B[[w]]$ with finite-dimensional closed-arc cokernel gives a finite free B-module. Once both an algebra and its cocenter are finite free, a symmetrizing trace on the closed algebra lifts and identifies the centre with the dual cocenter. We prove these facts before applying them to contraction quotients.

4.1.1 Freeness over a possibly nonreduced base

On the closed classifying arc, Smith normal form computes the cokernel length from the determinant. Freeness over B requires control through every power of its maximal ideal, including its nilpotent directions.

Lemma 4.2 (Freeness of the cokernel over the coefficient ring)**.** *Let $(B, \mathfrak{m}, k)$ be a complete Noetherian local ring, put $D_B = B[[w]]$, and let $\Phi : D_B^a \to D_B^a$ be a square matrix. If $\det \overline{\Phi} \neq 0$ in $k[[w]]$, then Φ is injective and its cokernel is finite free over B, of rank $\operatorname{ord}_w \det \overline{\Phi}$.*

Proof. Put $M = \operatorname{coker} \Phi$. Smith normal form over $k[[w]]$ makes its closed fibre finite-dimensional. Lift a basis to $m_1, \dots, m_b \in M$. Successive reduction modulo $\mathfrak{m}M$ writes every $m \in M$ as

$$m = \sum_i b_i^{(n)} m_i + r_n, \qquad r_n \in \mathfrak{m}^n M, \quad b_i^{(n+1)} - b_i^{(n)} \in \mathfrak{m}^n.$$

Completeness of B gives limits for the coefficients. Since M is finite over the Noetherian local ring D_B, Krull intersection gives $\bigcap_n \mathfrak{m}^n M = 0$. The limiting coefficients express m as a B-linear combination of the m_i. Thus M is finite over B.

If $\Phi v = 0$, injectivity of $\overline{\Phi}$ gives $v \in \mathfrak{m} D_B^a$. On each associated graded piece the map is

$$1 \otimes \overline{\Phi} : (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_k k[[w]]^a \longrightarrow (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_k k[[w]]^a,$$

which is injective. The identification of each graded piece uses the flatness of $B[[w]]$ over the Noetherian ring B. Induction places v in every power of $\mathfrak{m}$, and Krull intersection gives $v = 0$.

The exact sequence just obtained is a flat resolution over B, so

$$\operatorname{Tor}_1^B(M, k) = \ker \overline{\Phi} = 0.$$

Take a minimal free surjection $B^b \to M$, with finite kernel L. The Tor vanishing makes the reduction of L inject into the kernel of the closed-fibre isomorphism. Thus $L/\mathfrak{m}L = 0$, and Nakayama gives $L = 0$. The rank is the closed-fibre dimension; Smith normal form gives $b = \dim_k(M/\mathfrak{m}M) = \operatorname{ord}_w \det \overline{\Phi}$. □

The finite-slice argument applied to a free resolution on the marked base also identifies the derived pullback with the ordinary one. In the following corollary, $(B, \mathfrak{m}, k)$ remains a complete Noetherian local ring as in Lemma 4.2.

Corollary 4.3 (Derived pullback along a classifying map)**.** *Suppose a finite module C over a Noetherian local ring R has an exact sequence $0 \to R^a \xrightarrow{\Theta} R^a \to C \to 0$. For a local map $F : R \to B[[w]]$, if $C \otimes_R k[[w]]$ is finite-dimensional, then*

$$C \otimes_R^L B[[w]] \simeq (C \otimes_R B[[w]])[0].$$

The ordinary pullback is finite free over B, with rank equal to the dimension of that closed fibre.

Proof. Compute the derived pullback with the displayed free resolution. Its closed-fibre matrix $\overline{\Theta(F)}$ has finite-dimensional cokernel, hence is invertible over $k((w))$. Lemma 4.2 makes $\Theta(F)$ injective and its cokernel finite free over B. The pulled-back resolution therefore has cohomology only in degree zero, as asserted. □

4.1.2 Symmetrizing traces and base change

The trace pairing identifies the centre with the dual cocenter of a symmetric algebra; see [10, Corollary 2.2] for the central extension. The relative assertion requires lifting a given closed-fibre trace and preserving its perfect pairing under scalar extension. Here the base need only be local.

Lemma 4.4 (Lifting a symmetrizing trace)**.** *Let $(B, \mathfrak{m}, k)$ be a commutative local ring, and let $\mathscr{A}$ be a finite free B-algebra. Suppose $M = HH_0(\mathscr{A}/B)$ is finite free and the closed fibre $A_0 = \mathscr{A} \otimes_B k$ of $\mathscr{A}$ is symmetric. Every symmetrizing trace of A_0 lifts to a perfect B-linear symmetrizing trace τ on $\mathscr{A}$. It induces an isomorphism*

$$Z(\mathscr{A}) \xrightarrow{\sim} M^\vee, \qquad z \longmapsto (\bar{a} \longmapsto \tau(za)).$$

The centre and cocenter commute with every commutative scalar extension $B \to B'$. The lifted trace determines the displayed duality, while the base-change maps are canonical.

Proof. As a commutator cokernel, M commutes with arbitrary scalar extension. The given closed trace is a functional on $M \otimes_B k$. Freeness of M lifts it to a functional $M \to B$, hence to a trace $\tau : \mathscr{A} \to B$. The determinant of the pairing $(a, b) \mapsto \tau(ab)$ has nonzero reduction, so it is a unit in the local ring B. The lifted trace is therefore symmetrizing.

To identify the centre, take the annihilator of the commutator submodule under this perfect pairing. We have

$$\tau(z[a, b]) = \tau((za - az)b).$$

This identity shows that a central element annihilates every commutator under the pairing. Conversely, if $\tau(z[a, b]) = 0$ for all a, b, perfection forces $za - az = 0$ for every a. The annihilator is thus the centre, giving the stated duality and centre freeness.

After any scalar extension to B', the pairing remains perfect. Base change for the cocenter and for the dual of a finite free module identifies the trace formula over B' with the natural centre comparison. The natural centre comparison is therefore an isomorphism for every such extension, including a nonflat one. □

4.2 One curve: presentations and pullback along classifying maps

At a transverse point of D_j, the matrix algebra M_j contributes j^2 to the algebra dimension and 1 to the cocenter dimension. These are the exponents in the two determinant formulas.

4.2.1 A square presentation of the universal quotient

Recall the notation of Subsection 3.8. The ring R_H is the completed marked base, $\widetilde{R}$ is its finite faithfully flat Cartan cover, and $\mathcal{C}_H$ is the universal contraction quotient. The equation $P_j = 0$ defines the reduced coefficient-j curve divisor. The algebra presentation is derived below from the same central extension that gives the cocenter presentation.

Proposition 4.5 (A square presentation of the universal algebra)**.** *The finite R_H-module $\mathcal{C}_H$ has a free resolution of the form*

$$0 \longrightarrow R_H^a \xrightarrow{\Theta_A} R_H^a \longrightarrow \mathcal{C}_H \longrightarrow 0 \tag{4.1}$$

with

$$\mathrm{Fitt}_0^{R_H}(\mathcal{C}_H) = (\det \Theta_A) = \left(\prod_{j=1}^{\ell} P_j^{j^2} \right). \tag{4.2}$$

In particular, $\mathcal{C}_H$ is a Cohen–Macaulay module of codimension one over R_H.

Proof. Let E_p be the central extension of Subsection 3.5. Theorem 3.2(i) of [9] makes $\Pi[z]$ finite free over $K_\rho = \mathbb{C}[x_i - \rho_i z]$. After flat completion, its retained corner is a direct summand of a finite free $\widetilde{R}$-module. It is therefore free, since $\widetilde{R}$ is local. Before the change of coefficient ring, z is a polynomial variable, so multiplication by z is injective. Flat completion and the corner projection preserve this injection. Its cokernel is obtained by setting $z = 0$:

$$0 \longrightarrow E_p \xrightarrow{z} E_p \longrightarrow \mathcal{D}_p \longrightarrow 0. \tag{4.3}$$

To compute the determinant, use the generic decomposition (3.13). The factor of $(E_p)_L$ indexed by a positive root with $m_p(\alpha) > 0$ is $M_{m_p(\alpha)}(L)$, on which z acts as $-(\lambda, \alpha)/(\rho, \alpha)$ [9, Theorem 3.2(v) and the following display]. This scalar occurs with multiplicity $m_p(\alpha)^2$, the dimension of the matrix algebra. Hence, in the domain $\widetilde{R}$,

$$\det(z \mid E_p) = \eta \prod_{\substack{\alpha \in \Phi^+ \\ m_p(\alpha) > 0}} (\lambda, \alpha)^{m_p(\alpha)^2}, \qquad \eta \in \mathbb{C}^\times.$$

The determinant is nonzero, so $\mathcal{D}_p$ is torsion and has projective dimension one over $\widetilde{R}$.

By (3.25), $\mathcal{C}_H \otimes_{R_H} \widetilde{R} \cong \mathcal{D}_p$. Choose a finite free surjection $R_H^a \twoheadrightarrow \mathcal{C}_H$ with kernel K. The resolution (4.3) makes $K \otimes_{R_H} \widetilde{R}$ projective. Finite projectivity descends along the faithfully flat map $R_H \to \widetilde{R}$, and projective modules over the local ring R_H are free. Thus K is free. Since $\mathcal{C}_H$ is torsion, $\operatorname{rk} K = a$, proving (4.1).

Finally, Fitting ideals commute with scalar extension. The root-orbit and reduced-divisor identifications in Lemmas 2.5 and 2.6 identify the displayed determinant ideal with the extension of $(\prod_j P_j^{j^2})$. Faithful flatness then gives (4.2) downstairs. Since R_H is regular local, Auslander–Buchsbaum shows that the nonzero torsion module $\mathcal{C}_H$, of projective dimension one, has depth and dimension $\dim R_H - 1$. □

The two universal presentations are supported on the same root divisors. Up to units, their determinants are $\prod_j P_j^{j^2}$ for the algebra and $\prod_j P_j$ for the cocenter, the latter by Theorem 3.15. Their exponents measure the dimension of a matrix block and the number of blocks, respectively.

4.2.2 Classifying an arbitrary geometric deformation

Fix a completed simple flop $f_0 : X_0 \to Y_0$ of a smooth complex threefold. Let $\mathcal{T}_0 = \mathcal{O} \oplus \mathcal{T}_0'$ be its selected basic tilting generator, as in Section 2, and A_0 its contraction algebra.

The geometric families and their relative quotients are those of Remark 4.1.

Proposition 4.6 (Families of classifying arcs)**.** *Let B be complete Noetherian local with residue field $\mathbb{C}$, and let f_B be such a completed deformation. A lift of a general surface-section parameter supplies a continuous local map*

$$F : R_H \longrightarrow D_B = B[[w]]$$

and identifies f_B with the completed pullback of the universal marked partial resolution. There is a relative tilting generator $\mathcal{T}_B$ lifting $\mathcal{T}_0$. For any such lift with the fixed summands, its relative contraction quotient satisfies

$$\mathscr{A}_B := \frac{\operatorname{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}}}{\operatorname{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}} e_0 \operatorname{End}_{X_B}(\mathcal{T}_B)^{\mathrm{op}}} \cong \mathcal{C}_H \otimes_{R_H,F} D_B. \tag{4.4}$$

Its reduction modulo $\mathfrak{m}$ is A_0.

Proof. Choose a general section parameter w_0 on the central target and lift it to w on the completed target. Topological nilpotence of w defines the action of $B[[w]]$. The central target is a cDV domain and the central source is smooth and integral, so w_0 is a non-zero-divisor on both. Their local rings are torsion-free, hence flat, over $\mathbb{C}[[w]]$. Apply the fibrewise criterion for flatness to $B \to D_B \to Q$, where Q is a local ring of the completed source or target. The module Q is flat over B, and $Q/\mathfrak{m}Q$ is flat over $D_B/\mathfrak{m}D_B$. Thus Q is flat over D_B [31, Tag 00MP].

The closed surface, over $(\mathfrak{m}, w)$, is the marked partial resolution of the ADE section. The formal versality statement of [20, Section 2.3] applies to arbitrary formal deformations of the closed partial-resolution surface, including those over nonreduced bases. The base of that formally semiuniversal deformation is $H = \mathfrak{h}/W_0$. Compatible applications to successive Artin quotients of D_B give $F : R_H \to D_B$ and identify the formal sources. To identify the targets, start with $Rf_{0*}\mathcal{O}_{X_0} = \mathcal{O}_{Y_0}$. On a square-zero thickening, flatness identifies the kernel sheaves with the central structure sheaves tensored by the square-zero ideal. Proper pushforward and the central higher-direct-image vanishing give the same identity on the thickening. Induction and formal functions yield $Rf_{B*}\mathcal{O}_{X_B} = \mathcal{O}_{Y_B}$. Relative affinization therefore identifies the affine targets and the contraction morphisms.

The universal generator pulls back by Lemmas 2.19 and 2.20. Its central restriction is $\mathcal{T}_0$, with its specified summands, by Proposition 2.22 and the same rank-and-determinant argument on the central arc. Vanishing of the first and second self-Ext groups of $\mathcal{T}_0$ lifts the summands and the isomorphisms between their lifts through each square-zero thickening. As in Proposition 3.14, the compatible isomorphisms pass to the formal limit and preserve e_0. Thus any other lift of the fixed generator gives an isomorphic marked quotient. Ordinary endomorphism base change (Proposition 2.21) and right exactness of the idempotent quotient now give (4.4).

The reduction F_0 modulo $\mathfrak{m}$ classifies the original flop, and Lemma 3.16 identifies the corresponding central-arc quotient with A_0. This closed fibre is finite-dimensional, so Proposition 4.5 and Corollary 4.3 make the right side of (4.4) finite over B. On this finite module, w acts nilpotently modulo $\mathfrak{m}$. Consequently the $(\mathfrak{m}, w)$-adic and $\mathfrak{m}$-adic topologies agree. The ordinary and completed quotient expressions therefore coincide, with all moving support points retained. □

4.2.3 The one-curve application

Proposition 4.7 (The one-curve application)**.** *Let f_0 and f_B be as in Proposition 4.6. Put*

$$d = \dim_{\mathbb{C}} A_0 = \sum_j j^2 n_j, \qquad N = \sum_j n_j.$$

Then $\mathscr{A}_B$ is finite free over B of rank d, and $HH_0(\mathscr{A}_B/B)$ and $Z(\mathscr{A}_B)$ are finite free of rank N. For every commutative B-algebra B', the canonical maps

$$HH_0(\mathscr{A}_B/B) \otimes_B B' \xrightarrow{\sim} HH_0(\mathscr{A}_B \otimes_B B'/B'),$$
$$Z(\mathscr{A}_B) \otimes_B B' \xrightarrow{\sim} Z(\mathscr{A}_B \otimes_B B')$$

are isomorphisms. Moreover

$$\mathcal{C}_H \otimes^L_{R_H,F} B[[w]] \simeq \mathscr{A}_B[0].$$

The conclusion holds for every complete Noetherian local base B and every geometric family specified in the hypotheses.

Proof. For the algebra, pull back (4.1) along F. The closed-fibre cokernel is A_0, so Corollary 4.3 gives algebra freeness, rank d, and the derived statement. In terms of the divisor equations,

$$d = \operatorname{ord}_w \det \Theta_A(F_0) = \sum_j j^2 \operatorname{ord}_w P_j(F_0) = \sum_j j^2 n_j,$$

where Proposition 2.16 identifies the divisor orders with the GV multiplicities.

For the cocenter, use (3.26) together with the commutator-cokernel description of the relative cocenter. Together, these give

$$HH_0(\mathscr{A}_B/B) \cong \operatorname{coker}\big(\Theta(F) : D_B^q \to D_B^q\big).$$

Here the relative cocenter can be computed over either B or D_B: the B-span of commutators is already a D_B-module, since $r[a,b] = [ra,b]$ for a central scalar r. Lemma 4.2 and $\operatorname{ord}_w \prod_j P_j(F_0) = N$ make the cocenter free. To obtain the centre, recall that A_0 is symmetric by Proposition 3.8. Lemma 4.4 then supplies the centre, the lifted trace, and arbitrary centre base change. Cocenter base change already follows from its cokernel presentation. □

The scalar presentations of the individual root divisors give a further consequence: each intersection multiplicity is conserved over the whole base.

Corollary 4.8 (Constant degree of intersections with each root divisor)**.** *For each j, the scheme*

$$\operatorname{Spec}\big(B[[w]]/(P_j(F))\big) \longrightarrow \operatorname{Spec} B$$

is finite flat of degree n_j.

Proof. Apply Lemma 4.2 to the one-by-one matrix $P_j(F)$, whose closed specialization has order n_j. □

For several curves, the remaining task is to construct the universal presentations with every retained projector kept in the comparison.

4.3 Idempotent subalgebras associated to several curves

Let $f_0 : X_0 \to Y_0$ be the completion at an exceptional image point of a projective flopping contraction of a smooth complex threefold, with any finite exceptional configuration. We use the relative-algebra convention of Remark 4.1. For one curve, a root contributes one matrix size. For several curves, it contributes a vector of ranks for the retained projectors. The universal algebra and its Peirce entries record these dimensions; the cocenter records the number of simple factors.

4.3.1 Restricted roots and the marked universal quotient

Let Δ be the ADE diagram of a general surface section, and let $J \subseteq \Delta$ be the nonempty set of retained vertices. We use the Katz–Morrison discriminant description in the restricted-root form of Nabijou–Wemyss [28, Theorem 2.6]. Here a retained vertex labels a curve that remains on the partial resolution. The complementary set I labels the curves contracted from the minimal resolution. Put

$$I = \Delta \setminus J, \qquad W_I = \langle s_i : i \in I \rangle, \qquad S = \widehat{\mathcal{O}}_{\mathfrak{h},0}, \qquad R_J = S^{W_I} = \widehat{\mathcal{O}}_{\mathfrak{h}/W_I,0}.$$

Write $S = \mathbb{C}[[\lambda_i : i \in \Delta]]$, and let $q : \mathfrak{h} \to \mathfrak{h}/W_I$ denote the quotient map. In the notation of [28], the contracted subset is I and the retained subset is J. The inclusion $R_J \to S$ is finite faithfully flat: both rings are regular local of the same dimension, and a system of parameters downstairs is a regular sequence upstairs, as in Subsection 3.8. For $\alpha = \sum_i m_i(\alpha)\alpha_i \in \Phi^+(\Delta)$, set

$$\pi_I(\alpha) = (m_j(\alpha))_{j \in J}, \qquad \mathcal{B}_J = \pi_I(\Phi^+(\Delta)) \setminus \{0\}, \qquad s_\beta = \sum_{j \in J} \beta_j.$$

Thus $\pi_I(\alpha)$ retains exactly the coefficients relevant to the partial resolution, while s_β will be the size of the corresponding generic matrix block. Proportional but distinct restricted roots remain distinct elements of $\mathcal{B}_J$.

Lemma 4.9 (Equations of the restricted-root divisors)**.** *For $\beta \in \mathcal{B}_J$, the product*

$$p_\beta = \prod_{\pi_I(\alpha) = \beta} \ell_\alpha, \qquad \ell_\alpha = \sum_i m_i(\alpha)\lambda_i,$$

is invariant under W_I. It descends to a reduced irreducible element $P_\beta \in R_J$, whose zero locus is the coefficient-β curve divisor of the marked partial-resolution family. Distinct β's give distinct divisors, even when they are proportional.

Proof. The fibres of π_I on non-Levi positive roots are single W_I-orbits [28, Lemma 2.3]. For $i \in I$, s_i preserves the retained coefficients and sends every positive root other than α_i to a positive root. It therefore permutes the factors of p_β without signs.

Those factors are pairwise nonassociated linear primes in the factorial ring S. A proper factorization in R_J would select a proper nonempty invariant subset of them, contrary to transitivity. The same argument in the formal power-series ring proves irreducibility after completion and reducedness. The geometric interpretation of the descended divisor, including the distinction between proportional restricted roots, is [28, Theorem 2.6 and Remark 2.7]. Finally, at a general point of a non-Levi root hyperplane the W_I-stabilizer is trivial. The quotient is therefore generically étale there, so the descended equation has multiplicity one. □

To identify the algebra supported on these divisors, let $\mathcal{X}_J \to \mathcal{Y}_J$ be the completed universal family over R_J. Write the normalized generator and contraction quotient of this completed universal family as

$$T_J = \mathcal{O} \oplus \bigoplus_{j \in J} T_j, \qquad \Lambda_J = \operatorname{End}(T_J)^{\mathrm{op}}, \qquad C_J = \Lambda_J / \Lambda_J e_0 \Lambda_J.$$

All projectors remain fixed throughout the comparison. Put $e_J = \sum_{j \in J} e_j$ and $e_{0J} = e_0 + e_J$.

On the closed surface, Iyama–Kalck–Wemyss–Yang identify the tilting bundle on a partial resolution with the retained summands on the full resolution [17, Theorem 4.6]. Karmazyn's uniqueness theorem for lifts of the bundles carries this comparison through the formal family.

Proposition 4.10 (The marked Cartan comparison)**.** *There is an isomorphism preserving every labelled projector*

$$\Lambda_J \widehat{\otimes}_{R_J} S \cong e_{0J} \widehat{\Pi}_{\widetilde{\Delta},\lambda} e_{0J}.$$

Killing the free projector gives

$$C_J \otimes_{R_J} S \cong D_J := e_J \Pi_{\Delta,\lambda} e_J. \tag{4.5}$$

The universal generator survives arbitrary Noetherian classifying pullback, and its ordinary endomorphism algebra commutes with that base change, including central localization and completion.

Proof. On the central ADE surface, let $q_0 : Z_0 \to X_{J,0}$ be the map from the minimal resolution to the partial resolution. The tilting-bundle construction in [17, Theorem 4.6 and its proof] gives a bundle isomorphism

$$q_0^* T_{J,0} \cong T_{\mathrm{full},0,0} \oplus \bigoplus_{j \in J} T_{\mathrm{full},0,j}.$$

In the positive-dual convention, the displayed bundle isomorphism is obtained by pulling back the line bundles of determinant degree one and the maximal extensions of those line bundles. The vanishing of $\operatorname{Ext}^1(-, \mathcal{O})$ makes the pulled-back extensions maximal. Dualizing gives the convention used in the statement.

After Cartan base change, the full simultaneous resolution maps to the partial resolution; write this morphism as q. Its surface fibres have rational singularities, so $Rq_*\mathcal{O} = \mathcal{O}$, also after completion. The same central-fibre vanishing and flatness prove the assertion on Artin thickenings. Formal functions then give the assertion on the formal family. The pullback of T_J and the retained summands of the full generator are therefore bundles on the same S-flat formal family with the same central restrictions. Uniqueness of the lifts of each labelled summand gives compatible isomorphisms on all thickenings and hence on the formal limit [21, Lemma 3.2]. They preserve the projectors and are determined up to automorphisms of the labelled summands.

Projection formula together with $Rq_*\mathcal{O} = \mathcal{O}$ identifies endomorphisms before and after pullback. The full universal McKay algebra is the extended-Dynkin deformed preprojective algebra [20, Proposition 4.15]; retaining the specified summands gives the displayed corner.

Ordinary base change uses the flatness of the universal source and target over the marked base [20, Section 2.3], together with flatness of the lifted endomorphism algebra [21, Theorem 3.4]. For local rings $R_J \to A \to Q$ of target and source, and a base B, the calculation

$$Q \otimes^L_A (A \otimes_{R_J} B) \simeq Q \otimes^L_{R_J} B \simeq Q \otimes_{R_J} B$$

gives geometric Tor-independence. The universal projective morphism is quasi-compact and quasi-separated. Apply [31, Tag 08IB] to $\mathcal{E}nd(T_J)$. Tilting and base-flatness concentrate both sides in degree zero. The evaluation surjection, higher-direct-image vanishing, ample determinant, and structure-sheaf summand used in Lemma 2.20 also survive for this multi-summand generator. Thus the pulled-back bundle remains a perverse projective generator.

Finally, scalar extension commutes with the idempotent quotient. The corner identity

$$\frac{e_{0J}\Lambda e_{0J}}{e_{0J}\Lambda e_0\Lambda e_{0J}} \cong e_J(\Lambda/\Lambda e_0\Lambda)e_J$$

shows that removing the affine vertex leaves the finite-Dynkin corner in (4.5). Since S is finite free over R_J, tensoring a complete module with S already gives a complete module. The completed tensor in this quotient is therefore the ordinary tensor product. □

4.3.2 The presentation of the algebra and its Fitting ideal

Adjoin a polynomial central variable z to the universal finite-Dynkin preprojective algebra Π_x. This makes the divisor multiplicities visible on the retained corner. Set $\rho_i = 1$, $\lambda_i = x_i - \rho_i z$, and

$$\mathscr{E} = \Pi_x[z] \otimes_{\mathbb{C}[\lambda_i]} S, \qquad \mathscr{E}_J = e_J \mathscr{E} e_J.$$

Etingof–Rains prove freeness over the shifted parameter ring and semisimplicity over its generic point [9, Theorem 3.2(i),(v)]. The simple factors of the generic central extension are indexed by positive roots, and the scalar action of z locates the corresponding root hyperplanes. The retained corner records the part of each factor seen by the projectors in J.

Lemma 4.11 (The central extension and its generic simple factors)**.** *The module $\mathscr{E}_J$ is finite free over S, and*

$$0 \longrightarrow \mathscr{E}_J \xrightarrow{z} \mathscr{E}_J \longrightarrow D_J \longrightarrow 0 \tag{4.6}$$

is exact. If $K = \operatorname{Frac} S$, then

$$(\mathscr{E}_J)_{\overline{K}} \cong \prod_{\substack{\alpha\in\Phi^+ \\ \pi_I(\alpha)\neq 0}} M_{s_{\pi_I(\alpha)}}(\overline{K}). \tag{4.7}$$

On the factor indexed by α, the central element z acts as $-\ell_\alpha/\langle\rho,\alpha\rangle$.

Proof. The whole extension is finite free over the shifted parameter ring [9, Theorem 3.2(i)]. The corner projection $a \mapsto e_J a e_J$ is S-linear and idempotent, so its image is projective and hence free over the local ring S. Multiplication by the polynomial variable z is injective before completion and remains injective after flat completion and on the corner. Its cokernel is obtained by setting $z = 0$, which proves (4.6).

The generic simple factors of the whole extension are $\operatorname{End}(V_\alpha)$, with $\dim e_i V_\alpha = m_i(\alpha)$ [9, Theorem 3.2(v) and its proof]. Orthogonality of the projectors gives

$$e_J V_\alpha = \bigoplus_{j\in J} e_j V_\alpha, \qquad \dim e_J V_\alpha = s_{\pi_I(\alpha)}.$$

Taking the corner of $\operatorname{End}(V_\alpha)$ gives $\operatorname{End}(e_J V_\alpha)$. Taking traces in the preprojective relation gives $0 = \ell_\alpha + z\langle\rho,\alpha\rangle$, determining the scalar action of z. □

Theorem 4.12 (Presentations and Fitting ideals for several retained vertices)**.** *There is a free resolution of the form*

$$0 \longrightarrow R_J^a \xrightarrow{\Theta_J} R_J^a \longrightarrow C_J \longrightarrow 0 \tag{4.8}$$

with

$$\operatorname{Fitt}_0^{R_J}(C_J) = \left(\prod_{\beta \in \mathcal{B}_J} P_\beta^{s_\beta^2} \right). \tag{4.9}$$

For $i, j \in J$, the module $e_i C_J e_j$ also has a free resolution with two free modules of equal rank, and

$$\operatorname{Fitt}_0^{R_J}(e_i C_J e_j) = \left(\prod_{\beta \in \mathcal{B}_J} P_\beta^{\beta_i \beta_j} \right).$$

For the zero module, the zeroth Fitting ideal is the unit ideal. The nonzero module C_J is Cohen–Macaulay of codimension one.

Proof. Over the algebraic closure of the fraction field, the factor belonging to α in (4.7) has algebra dimension $s^2_{\pi_I(\alpha)}$. Therefore

$$\det(z \mid \mathscr{E}_J) = u \prod_{\pi_I(\alpha) \neq 0} \ell_\alpha^{s^2_{\pi_I(\alpha)}} = u \prod_\beta (q^* P_\beta)^{s_\beta^2}, \qquad u \in \mathbb{C}^\times.$$

The determinant is nonzero, so D_J is a finite torsion S-module with a free resolution consisting of two free modules of equal rank. Apply the faithfully flat syzygy descent from the proof of Proposition 4.5 to $R_J \to S$ and (4.5). After base change to S, the first syzygy is projective. Faithfully flat descent makes it projective over R_J, hence free because R_J is local; torsion forces the two ranks to agree. This proves (4.8). Base change and faithful contraction of Fitting ideals give (4.9).

For the Peirce entries, restrict multiplication by z to the free direct summand $e_i \mathscr{E} e_j$. The cokernel of this restricted map is $e_i D_J e_j$, whose square presentation descends by the same syzygy construction. The α-summand of the generic fibre of $e_i \mathscr{E} e_j$ is $\operatorname{Hom}(e_j V_\alpha, e_i V_\alpha)$, of dimension $m_i(\alpha) m_j(\alpha) = \beta_i \beta_j$, giving the entrywise formula. Finally, regularity of R_J and Auslander–Buchsbaum give $\operatorname{depth}_{R_J} C_J = \dim R_J - 1 = \dim C_J$. □

Proposition 4.13 (The reduced annihilator)**.** *Put $\delta_J = \prod_{\beta \in \mathcal{B}_J} P_\beta$. Then*

$$\operatorname{Ann}_{R_J}(C_J) = (\delta_J).$$

There is a matrix Ξ_J such that $\Theta_J \Xi_J = \Xi_J \Theta_J = \delta_J I_a$.

Proof. Consider the polynomial

$$Q(Z) = \prod_{\pi_I(\alpha) \neq 0} (\langle \rho, \alpha \rangle Z + \ell_\alpha)$$

which annihilates $\mathscr{E}_J$ over the fraction field, by Lemma 4.11. Freeness over the domain S therefore implies $Q(z)\mathscr{E}_J = 0$. Setting $z = 0$ gives $(q^*\delta_J) D_J = 0$. The Fitting ideal shows that the support is exactly the union of the indicated hyperplanes, whose reduced equation is $q^*\delta_J$. Hence

$$(q^*\delta_J) \subseteq \operatorname{Ann}_S(D_J) \subseteq \sqrt{\operatorname{Ann}_S(D_J)} = (q^*\delta_J).$$

Faithful flatness gives the annihilator equality downstairs. Since δ_J annihilates the cokernel of Θ_J, lift $\delta_J I_a$ through Θ_J to obtain Ξ_J. Injectivity of Θ_J gives the second matrix identity. □

4.3.3 An exact sequence for cocenters after deleting vertices

Each matrix factor contributes one dimension to the cocenter. To obtain the cocenter presentation over the Cartan base, we compute the closed-fibre dimension of the retained corner. Write $P_\Delta = \Pi_0^\rho$ for the ordinary central extension specialized at $\lambda = 0$, with $\rho_i = 1$; the element z remains part of this algebra. Put

$$Q_I = P_\Delta / P_\Delta e_J P_\Delta.$$

The quotient deletes the retained vertices, leaving the components of I. Etingof–Latour–Rains compute the full-algebra trace count for each of these diagrams [10, Theorem 3.1]. The following exact sequence identifies the cocenter of the retained corner with the kernel of the quotient map.

Theorem 4.14 (An exact sequence for cocenters of ADE central extensions)**.** *For every nonempty $J \subseteq \Delta$, there is an exact sequence*

$$0 \longrightarrow HH_0(e_J P_\Delta e_J) \longrightarrow HH_0(P_\Delta) \longrightarrow HH_0(Q_I) \longrightarrow 0. \tag{4.10}$$

In particular, if $N_J = |\Phi^+(\Delta)| - |\Phi^+(I)|$, then

$$\dim_{\mathbb{C}} HH_0(e_J P_\Delta e_J) = N_J. \tag{4.11}$$

With arrows of internal degree one and z of degree two, the odd pieces of $HH_0(e_J P_\Delta e_J)$ vanish and

$$\dim HH_0(e_J P_\Delta e_J)_{2p} = \#\{\alpha \in \Phi^+(\Delta) : \pi_I(\alpha) \neq 0, \mathrm{ht}(\alpha) = p+1\}.$$

The main point is injectivity of the corner map. In the exact sequence of Lemma 4.16, the kernel of that corner map is the image of the boundary from $HH_1(Q_I)$. This boundary vanishes once $HH_1(P_\Delta) \to HH_1(Q_I)$ is surjective. The central element z provides the cycles used to prove that surjectivity.

Keller's localization theorem gives a sequence for dg Hochschild homology. The following comparison relates the low-degree dg Hochschild groups in that sequence to those of the ordinary Jacobi algebras. We use polynomial Ginzburg dg algebras before completion over the Cartan base, with cohomological grading for dg algebras and homological grading for Hochschild homology.

Lemma 4.15 (A low-degree comparison for self-injective Jacobi algebras)**.** *Let Γ be a nonpositive, homologically smooth, bimodule 3-Calabi–Yau dg algebra over $\mathbb{C}$, with $A = H^0(\Gamma)$ finite-dimensional, basic, and self-injective. Then $H^{-1}(\Gamma) = 0$, and the canonical projection gives natural isomorphisms*

$$HH_i(\Gamma) \cong HH_i(A), \qquad i = 0, 1.$$

For any nonpositive dg algebra, the degree-zero comparison holds and its negative Hochschild homology vanishes.

Proof. Amiot's theorem makes $\mathcal{C} = \mathrm{per}\,\Gamma / \mathcal{D}_{\mathrm{fd}}\Gamma$ a Hom-finite 2-Calabi–Yau category. The image T of Γ is cluster tilting, with endomorphism algebra A, [1, Theorem 2.1]. The projection to $\mathcal{C}$ is fully faithful on the fundamental domain

$$\mathcal{F} = \mathrm{per}\,\Gamma \cap \mathcal{D}^{\leq 0} \cap {}^{\perp}\mathcal{D}^{\leq -2} \qquad [1, \text{Proposition 2.9}].$$

Both Γ and $\Gamma[1]$ lie in this domain. Indeed, they are perfect and have cohomology in nonpositive degrees. For $M \in \mathcal{D}^{\leq -2}$, the two groups $\mathrm{Hom}_{\mathcal{D}(\Gamma)}(\Gamma, M) = H^0(M)$ and $\mathrm{Hom}_{\mathcal{D}(\Gamma)}(\Gamma[1], M) = H^{-1}(M)$ vanish, giving the required left orthogonality. Since A is basic and self-injective, the criterion of Iyama–Oppermann [35, Proposition 3.6], in the formulation [15, Proposition 4.4], gives $T \cong T[2]$. Hence

$$H^{-1}(\Gamma) = \mathrm{Hom}_{\mathrm{per}\,\Gamma}(\Gamma[1], \Gamma) \cong \mathrm{Hom}_{\mathcal{C}}(T[1], T) \cong \mathrm{Hom}_{\mathcal{C}}(T, T[1]) = 0.$$

The last equality holds because T is cluster tilting.

For the Hochschild comparison, use the reversed internal grading $H_i\Gamma = H^{-i}(\Gamma)$ and filter the Hochschild bicomplex by chain length. Over $\mathbb{C}$, this produces the first-quadrant spectral sequence

$$E^2_{p,q} = HH_p(H_*\Gamma)_q \Longrightarrow HH_{p+q}(\Gamma).$$

Under this reversed convention, the vanishing just proved is $H_1\Gamma = 0$, so the row of internal degree one is zero. In total degrees zero and one, the only possible terms are $HH_0(H_0\Gamma)$ and $HH_1(H_0\Gamma)$, at $(0,0)$ and $(1,0)$, and no higher differential enters or leaves them. The edge maps are induced by the canonical projection, which proves naturality and both comparisons. The degree-zero statement does not use the H_1-vanishing. Finally, there are no chains of negative total degree, so negative Hochschild homology vanishes. □

Lemma 4.16 (The vertex-deletion trace sequence)**.** *Let $P = \mathrm{Jac}(Q, W)$ be the ordinary Jacobi algebra of a finite quiver with polynomial potential. Delete a set of vertices, with sum of projectors e, and denote the resulting Jacobi algebra by $Q' = P/PeP$. Suppose P and Q' are finite-dimensional, basic, and self-injective. There is a natural exact sequence*

$$HH_1(P) \longrightarrow HH_1(Q') \longrightarrow HH_0(ePe) \longrightarrow HH_0(P) \longrightarrow HH_0(Q') \longrightarrow 0. \tag{4.12}$$

The map from the corner sends $[a]$ to $[a]$.

Proof. Let Γ and Γ' denote the polynomial Ginzburg dg algebras before and after the deletion. Both are nonpositive, homologically smooth, and bimodule 3-Calabi–Yau [23, Theorem 6.3]. The potential supplies the negative cyclic lift of the deformation class required in [24]. Vertex deletion is a dg localization [23, Corollary 7.3]. Its kernel is generated by the deleted representable projectives by [23, Theorem 5.8(b)]. Deleting several vertices one at a time shows that, on compact objects and up to idempotent completion, the kernel is $\mathrm{thick}(e\Gamma)$, which is dg-Morita equivalent to $e\Gamma e$.

Morita invariance and localization of Hochschild mixed complexes [25, Theorem 5.2] give a triangle

$$C_*(e\Gamma e) \longrightarrow C_*(\Gamma) \longrightarrow C_*(\Gamma') \longrightarrow C_*(e\Gamma e)[1].$$

Lemma 4.15 identifies the two degree-one terms with $HH_1(P)$ and $HH_1(Q')$, and the degree-zero terms with the respective ordinary cocenters. For the corner, $H^0(e\Gamma e) = ePe$ and $HH_{-1}(e\Gamma e) = 0$. Thus the corresponding segment of the long exact sequence is (4.12). Under Morita invariance, the trace of an endomorphism of the projective $e\Gamma$ is its cyclic class in Γ. The corner map is therefore the stated inclusion on cyclic classes, and this construction remains meaningful for the generally nonunital corner inclusion. □

Lemma 4.17 (Central cycles generate first Hochschild homology)**.** *Let Q be a finite-dimensional symmetric complex algebra and $c \in Z(Q)$. If every derivation of Q annihilating c is inner, then the map*

$$Q \longrightarrow HH_1(Q), \qquad q \longmapsto [q \otimes c]$$

is surjective. If $P \twoheadrightarrow Q$ lifts c to a central element $z \in P$, then $HH_1(P) \to HH_1(Q)$ is surjective.

Proof. The cycles in the statement are available because the Hochschild boundary of $q \otimes c$ is $qc - cq = 0$. A symmetrizing trace τ identifies the dual chain complex with the Hochschild cochain complex and gives a perfect pairing

$$HH_1(Q) \times HH^1(Q) \longrightarrow \mathbb{C}, \qquad \langle [q \otimes c], [D] \rangle = \tau(qD(c)).$$

Inner derivations kill c, so this formula depends only on the cohomology class of D. If a class pairs trivially with every displayed cycle, then $\tau(qD(c)) = 0$ for every $q \in Q$. Nondegeneracy of the symmetrizing form gives $D(c) = 0$. The hypothesis then makes D inner, so its cohomology class is zero. Finite-dimensional duality proves the asserted surjectivity. For the second assertion, a lift a of q gives the cycle $a \otimes z$, which maps to $q \otimes c$. □

Proof of Theorem 4.14. For the vertex-deletion sequence, use the following polynomial potential. Fix an orientation of Δ, use left-to-right path composition, and put

$$r_i = e_i \left(\sum_a [a, a^*] \right) e_i, \qquad W_\Delta = \frac{1}{2} \sum_i r_i^2.$$

For an arrow $a : i \to j$, its two cyclic derivatives are

$$\partial_{a^*} W_\Delta = r_i a - a r_j, \qquad \partial_a W_\Delta = a^* r_i - r_j a^*.$$

It follows that in the Jacobi algebra $z = \sum_i r_i$ commutes with every arrow and that $r_i = z e_i$. Conversely, these central-extension relations recover both displayed cyclic derivatives. The resulting polynomial algebra maps are inverse, and hence

$$\mathrm{Jac}(\overline{Q}, W_\Delta) \cong P_\Delta.$$

Deleting J removes the incident arrows. The potential restricts to the sum of the analogous potentials on the connected components I_ν of I, and consequently

$$Q_I \cong \prod_{I_\nu} P_{I_\nu},$$

compatibly with the common central element z.

For every connected ADE diagram, Etingof–Latour–Rains prove symmetry [10, Proposition 2.1], the full-algebra trace formula [10, Theorem 3.1], and that derivations annihilating z are inner [10, Corollary 3.5 and the following remark]. The normalization $\rho_i = 1$ induces the same normalization on every component. The one-vertex algebra is $\mathbb{C}$, where the corresponding claims are verified directly. Hence P_Δ and Q_I are finite, basic, symmetric Jacobi algebras. A derivation of a finite product preserves its central idempotents, so the condition on derivations holds componentwise for Q_I. Apply Lemma 4.17 to $P_\Delta \twoheadrightarrow Q_I$, with c the image of z. It

gives $HH_1(P_\Delta) \twoheadrightarrow HH_1(Q_I)$. In (4.12), this makes the boundary from $HH_1(Q_I)$ vanish. The remaining sequence is (4.10). If I is empty, it is the identity sequence on the full algebra.

The full trace dimensions in [10, Theorem 3.1] are the numbers of positive roots, and degree $2p$ counts the roots of height $p+1$. Subtract the formula for the components of I from the formula for Δ. The roots removed in this subtraction are precisely those with zero retained coefficients, equivalently those supported on I. This gives both asserted dimensions. Since the ordinary corner and quotient maps preserve the internal grading, the same subtraction gives the degreewise statement. □

The dimension of the corner cocenter is therefore the number of positive roots with at least one nonzero retained coefficient. This also equals the generic cocenter rank of the retained central extension.

4.3.4 The universal cocenter and its reduced divisor

Theorem 4.18 (The universal multi-curve cocenter)**.** *The module $M_J = HH_0(\mathscr{E}_J/S)$ is free of rank N_J. For $\mathcal{H}_J = HH_0(C_J/R_J)$, there is a free resolution of the form*

$$0 \longrightarrow R_J^b \xrightarrow{\Theta_{\mathrm{tr},J}} R_J^b \longrightarrow \mathcal{H}_J \longrightarrow 0 \tag{4.13}$$

with

$$\mathrm{Fitt}_0^{R_J}(\mathcal{H}_J) = (\delta_J) = \left(\prod_{\beta \in \mathcal{B}_J} P_\beta \right). \tag{4.14}$$

Proof. Because M_J is a commutator cokernel, it commutes with scalar extension. Its closed fibre is $HH_0(e_J P_\Delta e_J)$, whose dimension is N_J by Theorem 4.14. On the other hand, (4.7) gives generic rank N_J, since each matrix algebra has a one-dimensional cocenter. Lift a basis of the closed fibre and use Nakayama to obtain $S^{N_J} \twoheadrightarrow M_J$. Its kernel is a submodule of a free module over the domain S; its generic fibre is zero, so the kernel itself is zero. Thus M_J is free over the Cartan parameter ring S.

Since z is central,

$$HH_0(D_J/S) = M_J/zM_J.$$

On the root factor α, z acts on the cocenter by the same scalar as on the algebra. Therefore

$$\det(z \mid M_J) = u \prod_{\pi_I(\alpha) \neq 0} \ell_\alpha = u\, q^* \delta_J, \qquad u \in \mathbb{C}^\times.$$

Its nonvanishing makes multiplication by z injective on the free module M_J, hence gives a square presentation of $HH_0(D_J/S)$. By (4.5), this module is the Cartan pullback of $\mathcal{H}_J$. As in the proof of Theorem 4.12, a first syzygy of a finite free surjection onto $\mathcal{H}_J$ becomes projective after faithfully flat base change. Faithfully flat descent makes it projective over R_J, hence free because R_J is local. Since $\mathcal{H}_J$ is torsion, the two free ranks agree, proving (4.13). Fitting ideals commute with base change and contract faithfully flatly, which gives (4.14). □

The product defining P_β accounts for all Cartan roots above β. The Fitting exponents of the three presentations are

universal module	exponent of P_β
$e_i C_J e_j$	$\beta_i \beta_j$
C_J	s_β^2
$HH_0(C_J/R_J)$	1

After pullback along the same family of classifying arcs, the closed-arc determinant orders give the ranks over the deformation base.

4.3.5 Expressing a geometric family as a pullback

Let $(B, \mathfrak{m}, \mathbb{C})$ be complete Noetherian local, and let $f_B : X_B \to Y_B$ be a deformation of f_0 as a projective contraction, with source and affine target flat over B. We complete along the central contraction, retaining the full moving support. Write $T_0 = \mathcal{O} \oplus \bigoplus_{j \in J} T_{0,j}$ for the fixed normalized basic generator.

The variable w is the surface-section parameter; B parametrizes the external deformation.

Proposition 4.19 (The multi-curve classifying pullback)**.** *There is a continuous local map $F : R_J \to D_B = B[[w]]$ and a relative tilting generator T_B lifting T_0, such that the relative contraction quotient satisfies*

$$\mathscr{A}_B := \operatorname{End}(T_B)^{\mathrm{op}} / \operatorname{End}(T_B)^{\mathrm{op}} e_0 \operatorname{End}(T_B)^{\mathrm{op}} \cong C_J \otimes_{R_J, F} B[[w]]. \tag{4.15}$$

All labelled projectors are preserved. Any other lift of the same marked generator gives an isomorphic marked quotient, and $\mathscr{A}_B / \mathfrak{m} \mathscr{A}_B \cong A_0$.

Proof. Lift a general surface-section parameter w_0 to w. The smoothness of the central source and the cDV property of the target allow the flatness argument of Proposition 4.6 to apply. That flatness argument gives D_B-flat source and target by the fibrewise criterion [31, Tag 00MP]. The closed surface is the partial resolution with retained set J, and its formally semiuniversal base is $\mathfrak{h}/W_I$ for arbitrary I [20, Section 2.3]. The compatible Artin-lifting and relative-affinization construction from the proposition produces F and identifies the relative contraction.

Before taking the idempotent quotient, Proposition 4.10 identifies the pulled-back universal generator and its endomorphism algebra by ordinary base change. The labelled summands of the pulled-back universal generator on the central threefold are basic: modulo w, these summands are the distinct indecomposable projectives on the partial surface. A nontrivial decomposition of one such summand would restrict to a nontrivial decomposition on that partial surface, because a summand whose reduction is zero vanishes by Nakayama in the completed neighbourhood. The determinant degrees distinguish the curve labels. Van den Bergh's classification of normalized perverse projectives [33, Proposition 3.5.4 and Theorem 3.5.5] consequently identifies the central generator with T_0.

The higher self-Ext groups of these summands vanish. Karmazyn's lifting lemma [21, Lemma 3.2] identifies their lifts through every external-base thickening and in the formal limit while preserving the chosen projectors. Ordinary endomorphism base change, followed by the right-exact idempotent quotient, gives (4.15).

The ordinary tensor expression must still be compared with the completed support. The central classifying arc lies in no curve divisor: otherwise the pulled-back family of exceptional

curves would give an exceptional surface in X_0. Choose a divisor equation with pullback $g(w) \neq 0$. Then $\mathbb{C}[[w]]$ is finite over $\mathbb{C}[[g(w)]]$, and hence over R_J. As in Lemma 3.16, tensoring the completed universal affine target ring along this central arc gives a finite algebra over $\mathbb{C}[[w]]$, with local Artinian closed fibre. This central-arc algebra is therefore already local and complete, and the ordinary central-arc quotient is the original marked contraction algebra A_0.

The ordinary pullback on the right-hand side of (4.15) is finite over D_B by Theorem 4.12, and its closed B-fibre is finite-dimensional. Corollary 4.3 applied to the pulled-back square presentation makes this ordinary pullback finite over B. On the resulting finite B-module w is nilpotent modulo $\mathfrak{m}$, so the $(\mathfrak{m}, w)$-adic and $\mathfrak{m}$-adic topologies agree. The ordinary pullback is already complete, which identifies the ordinary tensor expression with the quotient carrying the full relative support. □

The closed classifying arc supplies the divisor orders m_β. They determine the ranks below; the perturbation count of Nabijou–Wemyss identifies them with the GV invariants afterwards.

Proposition 4.20 (Ranks and intersection multiplicities after pullback)**.** *In the preceding setting, $\mathscr{A}_B$ is finite free over B of rank $\dim_{\mathbb{C}} A_0$, and*

$$C_J \otimes^L_{R_J,F} B[[w]] \simeq \mathscr{A}_B[0].$$

Write $f = F \bmod \mathfrak{m}$ and $m_\beta = \operatorname{ord}_w P_\beta(f(w))$. These integers are positive and finite, and

$$\operatorname{rank}_B \mathscr{A}_B = \sum_{\beta \in \mathcal{B}_J} m_\beta s_\beta^2, \tag{4.16}$$

$$\operatorname{rank}_B (e_i \mathscr{A}_B e_j) = \sum_{\beta \in \mathcal{B}_J} m_\beta \beta_i \beta_j. \tag{4.17}$$

Moreover $\operatorname{Spec} B[[w]]/(P_\beta(F))$ is finite flat over B of degree m_β.

Proof. Use the square algebra presentation. Its reduction modulo $\mathfrak{m}$ has cokernel A_0, which is finite-dimensional, so its matrix is invertible over $\mathbb{C}((w))$. Lemma 4.2 gives finite freeness over B and injectivity of the pulled-back presentation. The resulting exact sequence proves the derived statement.

Smith normal form on the closed classifying arc and the Fitting ideal give (4.16). Each $P_\beta(f)$ is nonzero because their product, with positive exponents, is nonzero. Since f is local, $P_\beta(f)$ vanishes at the origin, and hence $m_\beta > 0$. Apply Lemma 4.2 to the presentations of the Peirce spaces and to the one-by-one matrices $P_\beta(F)$. This gives the remaining rank formula and the stated finite flat schemes cut out by the divisor equations. Thus the ranks are determined by determinant orders on the closed classifying arc. □

Corollary 4.21 (The Cartan matrix and intersection multiplicities)**.** *The Cartan matrix with respect to the idempotents $(e_i)_{i \in J}$ is*

$$\left(\dim_{\mathbb{C}} e_i A_0 e_j\right)_{i,j \in J} = \sum_{\beta \in \mathcal{B}_J} m_\beta \, \beta \beta^{\mathsf{T}}.$$

It is symmetric positive definite. For the genus-zero invariants of the central flopping contraction, $m_\beta = n_\beta$.

Proof. Restricting (4.17) to the closed fibre gives the displayed matrix formula. Every retained simple root projects to a standard basis vector and every m_β is positive, so the sum is positive definite.

Choose a transverse perturbation of the central classifying arc as in [28, Section 2.6]. Conservation of the local intersection number with the reduced divisor $P_\beta = 0$ shows that the number of transverse intersections of the perturbation with this divisor is m_β. These intersections count curves of class β by [28, Corollary 2.10 and Section 3.1], giving $m_\beta = n_\beta$. Thus the determinant orders on the central arc recover the GV multiplicities of the central contraction. □

4.3.6 Hua–Toda flatness over general bases

Kawamata proved algebra flatness over a disk [22, Theorem 3.1(2)]. In the present setting, the rank calculation above treats the algebra and its Peirce spaces, the cocenter has a corresponding pulled-back presentation to which Lemma 4.2 applies, and a relative symmetrizing trace identifies the centre. The theorem records the three flatness assertions and their base-change consequences over the complete local bases in its statement.

Theorem 4.22 (Hua–Toda flatness and base change)**.** *Let f_B, T_B, and $\mathscr{A}_B$ be as in Proposition 4.19, with arbitrary complete Noetherian local complex base $(B, \mathfrak{m}, \mathbb{C})$. Let $n_\beta = m_\beta$ be the central curve multiplicities of Corollary 4.21, and put $N = \sum_{\beta \in \mathcal{B}_J} n_\beta$. Then*

$$\mathscr{A}_B \cong B^{\sum_\beta n_\beta s_\beta^2}, \qquad HH_0(\mathscr{A}_B/B) \cong B^N, \qquad Z(\mathscr{A}_B) \cong B^N$$

as B-modules. For every commutative B-algebra B', the canonical maps

$$HH_0(\mathscr{A}_B/B) \otimes_B B' \xrightarrow{\sim} HH_0(\mathscr{A}_B \otimes_B B'/B'),$$
$$Z(\mathscr{A}_B) \otimes_B B' \xrightarrow{\sim} Z(\mathscr{A}_B \otimes_B B')$$

are isomorphisms. Every symmetrizing trace of A_0 lifts to a perfect B-linear symmetrizing trace of $\mathscr{A}_B$. Moreover,

$$C_J \otimes^L_{R_J,F} B[[w]] \simeq \mathscr{A}_B[0],$$
$$\mathcal{H}_J \otimes^L_{R_J,F} B[[w]] \simeq HH_0(\mathscr{A}_B/B)[0].$$

The free bases and lifted trace can be chosen, and the base-change maps are canonical.

Proof. Proposition 4.20 gives the algebra assertion and its derived pullback. Put $H_B = HH_0(\mathscr{A}_B/B)$. The B-span of commutators is already a D_B-module, since $d[a,b] = [da, b]$ for a central scalar $d \in D_B$. Hence

$$H_B \cong \mathcal{H}_J \otimes_{R_J,F} D_B \cong \operatorname{coker} \Theta_{\mathrm{tr},J}(F).$$

On the closed classifying arc, the determinant of this presentation is a unit times $\prod_\beta P_\beta(f(w))$. It is nonzero and has order N. Lemma 4.2 gives freeness of rank N and injectivity of the pulled-back presentation. This also proves the second derived statement over B, without assuming that B is reduced.

Finally, the central contraction algebra is symmetric: the stable endomorphism algebra of the entire maximal Cohen–Macaulay generator is covered by [2, Theorem 2.6]. For any

symmetrizing trace of A_0, apply Lemma 4.4 to the finite free algebra and cocenter just obtained. It supplies a perfect relative lift. The lift is a choice; it proves centre freeness, whereas the two base-change maps are canonical. □

For the local consequences below, write $d = \dim_{\mathbb{C}} A_0$.

Corollary 4.23 (Local bases, geometric base change, and saturation)**.** *The commutator submodule of $\mathscr{A}_B$ is a B-direct summand. For every ideal $L \subseteq B$,*

$$[\mathscr{A}_B, \mathscr{A}_B] \cap L\mathscr{A}_B = L[\mathscr{A}_B, \mathscr{A}_B].$$

For a continuous local map $B \to B'$ of complete Noetherian local complex bases, $\mathscr{A}_B \otimes_B B'$ is the geometric relative quotient formed from the pulled-back fixed generator and full finite support. Over a Noetherian algebraic or analytic base with a finite relative quotient, the algebra, centre, and cocenter are locally free near a fibre of the stated kind, with ranks d, N, N, and the centre and cocenter commute with arbitrary base change. Several exceptional image points are treated by their completed pieces.

Proof. The cokernel of the commutator submodule is finite free, so the commutator sequence splits. Intersecting the resulting direct-sum decomposition with $L\mathscr{A}_B$ proves the saturation formula.

By Proposition 4.19 and the ordinary endomorphism comparison, geometric base change reduces to comparison of the power-series pullbacks. Apply Weierstrass preparation to $\det \Theta_J(F)$. This produces a distinguished polynomial $W(w)$ that annihilates the finite quotient. On modules killed by W, base change is computed through $B[w]/(W)$. After a continuous local base change, Weierstrass division identifies $B[w]/(W) \otimes_B B'$ with $B'[[w]]/(W')$, so the finite polynomial quotient computes the required power-series base change.

For a noncomplete local base, complete it, apply the theorem, and descend finite projectivity through the faithfully flat completion. Locally choose a lift of a central symmetrizing trace through the locally free cocenter and shrink so that its pairing is perfect. The duality argument then gives arbitrary centre base change there as well. Finally, disjoint completed pieces at the exceptional image points give the corresponding finite products. □

4.3.7 The exponents in two examples

For $\Delta = A_3$, with chain $1 - 2 - 3$, take $J = \{1, 3\}$. The restricted roots are $(1, 0), (0, 1), (1, 1)$, and

$$P_{10} = \lambda_1(\lambda_1 + \lambda_2), \qquad P_{01} = \lambda_3(\lambda_2 + \lambda_3), \qquad P_{11} = \lambda_1 + \lambda_2 + \lambda_3.$$

Thus the algebra and cocenter Fitting ideals are respectively

$$(P_{10}P_{01}P_{11}^4), \qquad (P_{10}P_{01}P_{11}).$$

If the three intersection orders are a, b, c, the algebra rank is $a + b + 4c$, its centre and cocenter have rank $a + b + c$, and its Cartan matrix with respect to the two curve idempotents is

$$\begin{pmatrix} a + c & c \\ c & b + c \end{pmatrix}.$$

This example distinguishes the number of Cartan roots above a restricted root from the exponent contributed by its matrix block.

For the coefficient-five vertex of E_8, deleting that vertex leaves $A_3 \sqcup A_4$. Theorem 4.14 gives

$$\dim HH_0(e_p \Pi_0^\rho e_p) = 120 - 6 - 10 = 104.$$

This recovers (3.18) through the full-algebra trace formula of Etingof–Latour–Rains. The explicit rank certificate in Subsection 3.6 gives the direct corner calculation used in the single-curve proof.

The Peirce dimensions and cocenter count computed here are the numerical quantities used in the orbit argument with fixed idempotents in Section 5.

5 Semisimple generic fibres with the curve projectors retained

The universal presentations extend the single-curve construction to every retained subset, with ranks giving the Peirce dimensions and the number of simple factors. For an analytic disk deformation satisfying the hypotheses of Kawamata's Theorem 3.1, the relative contraction algebra is flat and its generic fibre is a product of matrix algebras; the multiplicities of the original summands are recorded by the integers $m_{j,i}$ [22, Theorem 3.1(2)–(3)]. A formal classifying arc keeps the separable algebra of labelled projectors fixed. After a finite extension, the lattice construction produces a polynomially split algebra. The orbit argument identifies the simple matrix factors and the ranks of those projectors in every geometrically semisimple generic fibre.

Throughout this section, $f_0 : X_0 \to Y_0$ is the completion at an exceptional image point of a projective flopping contraction of a smooth complex threefold. We use the marked generator, retained set J, restricted roots $\mathcal{B}_J$, and quotient C_J of Subsection 4.3. The original curves are labelled $(C_i)_{i\in J}$. In the positive-dual convention the generator is

$$P_0 = \mathcal{O} \oplus \bigoplus_{i\in J} P_{0,i}, \qquad \deg_{C_j} \det P_{0,i} = \delta_{ij}, \qquad A = \operatorname{End}(P_0)/\operatorname{End}(P_0)e_0 \operatorname{End}(P_0).$$

Dualizing identifies this quotient with the convention $\operatorname{End}(T_0)^{\mathrm{op}}/(e_0)$ used earlier, preserving all projectors. Put

$$E = \bigoplus_{i\in J} \mathbb{C}e_i, \qquad V_\beta = \bigoplus_{i\in J} \mathbb{C}^{\beta_i}, \qquad S_E = \prod_{\beta\in\mathcal{B}_J} \operatorname{End}_{\mathbb{C}}(V_\beta)^{\times n_\beta}. \tag{5.1}$$

On each β-factor, e_i acts as the projection onto $\mathbb{C}^{\beta_i}$. These labelled idempotents are generally noncentral. A marking consists of the fixed embedding $E \hookrightarrow A$, which prescribes the projectors and their left and right actions. Equivalently, the simple module of a β-factor has dimension vector $(\dim_{\mathbb{C}} e_i V_\beta)_{i\in J} = \beta$ relative to the idempotents $(e_i)_{i\in J}$, and this dimension vector occurs n_β times. Write $s_\beta = \sum_i \beta_i$, keeping distinct proportional restricted roots separate.

5.1 Constructing a semisimple generic fibre with fixed projectors

Theorem 5.1 (Semisimple generic fibre with fixed projectors)**.** *There is a finite free unital $\mathbb{C}[[s]]$-algebra $\mathscr{A}$, obtained from the universal marked geometric family, with a marked identification*

$\mathscr{A}/(s) \cong A$. Its labelled orthogonal idempotents e_i sum to its identity. For a finite extension $K'/\mathbb{C}((s))$, there is an isomorphism preserving these idempotents

$$\mathscr{A} \otimes_{\mathbb{C}[[s]]} K' \cong S_E \otimes_{\mathbb{C}} K'.$$

In particular, e_i has rank β_i on each of the n_β factors of size s_β. After writing $K' = \mathbb{C}((u))$, $s = u^e$, there is also a finite free marked $\mathbb{C}[u]$-algebra $\mathscr{P}$ such that

$$\mathscr{P}/(u) \cong A, \qquad \mathscr{P}[u^{-1}] \cong S_E \otimes_{\mathbb{C}} \mathbb{C}[u, u^{-1}].$$

Thus the deformation can be chosen polynomially split, with all projector ranks preserved.

Lemma 5.2 (Transverse translation of a formal arc)**.** *Let $R = \mathbb{C}[[t_1, \dots, t_r]]$, and let $f(w) = (f_1(w), \dots, f_r(w))$, with $f_\nu(w) \in w\mathbb{C}[[w]]$. Let $P_1, \dots, P_q \in R$ be pairwise nonassociated prime elements such that $d_b = \operatorname{ord}_w P_b(f(w))$ is positive and finite. Let Σ be a reduced closed formal subscheme of $\bigcup_b V(P_b)$, of codimension at least two in* $\operatorname{Spec} R$*. There is a nonempty Zariski open set of vectors $v \in \mathbb{C}^r$ for which $F(w, s) = f(w) + sv$ has the following properties. Each $P_b(F)$ is a unit times a distinguished polynomial in w of degree d_b; over $\mathbb{C}((s))$ these polynomials are separable, pairwise coprime, and have no root mapping into Σ. After a finite ramified extension their roots lie in $u\mathbb{C}[[u]]$.*

Proof. Introduce independent parameters $a_1, \dots, a_r$, and put $B_{\mathrm{tr}} = \mathbb{C}[[a_1, \dots, a_r]]$ and $D = B_{\mathrm{tr}}[[w]]$. By Weierstrass preparation, $G_b = P_b(f(w) + a)$ is a unit times a monic distinguished polynomial $W_b(a, w)$ of degree d_b. Thus $D/(G_b)$ is finite free over B_{tr}. The coordinate change $t_\nu = a_\nu + f_\nu(w)$ gives

$$D/(G_b) \cong (R/(P_b))[[w]],$$

which is a domain. Its generic fibre over B_{tr} is therefore a finite field extension. In characteristic zero that extension is separable, so the discriminant of W_b is nonzero.

For $b \neq c$, the incidence module $D/(G_b, G_c)$ is finite over B_{tr}. The coordinate change identifies it with $(R/(P_b, P_c))[[w]]$, of dimension at most $r - 1$, whereas $\dim B_{\mathrm{tr}} = r$. It is consequently torsion, so the resultant of W_b, W_c is nonzero. To impose avoidance of Σ, let K_Σ be its radical ideal. Since $\prod_b P_b \in K_\Sigma$, the module

$$D/K_\Sigma(f(w) + a)$$

is finite over B_{tr}. After the coordinate change its dimension is at most $r-1$, whereas $\dim B_{\mathrm{tr}} = r$; it is therefore torsion. Choose a nonzero annihilator, taking 1 when the module is zero. Multiply it by the discriminants and resultants already obtained, and call the resulting nonzero element $b(a)$.

Let b_h be the lowest nonzero homogeneous term of b. For every v with $b_h(v) \neq 0$, $b(sv) = s^h(b_h(v) + O(s)) \neq 0$. This element is invertible over $\mathbb{C}((s))$, proving the three generic assertions. Division by the monic polynomial $\prod_b W_b$ shows that the power-series quotients used to test avoidance of Σ can be computed in $B_{\mathrm{tr}}[w]/(\prod_b W_b)$. Thus ordinary substitution $a = sv$ agrees with completed substitution and preserves the annihilation relation for $D/K_\Sigma(f(w) + a)$, with the annihilating element $b(a)$ specialized to $b(sv)$. The roots of a distinguished monic polynomial are integral over $\mathbb{C}[[s]]$ and reduce to zero. A finite splitting extension is a complete discretely valued field with residue field $\mathbb{C}$, so its valuation ring is $\mathbb{C}[[u]]$. Absorbing a unit into u gives $s = u^e$, and every root lies in $u\mathbb{C}[[u]]$. □

Katz–Morrison's simultaneous partial resolution supplies the geometry at the roots. Nabijou–Wemyss identify its discriminant components with restricted roots and the unique curve class at a general point of each component [28, Theorem 2.6]. Apply this description to the translated slice of Lemma 5.2.

Proposition 5.3 (A formal deformation separating the curve divisors)**.** *Let $f : R_J \to \mathbb{C}[[w]]$ classify f_0. There is a translated arc $F : R_J \to \mathbb{C}[[s, w]]$ reducing to f, such that its geometric generic contraction has exactly n_β disjoint split $(-1,-1)$-curves of each class β. At each such curve the completed target is a split ordinary double point and the source is its chosen small resolution. The generic fibre is formed from the full finite relative family.*

Proof. The marked base R_J is regular, and its curve-divisor equations P_β are distinct primes by Lemma 4.9. Their orders on the central arc are

$$\operatorname{ord}_w P_\beta(f(w)) = n_\beta$$

by Proposition 4.20 and Corollary 4.21. Transversality is imposed on every component of the surface discriminant, including inactive components, because the full threefold slice must be controlled rather than only its active curves.

The central arc is contained in none of these components. Write the central isolated cDV hypersurface as $G(x_1, x_2, x_3, w) = 0$. The relative critical scheme defined by $G, G_{x_1}, G_{x_2}, G_{x_3}$ is finite over $\mathbb{C}[[w]]$, because the central surface of the cDV hypersurface is ADE; this is the topological finiteness argument of Lemma 2.29. If the relative critical scheme had a component dominating the w-axis, the normalization of that component would have a parameter u with $w = u^e$. Differentiating $G(x(u), u^e) = 0$, while using the three vanishing partial derivatives, would give $G_w = 0$ along that curve. This contradicts the isolated singularity of the threefold.

Take Σ to be the reduced image of the intersections of two distinct root hyperplanes on the Cartan cover. The finite quotient preserves the codimension of the Cartan-cover intersection locus, so Σ has codimension at least two; Σ contains the singular and intersection loci of the full discriminant components. Apply Lemma 5.2 to the reduced equations of the full discriminant components. At an active point of the resulting slice, exactly one root α vanishes, with $\pi_I(\alpha) = \beta$. Since $s_\alpha \notin W_I$, the Cartan-to-marked map is étale there. The map to the unmarked surface base instead has normal parameter $\tau = \zeta^2$, and the transverse surface model is

$$xy - \xi^2 = \zeta^2.$$

The simple root of $P_\beta(F)$ makes ζ a parameter along the slice. Over a splitting field, the displayed equation is $xy - p_+p_- = 0$, and the chosen partial resolution is its small resolution, with normal bundle $\mathcal{O}(-1)^{\oplus 2}$. Its curve class is the restricted root β, by the marked root-divisor identification [28, Theorem 2.6].

At an inactive root, $s_\alpha \in W_I$. Here the marked normal parameter is already τ, so the transverse model $xy - \xi^2 = \tau$ has smooth threefold slice and no contracted curve. Away from the full discriminant the surface family is smooth. Hence the active roots account for every exceptional curve. The distinguished polynomial for $P_\beta(F)$ has degree n_β and simple roots, which gives the stated multiplicity. Forming the finite relative family before inverting s retains every moving root, as in Remark 2.18. □

Proposition 5.4 (The individual projector ranks)**.** *At a split ordinary double point of class β in Proposition 5.3, the completed relative contraction quotient is*

$$\operatorname{End}_{K'}\left(\bigoplus_{i\in J}(K')^{\beta_i}\right),$$

and e_i is the projection onto $(K')^{\beta_i}$. The maximal ideal of the completed target acts trivially on this quotient.

Proof. Pull back the fixed positive generator and complete at the chosen curve. It remains a projective generator in the fixed perverse heart by Proposition 4.10. The determinant line bundle of its i-th summand has degree zero at the deleted vertices and degree δ_{ij} at the retained vertices. Pairing this degree vector with α gives

$$\deg_{C_\beta}\det P_i=\beta_i.$$

This is the labelled version of Lemma 2.8; the line bundles and their pairings descend because the degree vector is W_I-invariant.

The split Atiyah calculation of Lemma 2.23 gives exactly two indecomposable positive perverse projectives, $\mathcal{O}$ and $\mathcal{Q}$, with determinant degrees zero and one. Consequently, if $q_i=\operatorname{rank}P_i$, Krull–Schmidt and the displayed degree give, separately for each label,

$$\widehat{P}_i\cong\mathcal{O}^{\oplus(q_i-\beta_i)}\oplus\mathcal{Q}^{\oplus\beta_i}.$$

After pushforward, the generator is a direct sum of free modules and $\bigoplus_i I^{\oplus\beta_i}$, where I is the rank-one Atiyah module. The original e_0 projects onto one free summand. The ideal it generates consists of all maps factoring through finite free modules, so it also kills every additional free summand in this decomposition. Lemma 2.25 identifies $\underline{\operatorname{End}}(I)$ with K', with the target maximal ideal acting by zero. The surviving matrix entries are therefore copies of K', with their usual composition. Grouping them by the original summands gives precisely the asserted projectors and multiplication. □

Lemma 5.5 (Polynomialization with fixed projectors)**.** *Let L be a finite free $\mathbb{C}[[u]]$-algebra with marked special fibre A, and suppose a marked generic isomorphism identifies $L[u^{-1}]$ with $S_E\otimes\mathbb{C}((u))$. There is a finite free marked $\mathbb{C}[u]$-algebra $\mathscr{P}$ whose completion is L and whose localization at u is $S_E\otimes\mathbb{C}[u,u^{-1}]$.*

Proof. The generic marked isomorphism lets us regard L as a lattice in $S_E((u))$. For some $m\geq 0$,

$$u^mS_E[[u]]\subseteq L\subseteq u^{-m}S_E[[u]].$$

Define $\mathscr{P}=L\cap S_E[u,u^{-1}]$. It is a unital subalgebra containing the constant labelled projectors, and

$$u^mS_E[u]\subseteq\mathscr{P}\subseteq u^{-m}S_E[u].$$

The sandwich makes $\mathscr{P}$ finite and torsion-free over the principal ideal domain $\mathbb{C}[u]$, hence free, and its localization is the stated split algebra. To compare completions, take an element of L and truncate its Laurent expansion far enough. The truncation lies in $\mathscr{P}$, because the discarded Laurent tail lies in $u^mS_E[[u]]\subseteq L$, and the truncation approximates the original element to arbitrarily high order. The same sandwich shows that the ambient u-adic topology agrees with the lattice topology. Completion therefore identifies $\mathscr{P}\otimes_{\mathbb{C}[u]}\mathbb{C}[[u]]$ with L, preserving products and projectors. Reduction modulo u gives the prescribed marked algebra A. □

For finite-dimensional algebras without distinguished idempotents, Jelisiejew–Smoktunowicz relate formal, polynomially split, and orbit-closure formulations of deformation [19, Theorems 2.19–2.20]. The lattice argument retains the chosen embedding of E and the constant projectors while passing from the formal family to the polynomial one.

Proof of Theorem 5.1. Choose the translated classifying map of Proposition 5.3, and form the relative quotient

$$\mathscr{A} = C_J \otimes_{R_J,F} \mathbb{C}[[s,w]].$$

The weighted square presentation in Theorem 4.12 and Lemma 4.2 make it finite free over $\mathbb{C}[[s]]$. Its rank is $\sum_\beta n_\beta s_\beta^2$, and its special fibre, with every projector, is A. Proposition 4.19 identifies it with the relative endomorphism quotient of the pulled-back generator.

The reduced curve equation $\delta_J = \prod_\beta P_\beta$ annihilates C_J by Proposition 4.13. Its pullback is a unit times the product of the distinguished polynomials above. After inverting s and taking a finite splitting extension, the Chinese remainder theorem separates their distinct roots. The free projector is full away from the exceptional curves, by Lemma 2.31. At each remaining point, Proposition 5.4 identifies the completed factor with the required matrix algebra and the prescribed projector ranks, and shows that the maximal ideal of the completed threefold target acts trivially. Faithful flatness of completion gives the same uncompleted factor by Lemma 2.32. The separated factors give the scalar extension of S_E to the splitting field, with the chosen idempotents preserved.

After identifying the generic factors, replace the formal family by a polynomial one. The finitely many coefficients still needed to split the generic matrix factors lie in a finite extension of $\mathbb{C}((s))$. Write the valuation ring of that extension as $\mathbb{C}[[u]]$, with $s = u^e$, and apply Lemma 5.5 to $L = \mathscr{A} \otimes_{\mathbb{C}[[s]]} \mathbb{C}[[u]]$. □

Remark 5.6 (Relation to Kawamata's existence result). Kawamata begins with an analytic disk deformation and proves flatness of the relative contraction algebra; when the noncentral fibres are disjoint $(-1,-1)$-curves, he obtains the generic matrix product [22, Theorem 3.1(2)–(3)]. His proof also decomposes the restriction of each original summand with multiplicities $m_{j,i}$, the geometric source of the projector-rank data. Kawamata's theorem has analytic disk scope. The formal classifying-arc statement needed here is established through the universal quotient with the embedding of E fixed, and is then polynomialized. The comparison after forgetting the idempotents with other deformation formulations is supplied by [19, Theorems 2.19–2.20]; the marked lattice step is written out here.

5.2 Fixed-idempotent laws and the orbit criterion

The existence theorem constructs a product of matrix algebras with fixed idempotents. The unmarked structure-constant and orbit method used here is reviewed by Jelisiejew–Smoktunowicz: semisimple orbits are open, and the formal, polynomially split, and orbit-closure deformation notions agree after the appropriate finite extensions [19, Section 2.1, Theorems 2.14 and 2.20]. With a fixed separable algebra E, the method shows that finite free associative formal deformations of A with geometrically semisimple generic fibre have the same simple matrix factors and the same dimension vectors relative to the lifted idempotents. Its relative tangent formula uses the centralizer of E, generally larger than the centre. Peirce dimensions and the number of simple factors are necessary numerical inputs, but they do not determine the marked factorization by themselves.

For a semisimple algebra whose simple modules have dimension vectors v_a relative to $(e_i)_{i\in J}$, the equalities $C=(c_{ij})=\sum_a v_a v_a^{\mathsf T}$ and $N=\#\{a\}$ do not, by themselves, determine the vectors. For example, the two multisets

$$\{(2,1),(2,1),(2,1),(0,3),(1,0)\},\qquad \{(3,0),(1,2),(1,2),(1,2),(1,0)\}$$

both have five vectors, Gram matrix $\left(\begin{smallmatrix}13 & 6\\ 6 & 12\end{smallmatrix}\right)$, and unmarked algebra $M_3(\mathbb{C})^4\times\mathbb{C}$. They are distinct even after exchanging the two labels. Thus the Gram matrix and the number of simple modules admit several multisets of dimension vectors. The orbit argument in Theorem 5.12 starts from the product constructed geometrically in Theorem 5.1 and proves that every geometrically semisimple generic fibre, together with the lifted idempotents, is isomorphic to it.

Let $V=\bigoplus_{i,j\in J}V_{ij}$ be a fixed E-bimodule, with $c_{ij}=\dim_{\mathbb{C}}V_{ij}$ and distinguished nonzero vectors $e_i\in V_{ii}$. The fixed-idempotent law scheme $X_E(V)$ parametrizes associative E-balanced products $V\otimes_E V\to V$ with these prescribed left and right E-actions. Thus $1=\sum_i e_i$ is fixed before the multiplication varies, and the associativity equations define the scheme structure. Its change-of-basis group is

$$G_E=\{g\in \mathrm{Aut}_{E\text{-}E}(V): g(e_i)=e_i \text{ for all } i\}.$$

This group is smooth and irreducible: its factors are $\mathrm{GL}(V_{ij})$ for $i\neq j$ and the stabilizers of the nonzero vectors e_i in $\mathrm{GL}(V_{ii})$. Set

$$b=\sum_i c_{ii},\qquad g=\dim G_E=\sum_{i,j}c_{ij}^2-b. \tag{5.2}$$

For a law $B\in X_E(V)$, its centralizer $B^E=\{x: xe=ex \text{ for all } e\in E\}$ is $\bigoplus_i e_iBe_i$, of dimension b.

Relative Hochschild cochains impose the fixed separable algebra data. They give the E-relative tangent dimension needed below, rather than merely the ordinary fixed-unit tangent space.

Lemma 5.7 (Relative cochains and marked tangent dimensions)**.** *For $B\in X_E(V)$, normalized relative Hochschild cochains compute ordinary Hochschild cohomology. If $Z^2_E(B)$ and $B^2_E(B)$ are their degree-two cocycles and coboundaries, then*

$$T_BX_E(V)=Z^2_E(B),\qquad T_B(G_E\cdot B)=B^2_E(B).$$

In particular,

$$\dim T_BX_E(V)=g-b+\dim Z(B)-\dim HH^1(B)+\dim HH^2(B). \tag{5.3}$$

Proof. The normalized relative complex has

$$C^0_E(B)=B^E,\qquad C^n_E(B)=\mathrm{Hom}_{E^e}((B/E)^{\otimes_E n},B)\quad (n\geq 1).$$

Insertion of the identity gives a contracting homotopy for the augmentation of the relative bar complex, so that complex is exact. Normalization removes its split degenerate subcomplex and leaves the terms $B\otimes_E(B/E)^{\otimes_E n}\otimes_E B$. Each is induced from an E^e-module. Since E^e is a product of copies of $\mathbb{C}$, every such module is projective, and induction makes each term B^e-projective. This is a projective resolution of B, so its cohomology is ordinary $HH^*(B)$.

Linearizing associativity gives the relative two-cocycle equation: fixed E-actions require the cochain to vanish when an input belongs to E. Linearizing a change of E-bimodule basis gives the coboundaries. The group of algebra automorphisms fixing E is smooth in characteristic zero, so its orbit has the asserted tangent space and Lie algebra $\mathrm{Der}_E(B)$. The degree-zero part of the same complex gives the exact sequence

$$0 \longrightarrow Z(B) \longrightarrow B^E \longrightarrow \mathrm{Der}_E(B) \longrightarrow HH^1(B) \longrightarrow 0.$$

Moreover, $\dim C^1_E(B) = g$, because these cochains are precisely the E-bimodule endomorphisms vanishing on the distinguished vectors. It follows that $\dim B^2_E(B) = g - b + \dim Z(B) - \dim HH^1(B)$. Adding $\dim HH^2(B)$ proves the formula. □

Lemma 5.8 (Semisimple laws with fixed idempotents)**.** *A finite-dimensional semisimple complex algebra with a fixed complete labelled system of orthogonal idempotents is classified, up to marked isomorphism, by a multiset of nonzero vectors $v_a = (v_{a,i})_{i\in J}$ of nonnegative integers. Its algebra and Peirce dimensions are*

$$S(\{v_a\}) = \prod_a \mathrm{End}_{\mathbb{C}}\left(\bigoplus_i \mathbb{C}^{v_{a,i}}\right), \qquad c_{ij} = \sum_a v_{a,i}v_{a,j}.$$

Every such orbit in $X_E(V)$ is open, has dimension $g - b + \dim Z(S)$, and is dense in an irreducible component.

Proof. On a simple matrix factor, simultaneous orthogonal projections are classified by the dimensions of their images. Inner automorphisms change bases in those images, and an algebra automorphism may permute factors carrying the same vector. This proves the classification and the Peirce formula; in particular $\dim Z(S)$ is the number of vectors. A marked isomorphism between laws on V is exactly an element of G_E.

Semisimplicity gives $HH^n(S) = 0$ for $n > 0$, so Lemma 5.7 makes the tangent and orbit dimensions equal to $g - b + \dim Z(S)$. The stabilizer is a product of groups $(\prod_i \mathrm{GL}_{v_{a,i}})/\mathbb{G}_m$, followed by finite permutations of identical factors, and is smooth. Apply Lemma 3.4(i) to the smooth irreducible group G_E. The orbit is open, so its closure is an irreducible component. □

Theorem 5.9 (The fixed-idempotent orbit criterion)**.** *Let $B \in X_E(V)$, and suppose a marked semisimple algebra $S \in X_E(V)$ satisfies*

$$B \in \overline{G_E \cdot S}, \qquad \dim Z(B) = \dim Z(S), \qquad \dim HH^1(B) = \dim HH^2(B).$$

Then B is a smooth point of $X_E(V)$, and $\overline{G_E \cdot S}$ is its unique irreducible component. If another marked semisimple law S' has $B \in \overline{G_E \cdot S'}$, then $S' \cong S$ as marked algebras.

Proof. The hypotheses and the preceding lemmas give

$$\dim T_B X_E(V) = g - b + \dim Z(S) = \dim(G_E \cdot S).$$

The semisimple orbits are open by Lemma 5.8. Applying Lemma 3.4(ii) proves smoothness, the unique-component assertion, and equality of the two marked orbits. □

5.3 Periodicity and general uniqueness

Return to the multi-curve contraction algebra A, and write

$$c_{ij} = \dim e_i A e_j = \sum_\beta n_\beta \beta_i \beta_j, \qquad N = \sum_\beta n_\beta = \dim Z(A) = \dim HH_0(A). \tag{5.4}$$

Corollary 4.21 and Theorem 4.22, restricted to the central fibre, place A and the geometrically constructed product on the same E-bimodule. The Peirce ranks specify the ambient fixed-idempotent law scheme, the cocenter count enters the tangent-dimension comparison, and the formal family supplies a semisimple orbit whose closure contains A.

Proposition 5.10 (The multi-curve Hochschild equality)**.** *The algebra A is symmetric and satisfies*

$$\Omega^4_{A^e}(A) \simeq A \quad \textit{in the stable bimodule category,} \qquad DHH^1(A) \cong HH^2(A).$$

Proof. The target is an isolated cDV hypersurface with a smooth crepant resolution, so August's symmetry theorem applies to the stable endomorphism algebra of the entire nonfree maximal Cohen–Macaulay generator [2, Theorem 2.6]. The four-term projective bimodule extension in [18, Setting 3.2.1 and Theorem 3.2.5] also applies to this generator and has the diagonal bimodule at both ends. Thus the untwisted four-periodicity used in Proposition 3.9 applies here. Lemma 3.10 then gives the asserted duality in positive Hochschild degrees. Each semisimple direct factor has zero positive Hochschild cohomology. □

Lemma 5.11 (Lifting and comparing the fixed idempotents)**.** *Let $(B, \mathfrak{m}, \mathbb{C})$ be complete Noetherian local, and let $\mathscr{D}$ be a finite free B-algebra with identified special fibre A. The system (e_i) lifts to orthogonal idempotents $(\widetilde{e}_i)$ summing to one. Any two such lifts are simultaneously conjugate by a unit congruent to one modulo $\mathfrak{m}\mathscr{D}$. Each $\widetilde{e}_i \mathscr{D} \widetilde{e}_j$ is free of rank c_{ij}, and an adapted trivialization defines a morphism $\operatorname{Spec} B \to X_E(V)$ whose closed point is A.*

Proof. Across a square-zero ideal, if $x^2 - x = q$ lies in that ideal, then $x + (1-2x)q$ is idempotent. Lift the first projector and work successively in its complementary corner to preserve orthogonality; take the last projector to be the remaining complement. Applying this construction to successive powers of $\mathfrak{m}$ and using completeness gives the system. For two lifts (p_i) and (q_i), the element

$$h = \sum_i q_i p_i \equiv 1 \pmod{\mathfrak{m}\mathscr{D}}$$

is invertible by completeness and satisfies $hp_i = q_i h$. This proves simultaneous conjugacy.

The maps $x \mapsto p_i x p_j$ are mutually orthogonal B-linear projectors. Their images are direct summands of the free module $\mathscr{D}$, hence free over the local ring, with ranks determined by the special fibre. Choose bases lifting fixed bases of $e_i A e_j$; in each diagonal block include p_i, which is unimodular because its reduction is nonzero. These choices give the claimed E-bimodule trivialization and classifying morphism. □

Theorem 5.12 (Geometric semisimple fibres with and without fixed idempotents)**.** *Let $\mathscr{D}$ be any finite free unital $\mathbb{C}[[t]]$-algebra with special fibre identified with A, and suppose its geometric generic fibre is semisimple. For any lifts of the labelled original idempotents, that fibre is marked-isomorphic to $S_E \otimes_{\mathbb{C}} \overline{\mathbb{C}((t))}$. Equivalently, the multiset of dimension vectors of its simple modules, relative to the lifted idempotents, is exactly*

$$\{\, \beta \text{ repeated } n_\beta \text{ times} : \beta \in \mathcal{B}_J \,\}. \tag{5.5}$$

Thus there is exactly one isomorphism class of semisimple geometric generic fibre together with the lifted idempotents. After forgetting the idempotents, this fibre is obtained by scalar extension from the complex semisimple algebra

$$\prod_{m\geq 1} M_m(\mathbb{C})^{\times N_m}, \qquad N_m = \sum_{s_\beta = m} n_\beta.$$

Moreover A is smooth in $X_E(V)$, with unique irreducible component $\overline{G_E \cdot S_E}$.

Proof. The Peirce formula in (5.4) places A and S_E on the same E-bimodule V. Theorem 5.1 places A in $\overline{G_E \cdot S_E}$: choose bases in every Peirce summand of the free family, including each projector, and use the generic marked isomorphism. Every equation of the orbit closure vanishes at the generic point after a finite field extension, hence on the whole parameter DVR, and therefore at A. Equation (5.4) gives $\dim Z(A) = N = \dim Z(S_E)$, while Proposition 5.10 gives the Hochschild equality. Theorem 5.9 now proves smoothness and the unique-component assertion.

Let $\mathscr{D}$ be an arbitrary family as in the statement. Lift the projectors and choose the adapted frame of Lemma 5.11. The geometric generic fibre of $\mathscr{D}$ is represented by a constant semisimple complex algebra S' with lifted idempotents, classified by the multiset of dimension vectors of its simple S'-modules. The finitely many coefficients of a marked isomorphism lie in a finite extension of $\mathbb{C}((t))$. The same orbit-closure argument gives $A \in \overline{G_E \cdot S'}$, so the orbit criterion forces $S' \cong S_E$. This proves (5.5). Changing the lifted idempotents conjugates them simultaneously, so the answer does not depend on their choice. Forgetting the projectors combines precisely the factors with the same total size s_β, giving the stated unmarked algebra. □

Corollary 5.13 (Smooth points of $X_E(V)$ and semisimple fibres)**.** *Put $d = \dim A$, and let b, g be as in (5.2). Then*

$$\widehat{\mathcal{O}}_{X_E(V),A} \cong \mathbb{C}[[x_1, \ldots, x_{g-b+N}]],$$
$$\widehat{\mathcal{O}}_{X_d,A} \cong \mathbb{C}[[y_1, \ldots, y_{d(d-1)-d+N}]].$$

The unique component of X_d through A is the closure of the unmarked orbit of S_E. In every finite free associative deformation of A over a complete Noetherian local complex base $(B, \mathfrak{m}, \mathbb{C})$, every semisimple geometric fibre has the multiset of dimension vectors of simple modules given in (5.5), with respect to any lifts of the original projectors.

Proof. Theorem 5.12 gives smoothness at A in $X_E(V)$. For ordinary smoothness, forget the fixed idempotents in the existence theorem and apply Theorem 3.6, using Proposition 5.10 and $\dim Z(A) = N$. The dimension formulas then follow from the corresponding orbit dimensions and Cohen's structure theorem for the completed local rings of $X_E(V)$ and X_d at A.

For an arbitrary complete local associative family, choose a smooth irreducible open neighbourhood U_E of A avoiding all other components. Its distinguished semisimple orbit is dense. Every semisimple orbit meeting U_E is open, hence meets that dense orbit and equals it. This remains true after an algebraically closed field extension, since irreducible complex varieties are geometrically irreducible and the tangent and orbit calculations commute with field extension. By Lemma 5.11, the family defines a morphism $\operatorname{Spec} B \to X_E(V)$. It factors through U_E: an open subset of a local spectrum containing the closed point is the whole spectrum. This proves the assertion for every semisimple geometric fibre. □

Geometry constructs the matrix factors and the ranks of the original idempotents on their simple modules; orbit theory makes the resulting multiset of dimension vectors unique. The smooth open neighbourhood of A in X_d is used in Section 6 for arbitrary finite free associative families. It is separate from the geometric relative quotients of Section 4.

Remark 5.14 (What the geometric generic fibre determines). Theorem 5.12 determines the multiset of dimension vectors of the simple modules after a splitting extension. Over $\mathbb{C}((t))$, Galois action may permute simple factors with the same full dimension vector, since every e_i is defined over the ground field. Different integral families can therefore have isomorphic geometric generic fibres. For the basic algebra A, complete primitive idempotent systems are conjugate after a permutation of their labels. Hence the multiset of dimension vectors is intrinsic to A up to such a permutation. The underlying semisimple algebra retains only the aggregated numbers N_m. In particular, keeping the idempotents retains more information than the matrix sizes after they are forgotten.

5.4 Projector ranks in examples

Return to the A_3, $J = \{1, 3\}$ example of Subsection 4.3.7, and let a, b, c be the divisor intersection orders of its classifying arc. The geometrically semisimple algebra together with the fixed idempotents is

$$\mathbb{C}^a \times \mathbb{C}^b \times M_2(\mathbb{C})^c, \qquad e_1 = (1^a, 0^b, E_{11}^c), \quad e_3 = (0^a, 1^b, E_{22}^c).$$

The three vectors $(1, 0), (0, 1), (1, 1)$ recover both the matrix sizes and the Cartan matrix computed from the same universal presentations.

Consider the truncated two-cycle algebra with basis e_1, e_2, a, b, ab, ba, in which every path of length three is zero. It has

$$C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad b = 4, \quad g = 6, \quad \dim HH^0 = 3, \quad \dim HH^1 = \dim HH^2 = 1.$$

The relative cochain spaces in degrees zero through three have dimensions $4, 6, 12, 24$, and their differentials have ranks $1, 4, 7$. Thus the tangent dimension in $X_E(V)$ is 5, equal to the orbit dimension for the algebra with its fixed idempotents $\mathbb{C} \times \mathbb{C} \times M_2(\mathbb{C})$, with

$$e_1 = (1, 0, E_{11}), \qquad e_2 = (0, 1, E_{22}).$$

The accompanying exact rational calculation records these ranks and the two Gram decompositions. The example shows that the relative tangent formula agrees with the orbit dimension for the dimension vectors supplied by the construction.

6 Consequences for centres and their deformations

Section 5 proves that the contraction algebra A defines a smooth point of X_d. On a smooth open neighbourhood of that point, distinguish two questions. For an already given finite free associative family, how do its centre and cocenter vary? For a prescribed commutative deformation of the centre, does an associative deformation of the whole algebra realize it? The first question has a uniform positive answer near a contraction algebra. The first-order form of the second is governed by the centre tangent map, and the type D family below supplies a direction outside its image.

Conjecture 6.1 (Jelisiejew–Smoktunowicz [19, Conjecture 1.4])**.** *Let A be a contraction algebra and $d = \dim A$.*

(1) *The algebra A defines a smooth point of the moduli space of d-dimensional unital associative algebras; in other words, A is unobstructed.*

(2) *The dimension of the centre is locally constant near this point, and the map sending an algebra to its centre is smooth.*

For the completed smooth-threefold contractions considered here, the smoothness already proved establishes Conjecture 6.1 (1). The bundle theorem gives local constancy and arbitrary base change for centres and cocenters in already given associative families. The centre tangent map gives the first-order lifting criterion for prescribed commutative centre deformations, with an explicit type D obstruction. Write $c = N = \sum_\beta n_\beta$ when the common rank is used as the dimension of a commutative algebra law. In this section, a marking of a centre deformation means an identification of its special fibre with $Z(A)$; it is distinct from the labelled projectors used in Section 5.

Recall the fixed-unit law scheme X_d and its change-of-basis group G from Subsection 3.2. The points of X_d are unital associative products on a fixed d-dimensional vector space. Write $\mathcal{V} = V \otimes \mathcal{O}_{X_d}$ for the tautological algebra and $\mathcal{V}_U$ for its restriction to an open subscheme U.

The structure-constant space and its Hochschild tangent description come from associative deformation theory [12]; the fixed-unit form used here is reviewed in [19, Section 2.1 and Proposition 2.8]. The preceding description of semisimple fibres gives the local bundle statement, from which the centre morphism and the map χ_B are constructed.

6.1 Unobstructedness

The smooth local lifting argument for Conjecture 6.1 is phrased in the fixed-unit deformation framework of [19].

Corollary 6.2 (Unobstructedness)**.** *Let A be the basic contraction algebra of a completed projective flopping contraction of a smooth complex threefold. Put $d = \dim A$ and $c = \sum_\beta n_\beta$; for one curve this is $c = \sum_j n_j$. Then*

$$\widehat{\mathcal{O}}_{X_d,A} \cong \mathbb{C}[[t_1, \dots, t_{d(d-1)-d+c}]]. \tag{6.1}$$

Consequently A is unobstructed. If $R' \to R$ is a surjection of local Artinian $\mathbb{C}$-algebras with residue field $\mathbb{C}$, every finite free unital R-algebra $\mathscr{B}$ with $\mathscr{B} \otimes_R \mathbb{C} \cong A$ lifts to a finite free unital R'-algebra $\mathscr{B}'$ with $\mathscr{B}' \otimes_{R'} R \cong \mathscr{B}$. In particular, Conjecture 6.1(1) holds for these algebras.

Proof. The completed local-ring description follows from Corollary 5.13; in the one-curve case it already follows from Theorem 3.1.

For the lifting statement, interpret the multiplication on $\mathscr{B}$ as a point of X_d specializing to the smooth point A. Choose an R-basis of $\mathscr{B}$ whose first vector is 1 and which lifts the fixed basis of A; the unit is unimodular because it reduces to the unit of A. The structure constants define a homomorphism $\mathcal{O}(X_d) \to R$. It sends the maximal ideal of the point A into the nilpotent maximal ideal of R, and hence factors through $\widehat{\mathcal{O}}_{X_d,A}$. Lift the images of the variables t_i arbitrarily to R'. The resulting homomorphism $\widehat{\mathcal{O}}_{X_d,A} \to R'$ gives structure constants over R' satisfying the associativity and unit equations, hence an algebra $\mathscr{B}'$ reducing to $\mathscr{B}$. □

Remark 6.3. Smoothness of X_d at A gives the lifting property even when $HH^3(A)$ is nonzero [19, Section 1]. The number $d(d-1) - d + c$ is the dimension of the framed space of algebra laws with fixed identity. The deformation functor of A up to isomorphism has tangent space $HH^2(A)$, of dimension $\dim HH^1(A)$ by Lemma 3.10; the second assertion of Corollary 6.2 says that this functor is unobstructed.

6.2 Centres and cocenters in associative families

Section 4 proves freeness for relative contraction quotients arising from geometric families. The result below concerns an arbitrary finite free associative family with special fibre A. The smooth open neighbourhood of A in X_d transfers rank information from the geometric construction to these families. Hua–Toda's generic matrix decomposition and flatness question concern geometric families [16, Theorem 4.2 and Conjecture 4.3]; the smooth neighbourhood in X_d extends the centre and cocenter conclusions to associative families.

Theorem 6.4 (Centre and cocenter near A)**.** *Let A, d, and c be as in Corollary 6.2. There is a smooth irreducible open neighbourhood U of A in X_d with the following properties.*

(i) *The centre $\mathsf{Z}_U = Z(\mathcal{V}_U)$ and the cocenter $\mathsf{M}_U = HH_0(\mathcal{V}_U/\mathcal{O}_U)$ are locally free of rank c, and their formation commutes with arbitrary base change: for every morphism $g\colon U' \to U$, the natural maps $g^*\mathsf{Z}_U \to Z(g^*\mathcal{V}_U)$ and $g^*\mathsf{M}_U \to HH_0(g^*\mathcal{V}_U/\mathcal{O}_{U'})$ are isomorphisms.*

(ii) *Let τ_0 be a symmetrizing trace of A. After shrinking U, there is an $\mathcal{O}_U$-linear map $\tau\colon \mathcal{V}_U \to \mathcal{O}_U$ that vanishes on commutators, restricts to τ_0 at A, and for which $(a,b) \mapsto \tau(ab)$ is a perfect pairing. Every fibre of $\mathcal{V}_U$ is then a symmetric algebra, and*

$$\mathsf{Z}_U \xrightarrow{\sim} \mathsf{M}_U^{\vee}, \qquad z \longmapsto (\bar{a} \mapsto \tau(za)).$$

In particular, the dimension of the centre is constant on U, as the first half of Conjecture 6.1(2) predicts.

Proof. By Corollary 5.13, A is smooth in X_d, and the unique component of X_d through A is the closure of the unmarked orbit of S_E. Choose a smooth irreducible open neighbourhood U meeting no other component. Its semisimple locus is dense, and each semisimple fibre has c simple factors.

For the centre, consider the commutator operator

$$\kappa\colon \mathcal{V}_U \longrightarrow \mathcal{E}nd_{\mathcal{O}_U}(\mathcal{V}_U), \qquad a \longmapsto (b \mapsto ab - ba),$$

a map of free modules of ranks d and d^2 with kernel Z_U. At a point B, the rank of κ is $d-\dim Z(B)$. This rank is $d-c$ both on the semisimple locus, because S_E has c simple factors, and at A, by (5.4). All minors of size $d-c+1$ consequently vanish on the dense semisimple locus and hence on the reduced scheme U. Some minor of size $d-c$ is nonzero at A; shrink U so that it is a unit. A matrix with an invertible $r\times r$ minor and vanishing $(r+1)$-minors is equivalent to $\left(\begin{smallmatrix}1_r & 0\\ 0 & 0\end{smallmatrix}\right)$. Thus $\ker\kappa$ is a direct summand of rank c, and $\ker(g^*\kappa)=g^*\ker\kappa$ for every g. Since the centre of $g^*\mathcal{V}_U$ is $\ker(g^*\kappa)$, this proves the assertion for Z_U.

For the cocenter, use the parallel cokernel calculation with

$$b\colon \mathcal{V}_U\otimes_{\mathcal{O}_U}\mathcal{V}_U\longrightarrow\mathcal{V}_U,\qquad a\otimes a'\longmapsto aa'-a'a,$$

whose cokernel is M_U. The rank of b at B is $d-\dim HH_0(B)$, which equals $d-c$ on the semisimple locus and, by (5.4), at A. Thus M_U is locally free of rank c. Formation of a cokernel commutes with base change by right exactness.

For (ii), choose an extension of the functional τ_0 to a local section of $\mathsf{M}_U^\vee$, and compose it with $\mathcal{V}_U\to\mathsf{M}_U$. At the local ring of A, Lemma 4.4 makes the pairing defined by the extended trace perfect. Shrink U so that the determinant remains invertible. The same trace formula gives $\mathsf{Z}_U\cong\mathsf{M}_U^\vee$ throughout U and after base change. Every fibre is therefore symmetric. □

Corollary 6.5 (Invariants of arbitrary local associative families)**.** *Let $(B,\mathfrak{m},\mathbb{C})$ be a Noetherian local complex algebra, and let $\mathscr{D}$ be a finite free unital B-algebra with special fibre A. Then $Z(\mathscr{D})$ and $HH_0(\mathscr{D}/B)$ are free of rank c, commute with every commutative scalar extension $B\to B'$, and every symmetrizing trace of A lifts to a perfect trace on $\mathscr{D}$. The assertion applies to every local associative family specified here, including all complete local bases in the preceding sections.*

Proof. Choose a basis beginning with the unit and lifting a fixed basis of A. The structure constants of $\mathscr{D}$ in this basis define $\operatorname{Spec}B\to X_d$. This map factors through U, because an open subset of a local spectrum containing its closed point is the whole spectrum. Pull back Theorem 6.4; for a prescribed closed trace, use Lemma 4.4. □

In particular, these conclusions apply when R is a local Artinian $\mathbb{C}$-algebra with residue field $\mathbb{C}$ and $\mathscr{B}$ is a finite free unital R-algebra with $\mathscr{B}\otimes_R\mathbb{C}\cong A$, including every local Artinian scalar extension $R\to R''$. The base-change isomorphism $Z(\mathscr{B})\otimes_R\mathbb{C}\cong Z(A)$ shows that every central element of A lifts to a central element of $\mathscr{B}$.

Lifting central elements inside a fixed associative family is weaker than lifting an arbitrarily prescribed multiplication on the centre. The latter problem begins with the centre morphism below. The bundle theorem also determines the difference of the first two positive Hochschild dimensions throughout the smooth neighbourhood.

Corollary 6.6. *For every $B\in U(\mathbb{C})$, $\dim HH^1(B)=\dim HH^2(B)$.*

Proof. The scheme U is smooth and irreducible of dimension $d(d-1)-d+c$, so $\dim T_BX_d=d(d-1)-d+c$. As in the proof of Corollary 3.7, $\dim T_BX_d=d(d-1)-\dim\operatorname{Der}(B)+\dim HH^2(B)$ and $\dim\operatorname{Der}(B)=d-\dim Z(B)+\dim HH^1(B)$. Since $\dim Z(B)=c$ by Theorem 6.4, the claim follows. □

Remark 6.7 (Relation to geometric flatness). The universal presentations in Theorem 4.22 give cocenter freeness on geometric slices and the numerical input for marked uniqueness. The resulting orbit geometry supplies the smooth neighbourhood used here. The bundle theorem on this neighbourhood extends the centre and cocenter properties to arbitrary associative families.

6.3 The centre morphism

The centre bundle has constant rank but no preferred basis. A unit-preserving frame turns its multiplication into a point of a fixed scheme of commutative algebra laws. The fixed-unit scheme X_d used above is the structure-constant scheme of [19, Section 2.1 and Proposition 2.8]; see also [12, 11]. The commutative analogue of X_d is as follows.

For $c \geq 1$, let Com_c be the affine scheme of commutative associative multiplications on $\mathbb{C}^c$ with identity e_1, and let $G_c = \mathrm{Stab}_{\mathrm{GL}_c}(e_1)$, a smooth connected group of dimension $c(c-1)$. Poonen constructs the scheme of based finite-rank commutative algebras and its fixed-identity locus [29, Definition 1.5 and Proposition 3.1]; Com_c is the corresponding complex fixed-unit law scheme used here. For a commutative algebra Z, write $Z^2_{\mathrm{com}}(Z)$ for the symmetric normalized Hochschild 2-cocycles of Z with values in Z, and $B^2_{\mathrm{com}}(Z) = \delta(C^1_{\mathrm{norm}}(Z,Z))$, where $(\delta f)(a,b) = af(b) - f(ab) + f(a)b$. Put

$$T^1_{\mathrm{com}}(Z) = Z^2_{\mathrm{com}}(Z)/B^2_{\mathrm{com}}(Z).$$

This is the Harrison cohomology group governing first-order commutative deformations of Z [14]. The tangent calculation identifies this classical commutative deformation space with the framed law scheme used by the centre morphism.

Lemma 6.8 (Commutative tangent spaces)**.** *Let $Z \in \mathrm{Com}_c(\mathbb{C})$. Then*

$$T_Z\mathrm{Com}_c = Z^2_{\mathrm{com}}(Z), \qquad T_Z(G_c \cdot Z) = B^2_{\mathrm{com}}(Z),$$

and $T^1_{\mathrm{com}}(Z)$ is a subspace of $HH^2(Z)$. The G_c-orbits are the isomorphism classes. The orbit $G_c \cdot \mathbb{C}^c$ of the split étale algebra is open in Com_c and has dimension $c(c-1)$.

Proof. The fixed-unit calculation is the commutative specialization of Lemma 3.2. A dual-number multiplication $m + \varepsilon\psi$ remains commutative exactly when ψ is symmetric. Since Z is commutative, every normalized coboundary is symmetric, so $B^2_{\mathrm{com}}(Z)$ is the space of all normalized coboundaries and $T^1_{\mathrm{com}}(Z) \subset HH^2(Z)$. The algebra $\mathbb{C}^c$ is separable, so $HH^2(\mathbb{C}^c) = 0$ and its tangent space equals the orbit tangent space. Its stabilizer is the finite group permuting the primitive idempotents. The proof of Proposition 3.5 then shows that the orbit is open, of dimension $\dim G_c = c(c-1)$. □

We call the closure of $G_c \cdot \mathbb{C}^c$, with its reduced structure, the *smoothable component* $\mathrm{Sm}_c \subset \mathrm{Com}_c$. It is the component reached by degenerating the split c-point algebra, and it is irreducible of dimension $c(c-1)$.

A unit-preserving frame places a commutative deformation of the centre in Com_c, the target relevant to Conjecture 6.1(2). This follows the motivation in [19, Section 1], relating algebra deformations to commutative deformations of $Z(A)$. The centre bundle produces such a law from a given algebra family, and smoothness of the resulting map would lift a prescribed centre deformation. We test smoothness on Com_c and on its smoothable component. At the type D examples, Corollary 6.17 gives an obstruction even after restriction to the scheme-theoretic image.

Definition 6.9 (The centre morphism)**.** Let $U \subset X_d$ be an open subscheme on which the centre $\mathsf{Z}_U = Z(\mathcal{V}_U)$ is a locally free $\mathcal{O}_U$-module of rank c whose formation commutes with arbitrary base change. Let $\mathscr{F}_U \to U$ be the scheme of unit-preserving frames: for a U-scheme U', the U'-points of $\mathscr{F}_U$ are the isomorphisms $\mathcal{O}_{U'}^c \to \mathsf{Z}_U \otimes \mathcal{O}_{U'}$ sending e_1 to 1. The unit section spans a subbundle of Z_U, so $\mathscr{F}_U$ is a G_c-torsor that is locally trivial for the Zariski topology. Transporting the multiplication of Z_U along the tautological frame defines a G_c-equivariant morphism

$$\mathrm{cent}\colon \mathscr{F}_U \longrightarrow \mathrm{Com}_c.$$

We say that the *centre morphism is smooth at* $B \in U(\mathbb{C})$ if cent is smooth at a point of $\mathscr{F}_U$ over B. By equivariance, this does not depend on the choice of the point.

The just-defined smoothness condition is equivalently the smoothness at B of the morphism $U \to [\mathrm{Com}_c/G_c]$ classifying Z_U: the frame bundle $\mathscr{F}_U$ is its base change along the smooth atlas $\mathrm{Com}_c \to [\mathrm{Com}_c/G_c]$. The unit is unimodular in every local trivialization of Z_U, so its span is a local direct summand; this is what makes the unit-preserving frame bundle a G_c-torsor.

6.4 The centre tangent map and the smooth locus

Let U be as in Theorem 6.4, so that the centre morphism $\mathrm{cent}\colon \mathscr{F}_U \to \mathrm{Com}_c$ of Definition 6.9 is defined. Fix $B \in U(\mathbb{C})$ and put $Z_B = Z(B)$. For $\phi \in Z^2_{\mathrm{norm}}(B,B)$, let B_ϕ be $B[\varepsilon]$ with multiplication $aa' + \varepsilon\phi(a,a')$, where $\varepsilon^2 = 0$. By Lemma 3.2, these are the tangent vectors to X_d at B. The centre bundle lifts central elements through B_ϕ; their induced first-order multiplication, modulo changes of the chosen frame, defines χ_B.

Lemma 6.10 (The centre tangent map)**.** *Let $\phi \in Z^2_{\mathrm{norm}}(B,B)$.*

(i) *There is a linear map $\tau\colon Z_B \to B$ with $\tau(1) = 0$ such that $z + \varepsilon\tau(z)$ is central in B_ϕ for all $z \in Z_B$; equivalently,*

$$[\tau(z), a] = \phi(a,z) - \phi(z,a) \qquad (a \in B). \tag{6.2}$$

Any two such maps differ by a linear map $Z_B \to Z_B$ vanishing at 1*.*

(ii) *For such τ, the bilinear map*

$$\psi(z,z') = \phi(z,z') + \tau(z)z' + z\tau(z') - \tau(zz') \tag{6.3}$$

takes values in Z_B and lies in $Z^2_{\mathrm{com}}(Z_B)$. The map $z + \varepsilon z'' \mapsto z + \varepsilon(\tau(z) + z'')$ is an isomorphism from $Z_B[\varepsilon]$, with multiplication $zz' + \varepsilon\psi(z,z')$, onto $Z(B_\phi)$.

(iii) *The class of ψ in $T^1_{\mathrm{com}}(Z_B)$ depends only on the class of ϕ in $HH^2(B)$. This defines a linear map*

$$\chi_B\colon HH^2(B) \longrightarrow T^1_{\mathrm{com}}(Z_B),$$

the centre tangent map *of B.*

Proof. (i) The algebra B_ϕ gives a dual-number point of X_d with closed point in the open subscheme U, so the corresponding morphism factors through U. By Theorem 6.4(i), $Z(B_\phi)$ is free of rank c over $\mathbb{C}[\varepsilon]$ and reduces to Z_B. Every z thus lifts to a central element $z + \varepsilon b$; choose such lifts on a basis containing 1 to define τ. Comparing $(z + \varepsilon\tau(z))a$ with $a(z + \varepsilon\tau(z))$

in B_ϕ gives (6.2). An element εb is central in B_ϕ exactly when $b \in Z_B$, which proves the last assertion.

(ii) Multiply these chosen central lifts. The product of $z + \varepsilon\tau(z)$ and $z' + \varepsilon\tau(z')$ is

$$zz' + \varepsilon\big(\phi(z,z') + \tau(z)z' + z\tau(z')\big) = \big(zz' + \varepsilon\tau(zz')\big) + \varepsilon\psi(z,z').$$

It is central, so $\varepsilon\psi(z,z')$ is central and $\psi(z,z') \in Z_B$. Central elements commute, so ψ is symmetric, and $\psi(1,z') = 0$ because ϕ is normalized and $\tau(1) = 0$. The displayed map is $\mathbb{C}[\varepsilon]$-linear and bijective, since both sides are free of rank c and it reduces to the identity of Z_B. By the computation above it is multiplicative for the stated product. That product is therefore associative, so ψ is a Hochschild cocycle.

(iii) Finally, changing τ by $f\colon Z_B \to Z_B$ with $f(1) = 0$ changes ψ by δf. If $\phi = \delta g$ with $g \in C^1_{\mathrm{norm}}(B,B)$, then $1 + \varepsilon g$ is an algebra isomorphism $B_\phi \to B[\varepsilon]$. Hence $z - \varepsilon g(z)$ is central in B_ϕ, and the choice $\tau = -g|_{Z_B}$ in (6.3) gives $\psi = 0$. Both constructions are linear in ϕ. □

Proposition 6.11 (The differential of the centre morphism)**.** *Let $(B,\mathfrak{f})$ be a $\mathbb{C}$-point of $\mathscr{F}_U$, and identify $\mathbb{C}^c$ with Z_B by the frame $\mathfrak{f}$. The image of $d\mathrm{cent}\colon T_{(B,\mathfrak{f})}\mathscr{F}_U \to T_{Z_B}\mathrm{Com}_c = Z^2_{\mathrm{com}}(Z_B)$ is spanned by $B^2_{\mathrm{com}}(Z_B)$ and the cocycles (6.3). Consequently*

$$\operatorname{rank} d\mathrm{cent}_{(B,\mathfrak{f})} = \dim B^2_{\mathrm{com}}(Z_B) + \operatorname{rank}\chi_B, \tag{6.4}$$

and $d\mathrm{cent}_{(B,\mathfrak{f})}$ is surjective if and only if χ_B is surjective.

Proof. A tangent vector to $\mathscr{F}_U$ at $(B,\mathfrak{f})$ is a first-order deformation B_ϕ together with a unit-preserving frame of $Z(B_\phi)$ lifting $\mathfrak{f}$. By Lemma 6.10(i), these frames are $z \mapsto z + \varepsilon\tau(z)$, where τ runs through a torsor under $C^1_{\mathrm{norm}}(Z_B, Z_B) = \operatorname{Lie} G_c$. By Lemma 6.10(ii), $d\mathrm{cent}$ sends this tangent vector to ψ. Varying τ gives all of $\psi + B^2_{\mathrm{com}}(Z_B)$, which proves the description of the image. The image of $d\mathrm{cent}$ contains $B^2_{\mathrm{com}}(Z_B)$. Modulo $B^2_{\mathrm{com}}(Z_B)$, the image of $d\mathrm{cent}$ is the image of χ_B in $T^1_{\mathrm{com}}(Z_B)$. □

The rank formula separates changes of frame from the induced commutative deformation class. After quotienting the centre tangent space by frame directions, the remaining first-order obstruction is the cokernel of χ_B.

To infer smoothness of cent from this criterion, its source must be smooth. The frame bundle is smooth because it lies over the smooth neighbourhood U. The following lemma applies even when the target is singular.

Lemma 6.12 (A tangent criterion for smoothness)**.** *Let $g\colon X \to Y$ be a morphism of schemes of finite type over $\mathbb{C}$, let $x \in X(\mathbb{C})$ and $y = g(x)$, and suppose that X is smooth at x. Then g is smooth at x if and only if $dg_x\colon T_xX \to T_yY$ is surjective. The target Y may be singular.*

Proof. If g is smooth at x, the infinitesimal lifting property lifts every tangent vector at y to one at x. Conversely, assume that dg_x is surjective, and let $e = \dim T_yY$ and $N = \dim_x X$. Choose a surjection $\mathbb{C}[[y_1,\dots,y_e]] \to \widehat{\mathcal{O}}_{Y,y}$ with kernel $I \subset (y_1,\dots,y_e)^2$. The cotangent map is injective, so the images of the y_i extend to a regular system of parameters of $\widehat{\mathcal{O}}_{X,x} \cong \mathbb{C}[[x_1,\dots,x_N]]$; take $x_i = y_i$ for $i \le e$. The composite $\mathbb{C}[[y_1,\dots,y_e]] \to \widehat{\mathcal{O}}_{Y,y} \to \widehat{\mathcal{O}}_{X,x}$ is then injective, and it kills I; hence $I = 0$. Thus $\widehat{\mathcal{O}}_{X,x}$ is a power series ring over $\widehat{\mathcal{O}}_{Y,y} \cong \mathbb{C}[[y_1,\dots,y_e]]$. The ring $\widehat{\mathcal{O}}_{X,x}$ is flat over $\mathcal{O}_{Y,y}$, and since $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ is faithfully flat, $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,y}$. The fibre through x has completed local ring $\mathbb{C}[[x_{e+1},\dots,x_N]]$, so it is smooth at x. A morphism that is flat at x with fibre smooth at x is smooth at x [13, Théorème 17.5.1]. □

The smooth-source hypothesis cannot be omitted. For example, the closed immersion

$$\operatorname{Spec}\mathbb{C}[t]/(t^2) \longrightarrow \operatorname{Spec}\mathbb{C}[t]/(t^3)$$

induces an isomorphism on tangent spaces and is not flat.

Theorem 6.13 (The smooth locus of the centre morphism)**.** *Let U be as in Theorem 6.4 and $B \in U(\mathbb{C})$. The centre morphism is smooth at B if and only if the centre tangent map*

$$\chi_B \colon HH^2(B) \longrightarrow T^1_{\mathrm{com}}(Z(B))$$

is surjective. These points form the $\mathbb{C}$-points of a dense open subscheme $U^{\mathrm{cent}} \subset U$, which contains every semisimple algebra in U.

Proof. The torsor $\mathscr{F}_U$ is locally trivial with smooth fibre G_c over the smooth scheme U, so it is smooth. Lemma 6.12 and Proposition 6.11 give the criterion. The smooth locus of cent is open and, by equivariance, G_c-stable. The torsor projection is smooth and surjective, hence open, so the image of this locus in U is open; call it U^{cent}. If B is semisimple, then $HH^2(B) = 0$, and $Z(B) \cong \mathbb{C}^c$ has $T^1_{\mathrm{com}} = 0$ by Lemma 6.8; so χ_B is surjective. The semisimple locus is dense in U. □

The cokernel of χ_B is the first-order obstruction to smoothness of the centre morphism. The type D family below gives an explicit nonzero class at the original contraction algebra.

6.5 The centre-lifting obstruction in type D

For $n \geq 2$, consider the algebra $D_{n,2}$ of [19, Definition 3.2 and Proposition 3.3],

$$A_n := D_{n,2} = \frac{\mathbb{C}\langle x, y\rangle}{\left(xy + yx,\ y^2 + x^2 + x^{2n-1},\ x^{2n+1}\right)}. \tag{6.5}$$

It is the completed Jacobi algebra of the potential $\frac{1}{2n}x^{2n} + \frac{1}{3}x^3 + xy^2$; the ideal displayed in (6.5) contains a power of (x, y), so completion changes nothing [19, Proposition 3.3]. By Brown–Wemyss, A_n is the contraction algebra of a simple flop of a smooth threefold, of type D and length two [5, Theorems 1.5 and 8.11]. It has dimension $4n$ and can only deform to $M_2(\mathbb{C})^{\times(n-1)} \times \mathbb{C}^4$ [19, Theorems 3.7 and 3.13]. Theorem 3.1 therefore identifies its GV invariants as $n_1 = 4$ and $n_2 = n - 1$, with $c = n + 3$.

These cited results provide the presentation and geometric realization of A_n, and determine the possible geometrically semisimple generic fibre of an associative deformation of A_n. We use the presentation to construct a polynomial family of centres and prove the non-lifting assertion directly.

Throughout this subsection we write

$$\mathsf{u} = x^2, \qquad \mathsf{v} = x^{2n-1}, \qquad \mathsf{w} = x^{2n-2}y,$$

using sans-serif letters for elements of the centre. The symbols u and w continue to denote geometric parameters.

The obstruction uses the contrast between v and w. The former is a power of the noncentral element x, whereas the latter lies outside the subalgebra $\mathbb{C}[x]$. A commutative centre family makes v^2 acquire a w-component. The calculation below excludes this symbolically for every $n \geq 2$; the exact $n = 2$ calculation later addresses the stronger unmarked statement.

Lemma 6.14 (Structure of $D_{n,2}$)**.** *Let $n \geq 2$.*

(i) *In A_n, $x^{2n-1}y = 0$, and the elements $1, x, \dots, x^{2n}$ and $y, xy, \dots, x^{2n-2}y$ form a basis. With $x^i = 0$ for $i > 2n$ and $x^i y = 0$ for $i > 2n-2$,*

$$x^i y \cdot x^j = (-1)^j x^{i+j} y, \qquad x^i y \cdot x^j y = (-1)^{j+1}(x^{i+j+2} + x^{i+j+2n-1}).$$

(ii) *The centralizer of x is $\mathbb{C}[x] \oplus \mathbb{C}\mathsf{w}$, where $\mathbb{C}[x] = \operatorname{span}(1, x, \dots, x^{2n})$.*

(iii) *The centre is*

$$Z_n := Z(A_n) = \operatorname{span}(1, \mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{w}) \cong \frac{\mathbb{C}[\mathsf{u}, \mathsf{v}, \mathsf{w}]}{(\mathsf{u}^{n+1}, \mathsf{uv}, \mathsf{uw}, \mathsf{v}^2, \mathsf{vw}, \mathsf{w}^2)},$$

of dimension $n + 3$.

(iv) *$\mathsf{w}x = x\mathsf{w} = 0$, and $\mathsf{v}A_n + A_n\mathsf{v} \subseteq \mathbb{C}[x]$.*

Proof. (i) Since $yx^i = (-1)^i x^i y$, multiplying $y^2 = -x^2 - x^{2n-1}$ by y on the left and on the right gives $-x^2 y + x^{2n-1}y = -x^2 y - x^{2n-1}y$, so $x^{2n-1}y = 0$. The listed elements span, and they are linearly independent by [19, Proposition 3.3]. The products follow from $yx^j = (-1)^j x^j y$ and the relation for y^2.

(ii) Write $a = f(x) + g(x)y$. Then $[a, x] = [g(x)y, x] = -2xg(x)y$. By (i), this vanishes exactly when g is a multiple of x^{2n-2}.

(iii) By (ii), a central element has the form $f(x) + \lambda\mathsf{w}$. Since $2n-2$ is even, w commutes with y. Moreover $[f(x), y] = 2\sum_{i \text{ odd}} f_i x^i y$, which vanishes exactly when $f_i = 0$ for all odd $i \leq 2n-3$. This gives the displayed basis. The relations hold because $x^{2n+1} = 0$, $x^{2n-1}y = 0$ and $\mathsf{w}^2 = x^{4n-4}y^2 = -x^{4n-2} - x^{6n-5} = 0$; comparing dimensions gives the presentation.

(iv) This follows from the products in (i). □

Lemma 6.15 (A polynomial family of centres)**.** *Let*

$$\mathscr{Z}_n = \frac{\mathbb{C}[t, \mathsf{u}, \mathsf{v}, \mathsf{w}]}{(\mathsf{u}^{n+1}, \mathsf{uv}, \mathsf{uw}, \mathsf{vw}, \mathsf{w}^2,\ \mathsf{v}^2 - t\mathsf{w})}. \tag{6.6}$$

(i) *The algebra $\mathscr{Z}_n$ is free over $\mathbb{C}[t]$ with basis $1, \mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{w}$, and $\mathscr{Z}_n/(t) \cong Z_n$.*

(ii) *For $t \neq 0$, the fibre is isomorphic to $\mathbb{C}[\mathsf{u}, \mathsf{v}]/(\mathsf{u}^{n+1}, \mathsf{uv}, \mathsf{v}^3)$, which is smoothable.*

(iii) *In the basis of (i), $\mathscr{Z}_n$ defines a morphism $\mathbb{A}^1 \to \mathrm{Com}_{n+3}$ that factors through Sm_{n+3}. Its tangent vector at $t = 0$ is the cocycle $\psi_0 \in Z^2_{\mathrm{com}}(Z_n)$ with $\psi_0(\mathsf{v}, \mathsf{v}) = \mathsf{w}$ and $\psi_0 = 0$ on all other pairs of basis vectors.*

Proof. (i) Let the powers of u multiply as in $\mathbb{C}[\mathsf{u}]/(\mathsf{u}^{n+1})$, put $\mathsf{v}^2 = t\mathsf{w}$, and let all other products of two basis vectors other than 1 vanish. The element w annihilates $\mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{w}$, and v annihilates the powers of u. Hence every product of three such basis vectors is zero in either order, unless all three are powers of u. The multiplication is therefore associative and commutative. It satisfies the relations of (6.6), and those relations reduce every monomial to the stated basis; this gives inverse isomorphisms. Setting $t = 0$ recovers Lemma 6.14(iii).

(ii) For $t \neq 0$, eliminate $\mathsf{w} = t^{-1}\mathsf{v}^2$. The family

$$\frac{\mathbb{C}[h, \mathsf{u}, \mathsf{v}]}{(\mathsf{uv},\ \mathsf{u}^{n+1} - h^n\mathsf{u},\ \mathsf{v}^3 - h^2\mathsf{v})}$$

is free over $\mathbb{C}[h]$ with basis $1, \mathsf{u}, \dots, \mathsf{u}^n, \mathsf{v}, \mathsf{v}^2$. Indeed, it maps onto the fibre product over $\mathbb{C}[h]$ of $\mathbb{C}[h][\mathsf{u}]/(\mathsf{u}^{n+1} - h^n\mathsf{u})$ and $\mathbb{C}[h][\mathsf{v}]/(\mathsf{v}^3 - h^2\mathsf{v})$ along $\mathsf{u} \mapsto 0$ and $\mathsf{v} \mapsto 0$; that fibre product is free with the images of these elements as a basis, and they span the source. For $h \neq 0$, its fibre has the $n+3$ distinct points $(\mathsf{u}, \mathsf{v}) = (h\xi, 0)$ with $\xi^n = 1$, and $(0,0)$, $(0,h)$, $(0,-h)$, so it is isomorphic to $\mathbb{C}^{n+3}$. In this basis the family is a morphism $\mathbb{A}^1 \to \mathrm{Com}_{n+3}$ mapping $h \neq 0$ into the irreducible set $G_{n+3} \cdot \mathbb{C}^{n+3}$. Its fibre at $h = 0$, which is the algebra in (ii), therefore lies in the closure Sm_{n+3}.

(iii) By (ii), the points with $t \neq 0$ map to Sm_{n+3}, which is closed. Since $\mathbb{A}^1$ is reduced, the whole morphism factors through Sm_{n+3}. The structure constants are affine-linear in t, and the only t-dependent one is that of $\mathsf{v} \cdot \mathsf{v}$. □

Theorem 6.16 (A centre deformation that does not lift)**.** *Let $n \geq 2$, let $R_\varepsilon = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, and let $\mathscr{Z}_{n,\varepsilon} = \mathscr{Z}_n \otimes_{\mathbb{C}[t]} R_\varepsilon$ with $t \mapsto \varepsilon$. There is no finite free R_ε-algebra $\mathscr{B}$ with $\mathscr{B}/\varepsilon\mathscr{B} \cong A_n$ together with a homomorphism of R_ε-algebras $\iota\colon \mathscr{Z}_{n,\varepsilon} \to \mathscr{B}$ whose reduction modulo ε is the inclusion $Z_n \subset A_n$.*

The statement rules out even a lift of the central inclusion $Z_n \subset A_n$. If the displayed commutative family were the centre of an associative deformation with that marking, the inclusion of the deformed centre into the total algebra would give this homomorphism. This stronger algebraic statement therefore yields the centre-lifting obstruction.

Proof. Suppose that $(\mathscr{B}, \iota)$ exists. The argument compares two descriptions of the square of a lift of v: the relation on x forces it into $\varepsilon\mathbb{C}[x]$, whereas the proposed centre family forces it to equal $\varepsilon\mathsf{w}$. Since $\mathscr{B}$ is free over R_ε, multiplication by ε induces an isomorphism of A_n-bimodules

$$A_n \xrightarrow{\sim} \varepsilon\mathscr{B}, \qquad a \longmapsto \varepsilon\tilde{a}, \tag{6.7}$$

where $\tilde{a}$ is any lift of a. We write εa for the image of a. Choose a lift $\tilde{x} \in \mathscr{B}$ of x, and put $\tilde{\mathsf{v}} = \iota(\mathsf{v})$ and $\tilde{\mathsf{w}} = \iota(\mathsf{w})$.

Since $x^{2n+1} = 0$, we have $\tilde{x}^{2n+1} = \varepsilon b$ for some $b \in A_n$. As $\tilde{x}$ commutes with its own powers, $0 = [\tilde{x}^{2n+1}, \tilde{x}] = \varepsilon[b, x]$, so $[b, x] = 0$ by (6.7). Lemma 6.14(ii) gives $b \in \mathbb{C}[x] \oplus \mathbb{C}\mathsf{w}$.

Now $\tilde{\mathsf{v}}$ lifts $\mathsf{v} = x^{2n-1}$, so $\tilde{\mathsf{v}} = \tilde{x}^{2n-1} + \varepsilon g$ for some $g \in A_n$. Since $\varepsilon^2 = 0$ and $4n - 2 \geq 2n + 1$,

$$\tilde{\mathsf{v}}^{\,2} = \tilde{x}^{2n+1}\tilde{x}^{2n-3} + \varepsilon(\mathsf{v}g + g\mathsf{v}) = \varepsilon(bx^{2n-3} + \mathsf{v}g + g\mathsf{v}).$$

Here $2n - 3 \geq 1$ and $\mathsf{w}x = 0$, so $bx^{2n-3} \in \mathbb{C}[x]$, and $\mathsf{v}g + g\mathsf{v} \in \mathbb{C}[x]$ by Lemma 6.14(iv). Hence $\tilde{\mathsf{v}}^{\,2} \in \varepsilon\,\mathbb{C}[x]$. On the other hand, $\tilde{\mathsf{v}}^{\,2} = \iota(\mathsf{v}^2) = \iota(\varepsilon\mathsf{w}) = \varepsilon\tilde{\mathsf{w}} = \varepsilon\mathsf{w}$. By (6.7), $\mathsf{w} \in \mathbb{C}[x]$, which contradicts Lemma 6.14(i). □

Corollary 6.17 (The centre morphism is not smooth at $D_{n,2}$)**.** *Let $n \geq 2$, $c = n+3$, and let U be a neighbourhood of A_n as in Theorem 6.4.*

(i) *The class $[\psi_0]$ is not in the image of χ_{A_n}. Hence the centre morphism is not smooth at A_n, and $A_n \notin U^{\mathrm{cent}}$.*

(ii) *The centre morphism factors through Sm_c, which is the scheme-theoretic image of $\mathrm{cent}\colon \mathscr{F}_U \to \mathrm{Com}_c$. The morphism $\mathrm{cent}\colon \mathscr{F}_U \to \mathrm{Sm}_c$ is not smooth at the points over A_n either.*

(iii) *The family $\mathscr{Z}_n$ does not lift near $t = 0$: there is no finite free algebra over $\mathbb{C}[t]_{(t)}$, over the convergent power series in t, or over $\mathbb{C}[[t]]$, with special fibre A_n and centre identified with $\mathscr{Z}_n$ compatibly with $Z_n \subset A_n$.*

In particular, the second half of Conjecture 6.1(2) fails for A_n, although the first half holds by Theorem 6.4.

Proof. (i) Suppose that $[\psi_0] = \chi_{A_n}([\phi])$. By Lemma 6.10, there is a map τ for which (6.3) equals ψ_0, after changing τ by an element of $C^1_{\mathrm{norm}}(Z_n, Z_n)$. The map $z \mapsto z + \varepsilon\tau(z)$ is then an algebra homomorphism $\mathscr{Z}_{n,\varepsilon} \to (A_n)_\phi$ lifting $Z_n \subset A_n$, contradicting Theorem 6.16. The smoothness conclusion then follows from Theorem 6.13.

(ii) Over the dense semisimple locus of U, the centres are isomorphic to $\mathbb{C}^c$, so cent maps a dense open subscheme of the reduced scheme $\mathscr{F}_U$ into $G_c \cdot \mathbb{C}^c$. Hence cent factors through Sm_c. By equivariance its image contains the open orbit, so the scheme-theoretic image is Sm_c. Let $\mathfrak{f}$ be the frame of Z_n given by the basis of Lemma 6.15(i). By part (iii) of that lemma, ψ_0 is tangent to Sm_c at $\mathrm{cent}(A_n, \mathfrak{f})$. By (i) and Proposition 6.11, it is not in the image of $d\mathrm{cent}_{(A_n,\mathfrak{f})}$. So the differential of $\mathrm{cent}\colon \mathscr{F}_U \to \mathrm{Sm}_c$ at $(A_n, \mathfrak{f})$ is not surjective, and cent is not smooth there by Lemma 6.12; by equivariance, the same holds at every point over A_n.

(iii) Reducing such a lift modulo t^2 would give a pair $(\mathscr{B}, \iota)$ as excluded by Theorem 6.16. □

Remark 6.18. The socle element $\mathsf{v} = x^{2n-1}$ of Z_n is a power of the noncentral element x. In every first-order deformation, the square of a lift of v stays in the subalgebra generated by x, because of the relation $x^{2n+1} = 0$ and the shape of the centralizer of x. The commutative centre deformation $\mathsf{v}^2 = t\mathsf{w}$ instead gives the square a component along $\mathsf{w} = x^{2n-2}y$, outside that subalgebra.

The obstruction is first order. A ramified base change $t = h^r$ with $r \geq 2$ has zero first-order class, so lifts of that pullback are a higher-order problem.

6.6 Exact tangent data for $D_{2,2}$

The preceding symbolic obstruction applies to every $n \geq 2$. For $n = 2$, exact linear algebra computes the full first-order image of χ_A. This image calculation is used only for the stronger conclusion without a chosen marking. Here $A = A_2$ has basis $1, x, x^2, x^3, x^4, y, xy, x^2y$, and Z_2 has basis $1, \mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q}$ with $\mathsf{u} = x^2$, $\mathsf{v} = x^3$, $\mathsf{w} = x^2y$ and $\mathsf{q} = \mathsf{u}^2 = x^4$. The only nonzero product of two of $\mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q}$ is $\mathsf{u}^2 = \mathsf{q}$.

Proposition 6.19 (Tangent data of $D_{2,2}$)**.** *For $A = D_{2,2}$ and $Z_2 = Z(A)$, the following hold.*

(i) $\dim \mathrm{Der}(A) = 6$, $\dim HH^1(A) = \dim HH^2(A) = 3$, *and* $\dim Z^2_{\mathrm{norm}}(A,A) = 53 = \dim_A X_8$.

(ii) $\dim B^2_{\mathrm{com}}(Z_2) = 10$, $\dim Z^2_{\mathrm{com}}(Z_2) = 26$, *and* $\dim T^1_{\mathrm{com}}(Z_2) = 16$.

(iii) *The centre tangent map χ_A has rank 3. Its image is spanned by three explicitly given cocycle classes $[\psi_1], [\psi_2], [\psi_3]$, recorded in Appendix C. The representatives satisfy $\psi_i(s,s) \in \mathbb{C}\mathsf{q}$ for every $s \in \mathrm{Soc}(Z_2)$.*

(iv) *At every point of $\mathscr{F}_U$ over A,* $\operatorname{rank} d\mathrm{cent} = 10+3 = 13$, *whereas* $\dim T_{Z_2}\mathrm{Sm}_5 \geq \dim \mathrm{Sm}_5 = 20$ *and* $\dim T_{Z_2}\mathrm{Com}_5 = 26$.

Proof. Parts (i)–(iii) are exact linear algebra over $\mathbb{Q}$, described in Appendix C. Part (iv) follows from (ii), (iii), Proposition 6.11, and Lemma 6.8. □

At $D_{2,2}$, the differential of the centre morphism misses at least seven dimensions of the tangent space of the scheme-theoretic image of the centre morphism. Theorem 6.16 fixes the special-fibre identification $\mathscr{Z}_{n,\varepsilon}/(\varepsilon) \cong Z_n$. For $n = 2$, the socle condition below is invariant under every automorphism of the centre, so the exact tangent calculation removes that marking from the conclusion.

Proposition 6.20 (Non-lifting without a marking)**.** *No first-order deformation of $A = D_{2,2}$ over R_ε has centre isomorphic to $\mathscr{Z}_{2,\varepsilon}$ as an R_ε-algebra. Equivalently, the* $\mathrm{Aut}(Z_2)$*-orbit of $[\psi_0]$ does not meet the image of χ_A.*

Proof. Let $\mathfrak{m} = (\mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q})$ be the maximal ideal of Z_2. Then $\mathfrak{m}^2 = \mathbb{C}\mathsf{q}$ and $\mathrm{Soc}(Z_2) = \mathbb{C}\mathsf{v} \oplus \mathbb{C}\mathsf{w} \oplus \mathbb{C}\mathsf{q}$. Consider the following linear condition on $\psi \in Z^2_{\mathrm{com}}(Z_2)$:

$$\psi(s,s) \in \mathbb{C}s + \mathfrak{m}^2 \qquad \text{for every } s \in \mathrm{Soc}(Z_2). \tag{6.8}$$

Condition (6.8) depends only on the class of ψ: for $s \in \mathrm{Soc}(Z_2)$ we have $s^2 = 0$ and $\mathfrak{m}s = 0$, so $(\delta f)(s,s) = 2sf(s)$ is a multiple of s for every $f \in C^1_{\mathrm{norm}}(Z_2, Z_2)$. The condition is invariant under $\mathrm{Aut}(Z_2)$, which preserves $\mathrm{Soc}(Z_2)$ and $\mathfrak{m}^2$. By Proposition 6.19(iii), the image of χ_A satisfies (6.8): indeed $\psi_i(s,s) \in \mathbb{C}\mathsf{q}$ for all $s \in \mathrm{Soc}(Z_2)$ and $i = 1,2,3$. The class $[\psi_0]$ violates (6.8), since $\psi_0(\mathsf{v},\mathsf{v}) = \mathsf{w} \notin \mathbb{C}\mathsf{v} + \mathbb{C}\mathsf{q}$.

Now let $\mathscr{B}$ be a first-order deformation of A, and suppose that $Z(\mathscr{B}) \cong \mathscr{Z}_{2,\varepsilon}$. By Corollary 6.5, $Z(\mathscr{B})$ reduces to Z_2, and with this marking it represents a class $\chi_A([\phi])$. The isomorphism reduces to an automorphism σ of Z_2 and identifies $[\psi_0]$ with the transport of $\chi_A([\phi])$ by σ. This contradicts the invariance of (6.8). □

Centres in families and the scheme of algebra laws. The smoothness proved in Section 5 gives an open neighbourhood of A in X_d on which centre and cocenter have constant rank. Corollary 6.2 and Theorem 6.4 control deformations that are already associative. A prescribed commutative centre deformation has the additional first-order condition that its class lie in the image of χ. Corollary 6.17 shows that at $D_{n,2}$, $n \geq 2$, the image of the centre tangent map misses a direction, even within the smoothable component and the scheme-theoretic image.

The type A and D algebras studied in [19] belong to this geometric class [5, Theorem 1.5]. The type D examples show that base-change compatibility of centres in associative families does not guarantee that a prescribed commutative centre deformation lifts.

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For this reason, I do not wish the existence or priority date of this manuscript to discourage a genuinely independent human derivation of these results. If a subsequent work obtains the same results through a mathematical argument developed independently and entirely by humans, I make no claim to special priority or intellectual credit for the corresponding proof ideas beyond the ordinary bibliographic acknowledgement of prior or concurrent results.

If, however, a subsequent argument makes substantive use of the results, constructions, or proof strategies contained in this manuscript, ordinary scholarly citation remains appropriate regardless of the tools used. In particular, I would appreciate explicit attribution when closely related arguments appear in AI-assisted work, since the provenance of ideas generated through overlapping language-model systems can otherwise be difficult to determine.

A Root enumeration for the primitive marked pairs

The root enumeration verifies the assertion used in Lemma 2.5: each nonzero marked-coefficient stratum is one Levi Weyl-group orbit. Karmazyn's primitive presentations determine the six marked pairs [20, Examples 4.22–4.27]. This appendix records the exact root-and-orbit calculation and the diagram-numbering convention it uses. Number the vertices in the diagrams of Figure 2 from left to right along the horizontal chain, and number the lower vertex last. Thus the retained vertices are $1, 2, 3, 3, 4, 3$ for lengths $1, \dots, 6$, respectively. The enumeration uses this diagram numbering. The table in the proof of Theorem 3.12 instead uses Karmazyn's numbering; the two conventions must therefore not be mixed.

Represent a root by its vector $v = (v_1, \dots, v_r)$ in the simple-root basis. If $i \sim h$ denotes adjacency in the diagram, the simple reflection acts by

$$s_i(v)_h = v_h \quad (h \neq i), \qquad s_i(v)_i = -v_i + \sum_{h \sim i} v_h.$$

Starting from the simple roots, repeatedly apply every reflection s_i and retain each new vector once. Since the ADE root system is finite, this terminates and gives the full Weyl orbits of the simple roots, hence the entire root system. Keeping the vectors with nonnegative coordinates produces Φ^+.

For each marked coefficient $j > 0$, choose a positive root with $v_p = j$ and repeat the procedure using only reflections with $i \neq p$. The resulting orbit agrees, in all six primitive cases, with the complete set of positive roots of coefficient j. The numbers of positive roots are

$$1, \quad 12, \quad 36, \quad 63, \quad 120, \quad 120,$$

and the non-Levi totals are

$$N_p = 1, \quad 9, \quad 29, \quad 53, \quad 104, \quad 106.$$

The individual orbit cardinalities are those of Lemma 2.5. The same calculation checks that the largest marked coefficient is the stated length.

The integer implementation is included in `tools/check_mathematics.py` in the source distribution. At every step it verifies

$$2\sum_i v_i^2 - 2\sum_{\{i,h\}\text{ an edge}} v_i v_h = 2.$$

Integer vector operations make the enumeration exact. Distinct coefficient orbits give disjoint sets of hyperplanes, because proportional roots in a reduced root system differ only by sign. The curve divisors descended from these disjoint sets of hyperplanes are therefore distinct.

B The coefficient-five E_8 calculation

B.1 The ρ-line relations

This appendix supplies the direct coefficient-five E_8 calculation. It derives the homogeneous relations and the relation matrices whose modular ranks bound the cocenter. Together with the tail and generic-rank arguments in Subsection 3.6, these bounds give cocenter dimension 104 for the remaining closed corner. The central extension is that of [9, Theorem 3.2], and the form of the nodal-corner relations comes from [10, Section 3]. The coefficient-five specialization, relation reduction, and exact row-rank certificate are carried out here. We use the normalization $\rho_i = (\rho, \alpha_i) = 1$ from Subsection 3.5. The algebra Π_0^ρ is the path algebra over $\mathbb{C}[z]$ of the double of a finite E_8 quiver Q, with z central, modulo

$$\sum_{a \in Q_1} [a, a^*] = z \sum_i e_i.$$

Different orientations of Q give isomorphic algebras [10, Section 2.1]. We compose paths from left to right. For an edge joining v and v', let $\ell_{v,v'}$ denote the summand of the left-hand side that is a loop at v through v'. It is aa^* or $-a^*a$, according to the orientation of the arrow a of Q joining them. The relation at v reads $\sum_{v' \sim v} \ell_{v,v'} = z\, e_v$. If $p\colon v \to v'$ and $q\colon v' \to v$ are the two arrows of the double quiver, then $\ell_{v,v'} = \pm pq$ and $\ell_{v',v} = \mp qp$; hence

$$\ell_{v,v'}\, p = -p\, \ell_{v',v}, \qquad q\, \ell_{v,v'} = -\ell_{v',v}\, q. \tag{B.1}$$

Lemma B.1 (Arms)**.** *Let $v_0, v_1, \dots, v_r$ be a chain in the Dynkin graph such that $v_1, \dots, v_r$ have no neighbours other than those in the chain. Then $X = \ell_{v_0,v_1}$ satisfies*

$$X(X+z)(X+2z)\cdots(X+rz) = 0.$$

Proof. Put $Y = \ell_{v_1,v_0}$. By (B.1), $qf(X) = f(-Y)q$ for every polynomial f over $\mathbb{C}[z]$; hence $Xf(X) = \pm pf(-Y)q$. It is therefore enough to find a degree-r polynomial f with $f(-Y) = 0$. If $r = 1$, the relation at v_1 is $Y = z$, and $f(T) = T + z$ has the required property. If $r > 1$, the relation at v_1 gives $\ell_{v_1,v_2} = z - Y$, and the induction hypothesis gives $\prod_{m=0}^{r-1}(\ell_{v_1,v_2} + mz) = 0$. Thus $\prod_{m=1}^{r}(mz - Y) = 0$, and $f(T) = \prod_{m=1}^{r}(T + mz)$ works. □

The nodal corners of Etingof–Latour–Rains are presented by relations of the same form [10, equation (3.1)]. We number the vertices of E_8 as in [20, Example 4.26]: the finite diagram is the chain $1,2,3,4,8,7,6$ with 5 attached to the branching vertex 8, and the retained vertex is $p=4$. Choose Q so that its arrow a between 4 and 8 points from 4 to 8, and put

$$d=-\ell_{4,3}, \qquad b=-\ell_{8,5}, \qquad c=-\ell_{8,7}.$$

Apply Lemma B.1 to the three arms, of lengths 3, 1, and 2, and then use the relations at vertices 4 and 8. This gives

$$\begin{aligned} d(d-z)(d-2z)(d-3z)=0, \qquad b(b-z)=0, \qquad c(c-z)(c-2z)=0,\\ aa^*=d+z\,e_4, \qquad a^*a+b+c+z\,e_8=0. \end{aligned} \tag{B.2}$$

These are the relations of Karmazyn's intermediate presentation [20, Example 4.26] at $t_i=z$, after the free vertex has been removed. The following change of generators is the form used for the closed-corner calculation. Put

$$B=b-2z\,e_8, \qquad C=c, \qquad H=a^*a=-B-C-3z\,e_8,$$

so that

$$B^2=-3zB-2z^2, \qquad C^3=3zC^2-2z^2C. \tag{B.3}$$

In the corner $B_5=e_4\Pi_0^{\rho}e_4$, define

$$\beta=aa^*+3z\,e_4=d+4z\,e_4, \qquad \gamma=aBa^*-2z^2e_4.$$

Lemma B.2 (Generators)**.** *The elements z,β,γ generate the algebra B_5.*

Proof. We reduce closed paths alternately at vertices 4 and 8. A closed path at 4 is a product of paths meeting 4 only at their ends. Those entering the arm $\{1,2,3\}$ are polynomials in d and z, by the arm relations and (B.1); the remaining paths have the form aqa^* with q closed at 8. Similarly, a closed path at 8 is a product of polynomials in b,z, polynomials in c,z, and elements a^*ra with r closed at 4. Since $da=a(H-z)$ by (B.2), we have $a^*d^ka=H(H-z)^k$. Induction on path length now shows that every closed path at 8 is a polynomial in B, C, and z. Substitute $C=-a^*a-B-3z$ in an element $aq(B,C)a^*$. Each occurrence of a^*a splits the expression into a product of two elements of the same form, while powers of B reduce by (B.3). Thus B_5 is generated by z, aa^*, and aBa^*, equivalently by z, β, and γ. □

Lemma B.3 (Relations)**.** *In B_5, the elements*

$$\begin{aligned} \mathsf{P}&=(\beta-4z)(\beta-5z)(\beta-6z)(\beta-7z),\\ \mathsf{R}_4&=(\gamma+\beta^2)^2+\beta\gamma\beta-5z^2(\gamma+\beta^2)+4z^4,\\ \mathsf{R}_5&=\gamma\beta\gamma+\gamma^2\beta+\gamma\beta^3-5z^2\gamma\beta \end{aligned}$$

vanish.

Proof. Since $\beta=d+4z$, the element P is $d(d-z)(d-2z)(d-3z)=0$. For the remaining two relations, multiply elements of the form $s\,e_4+aPa^*$, with $s\in\mathbb{C}[z]$ and P a noncommutative polynomial in B and C, using the rule

$$(s\,e_4+aPa^*)(s'e_4+aP'a^*)=ss'e_4+a(sP'+s'P+PHP')a^*.$$

Only (B.3) and $H = -B - C - 3z$ enter the reduction. They give $HB = 2z^2 - CB$, $BH = 2z^2 - BC$ and

$$\gamma + \beta^2 = 7z^2 + a(3z - C)a^*, \qquad \beta\gamma = -6z^3 + a(3z - C)Ba^*, \qquad \gamma\beta = -6z^3 + aB(3z - C)a^*.$$

For R_5, they give $\beta\gamma + (\gamma + \beta^2)\beta - 5z^2\beta = a(C - z)(C - 2z)a^*$. Hence

$$\mathsf{R}_5 = \gamma(\beta\gamma + (\gamma + \beta^2)\beta - 5z^2\beta) = a(BH - 2z^2)(C - z)(C - 2z)a^* = -aBC(C - z)(C - 2z)a^*,$$

which vanishes by (B.3). For R_4, put $g = \gamma + \beta^2$, so that $\mathsf{R}_4 = (g - z^2)(g - 4z^2) + \beta\gamma\beta$. The same rules give

$$\begin{aligned}(g - z^2)(g - 4z^2) &= 18z^4 + a(27z^3 - 9z^2C + (3z - C)H(3z - C))a^*,\\ (3z - C)H(3z - C) &= -27z^3 - 9z^2B + 11z^2C + 3zBC + 3zCB - CBC,\\ \beta\gamma\beta &= -18z^4 + a(9z^2B - 2z^2C - 3zBC - 3zCB + CBC)a^*,\end{aligned}$$

where the second line uses $C^3 = 3zC^2 - 2z^2C$. The sum is zero. □

The program `tools/check_e8_rho_relations_formal.py` independently verifies these reductions over the polynomial ring $\mathbb{Z}[z]$, with z retained as a variable.

Remark B.4 (Sign conventions in [20, Example 4.26]). This convention check matters because the rank matrices below use $\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5$ exactly as displayed in Lemma B.3. Those relations agree with the last two relations of Karmazyn's intermediate presentation, written in the invariants R_1, R_2, R_3, S_2 of the five-arm parameters $\tau_1, \dots, \tau_5$. On the ρ-line, $(\tau_1, \dots, \tau_5) = (2, 1, 0, -1, -2)z$ and $(R_1, R_2, R_3, S_2) = (3z, 2z^2, 0, 2z^2)$; here $\beta = a_4a_4^* + R_1$ and $\gamma = a_4Ba_4^* - S_2$. For the quadratic and cubic relations displayed there, the shifts must be $B = b + \tau_5$ and $C = c + \tau_3$. The final display of [20, Example 4.26] rewrites these relations through $T_3 = R_2 + S_2 + R_1S_1$, $T_2 = R_3 + R_2S_1 + R_1S_2$, $T_1 = -R_3R_1 + R_2S_2$ and $T_0 = R_3S_2$. The expressions T_3, T_2, T_1, T_0 are $+\sigma_2, \dots, +\sigma_5$ in the parameters τ_i, whereas Karmazyn's list of invariant generators sets $T_i = -\sigma(\tau)$. Substituting the sign convention $T_i = -\sigma(\tau)$ into Karmazyn's final display reverses the signs of the parameter terms and gives

$$\mathsf{R}_4' = (\gamma + \beta^2)^2 + \beta\gamma\beta + 5z^2(\gamma + \beta^2) - 4z^4, \qquad \mathsf{R}_5' = \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 + 5z^2\gamma\beta.$$

These are not relations of B_5. In degree four, the relation rows of Q_ρ are $[\mathsf{P}]$ and $[\mathsf{R}_4]$. By Subsection 3.6.1, the surjection $HH_0(Q_\rho)_4 \to HH_0(B_5)_4$ is an isomorphism, so the classes of degree four that vanish in $HH_0(B_5)$ are exactly the span of $[\mathsf{P}]$ and $[\mathsf{R}_4]$. In cyclic words,

$$\begin{aligned}[\mathsf{P}] &= [\beta^4] - 22z[\beta^3] + 179z^2[\beta^2] - 638z^3[\beta] + 840z^4,\\ [\mathsf{R}_4] &= [\gamma^2] + 3[\beta^2\gamma] + [\beta^4] - 5z^2[\gamma] - 5z^2[\beta^2] + 4z^4,\end{aligned}$$

and $[\mathsf{R}_4'] - [\mathsf{R}_4] = 10z^2[\gamma] + 10z^2[\beta^2] - 8z^4$. If $[\mathsf{R}_4']$ were in the span, comparing the coefficients of $[\gamma^2]$ and of $z[\beta^3]$ would force $[\mathsf{R}_4'] = [\mathsf{R}_4]$, which is false. Hence $\mathsf{R}_4' \neq 0$ in B_5.

The rank table below is computed from $\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5$. The companion repository at commit `5a2001f` documents the earlier R_4', R_5' convention in the archived E_8 checker sources, the recorded E_8 outputs, and the E_8 computation and reproduction guide.

B.2 Relation matrices and modular rank bounds

The C++17 program `tools/check_e8_corrected_rho.cpp` constructs the finite relation matrices used in Proposition 3.13. In each half-path degree $m \le 23$, it enumerates the cyclic-word columns $z^a[w]_{\mathrm{cyc}}$ and the rows $[\mathsf{R}w]_{\mathrm{cyc}}$ for $\mathsf{R} \in \{\mathsf{P}, \mathsf{R}_4, \mathsf{R}_5\}$ and words w, then performs exact Gaussian elimination over $\mathbb{F}_{1000003}$. The command `make math` compiles the program, repeats the calculation over $\mathbb{F}_{1000033}$ and compares both outputs with the table below. Agreement of the two ranks is a reproducibility check. For the characteristic-zero estimate, a nonzero minor modulo either prime remains nonzero over $\mathbb{Q}$. Hence r_m is a lower bound for the characteristic-zero relation rank and $u_m = c_m - r_m$ is an upper bound for the characteristic-zero cocenter nullity. The tail and generic-rank arguments in Subsection 3.6 establish the exact total dimension. The table lists m, c_m, r_m, and u_m.

Degree m	Columns c_m	Rank r_m over $\mathbb{F}_{1000003}$, $\mathbb{F}_{1000033}$	Nullity bound u_m
0	1	0	1
1	2	0	2
2	4	0	4
3	6	0	6
4	9	2	7
5	12	5	7
6	17	10	7
7	22	16	6
8	30	24	6
9	40	34	6
10	55	49	6
11	74	69	5
12	105	100	5
13	146	142	4
14	210	206	4
15	304	300	4
16	447	443	4
17	658	655	3
18	987	984	3
19	1480	1478	2
20	2246	2244	2
21	3416	3414	2
22	5227	5225	2
23	8014	8013	1

The SHA-256 checksums of the two E_8 programs are:

File	SHA-256
`check_e8_corrected_rho.cpp`	`2545d84e337ea8b4ac650c428b90880a36ab47cd168e8935e0d4598ac1dbabd9`
`check_e8_rho_relations_formal.py`	`208305e7e5a7613b75b66959b4ac7022c31644b54c988cbd642baad7382d5f1a`

C Exact linear algebra for $D_{2,2}$

Theorem 6.16 proves the centre-lifting obstruction symbolically for every $D_{n,2}$. This appendix gives an independent exact-rational calculation for $D_{2,2}$: it computes the tangent data in Proposition 6.19 and the automorphism-invariant socle condition (6.8) used in Proposition 6.20. The program `tools/verify_conjecture_1_4_counterexample.py` computes the cochain ranks, and `tools/check_d22_tangent_image.py` verifies the tangent image and socle condition. Both

use exact rational arithmetic (`fractions.Fraction` in Python) and are run by `make math`. They provide the finite $n = 2$ certificate; Theorem 6.16 and Corollary 6.17 remain the uniform symbolic arguments for every $n \geq 2$.

Cochains and ranks. Fix the bases and normalized cochain complexes. The algebra $A = D_{2,2}$ is given in the basis $(1, x, x^2, x^3, x^4, y, xy, x^2y)$ by the multiplication table of Lemma 6.14(i), and the first program checks associativity and the unit on all triples of basis vectors. The centre Z_2 has the basis $(1, \mathsf{u}, \mathsf{v}, \mathsf{w}, \mathsf{q})$, embedded as $(1, x^2, x^3, x^2y, x^4)$, and the same program checks that the embedding is multiplicative. Normalized cochains are multilinear maps on the span of the basis vectors other than 1; for Z_2, the second cochains are symmetric. The coboundary maps are sparse rational matrices. Exact row reduction gives the following dimensions.

	A, associative	Z_2, commutative
normalized 1-cochains	56	20
rank of δ on 1-cochains	50	10
normalized 2-cochains	392	50 (symmetric)
rank of δ on 2-cochains	339	24
2-cocycles	53	26
cocycles modulo coboundaries	$HH^2(A)$: 3	$T^1_{\mathrm{com}}(Z_2)$: 16

The derivations of A form the kernel on 1-cochains, of dimension $56 - 50 = 6$; inner derivations have dimension $8 - 5 = 3$, so $\dim HH^1(A) = 3$.

The centre tangent map. For each of the 53 basis cocycles ϕ of $Z^2_{\mathrm{norm}}(A, A)$, the first program solves the linear system (6.2) for τ with values in the chosen complement $\mathrm{span}(x, y, xy)$ of Z_2. This complement fixes a representative for the calculation. The first program checks that the solution exists and is unique, forms ψ by (6.3), and checks that ψ is symmetric, takes values in Z_2, and is a cocycle. It also verifies that the coboundaries of A map to commutative coboundaries and that the 53 cocycles ψ have rank 3 modulo $B^2_{\mathrm{com}}(Z_2)$. Finally, the coefficient of w in $\psi(\mathsf{v}, \mathsf{v})$ vanishes on all these cocycles and on $B^2_{\mathrm{com}}(Z_2)$, but equals 1 on ψ_0. This is an independent exact verification of Corollary 6.17(i) for $n = 2$.

Representatives and the socle condition. The representatives in Proposition 6.19(iii) are normalized and vanish on every pair not displayed below, apart from the values imposed by symmetry:

$$\begin{aligned}
&\psi_1(\mathsf{u}, \mathsf{w}) = -\mathsf{w}, \quad \psi_1(\mathsf{u}, \mathsf{q}) = -2\mathsf{q}, \quad \psi_1(\mathsf{v}, \mathsf{v}) = \mathsf{q};\\
&\psi_2(\mathsf{u}, \mathsf{v}) = -\tfrac{1}{2}\mathsf{w}, \quad \psi_2(\mathsf{u}, \mathsf{w}) = \tfrac{1}{2}\mathsf{v}, \quad \psi_2(\mathsf{v}, \mathsf{w}) = \tfrac{1}{2}\mathsf{q};\\
&\psi_3(\mathsf{u}, \mathsf{w}) = -2\mathsf{w}, \quad \psi_3(\mathsf{u}, \mathsf{q}) = -2\mathsf{q}, \quad \psi_3(\mathsf{w}, \mathsf{w}) = 2\mathsf{q},
\end{aligned}$$

These formulas also give $\psi_i(s, s) \in \mathbb{C}\mathsf{q}$ for every $s \in \mathrm{Soc}(Z_2)$.

The second program checks that ψ_1, ψ_2, ψ_3 of Proposition 6.19(iii) are commutative cocycles whose classes lie in the image of χ_A and span it. Reducing them modulo $B^2_{\mathrm{com}}(Z_2)$ by the reduced row-echelon basis of the coboundaries leaves the coefficients of w in $\psi(\mathsf{u}, \mathsf{v})$, of w in $\psi(\mathsf{u}, \mathsf{w})$, and of q in $\psi(\mathsf{u}, \mathsf{q})$ as non-pivot coordinates. The coefficient vectors of ψ_1, ψ_2, ψ_3 in these three coordinates form the matrix

$$\begin{pmatrix} 0 & -1 & -2 \\ -\frac{1}{2} & 0 & 0 \\ 0 & -2 & -2 \end{pmatrix}, \qquad \det = -1,$$

whose rows correspond to ψ_1, ψ_2, ψ_3. The program then verifies (6.8) for all 53 image cocycles and for the 20 generators of $B^2_{\mathrm{com}}(Z_2)$, as a polynomial identity in the coordinates of $s = \alpha\mathsf{v}+\beta\mathsf{w}+\gamma\mathsf{q}$, and checks that ψ_0 violates (6.8).

Uniform checks. For $n = 2, \dots, 6$, the first program also checks the identities of Lemma 6.14 used in the proof of Theorem 6.16: $\dim A_n = 4n$, $\dim Z(A_n) = n+3$, the centralizer of x has dimension $2n+2$, $\mathsf{w}x = x\mathsf{w} = 0$, and $\mathsf{v}A_n + A_n\mathsf{v} \subseteq \mathbb{C}[x]$. It also checks that the multiplication of (6.6) for $n = 2$ is associative over $\mathbb{Q}[t]$. These are finite checks of identities whose general proof is already given in Lemma 6.14.

The SHA-256 checksums of the two programs are:

File	SHA-256
`verify_conjecture_1_4_counterexample.py`	`e51dd6b0816f019f3cd2f38c9fb12312ec4b9f49d18323a18748055bdbb7ccef`
`check_d22_tangent_image.py`	`2140afdf251c2fb1f123a02cdb92fac87b4d029e822f336a4a2ca1e9e92dc9ae`

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